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FMCW Radar Target Detection

CFAR vs Deep Learning on Range–Doppler Maps

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Abstract

Constant false-alarm rate (CFAR) is a standard method for radar target detection, while deep-learning detectors are increasingly reported to outperform it. However, these comparisons are often difficult to interpret because the methods may use different inputs, different spatial context, or different operating points. More importantly, a gain measured in simulation may not survive on real radar data. This thesis therefore asks a broader question: under what conditions can a learned detector outperform CFAR?

A small 5,041-parameter convolutional network, TinyCNN, is compared with four CFAR variants on FMCW range–Doppler maps. The detectors use comparable local context and are evaluated through a common pipeline. The study then follows the learned detector through five stages: controlled synthetic comparison, attribution experiments, zero-shot transfer to real CARRADA radar data, domain-informed synthetic training, and real-data fine-tuning.

On synthetic data, TinyCNN significantly outperforms CFAR in five challenging conditions, while homogeneous clutter is effectively a tie. The attribution experiments show that this gain is not simply caused by integrating more target energy, and that the network's localisation behaviour depends strongly on how it is trained.

The advantage does not transfer automatically. On CARRADA, the original synthetic-trained TinyCNN and CA-CFAR achieve statistically comparable recall, while TinyCNN produces almost twice as many unmatched detections. Making the synthetic training data more similar to CARRADA substantially improves recall, but also produces much denser output.

Finally, after fine-tuning on real CARRADA training data, TinyCNN reaches 0.9611 recall compared with 0.8725 for CA-CFAR, while producing slightly fewer unmatched detections per frame (11.17 versus 11.64). The improvement over the previous learned model is mainly in selectivity rather than recall.

The results therefore support a conditional conclusion: CFAR is the stronger training-free baseline, while the learned detector achieves a higher performance ceiling when its training data represent the deployment domain.

Keywords: FMCW radar, CFAR, convolutional neural networks, range–Doppler maps, domain adaptation.

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List of Symbols and Abbreviations

Abbreviations

FMCW	Frequency-Modulated Continuous Wave
CFAR	Constant False Alarm Rate
CA	Cell-Averaging CFAR
GO	Greatest-Of CFAR
SO	Smallest-Of CFAR
OS	Order-Statistic CFAR
RD	Range–Doppler
CNN	Convolutional Neural Network
CPI	Coherent Processing Interval
NMS	Non-Maximum Suppression
CUT	Cell Under Test
SNR	Signal-to-Noise Ratio
CNR	Clutter-to-Noise Ratio
ROC	Receiver Operating Characteristic
CI	Confidence Interval

Symbols

Symbol	Meaning	Unit
B	Chirp bandwidth	Hz
T_c	Chirp duration	s
N	Samples per chirp	–
M	Chirps per coherent processing interval	–
f_b	Beat frequency	Hz
ΔR	Range resolution	m
Δv	Velocity resolution	m/s
P_d	Probability of detection	–
P_{fa}	Probability of false alarm	–
α	CFAR threshold multiplier	–
k	Effective Gamma shape of the background	–
w	`pos_weight`, positive-class loss weight	–

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Chapter 1: General Introduction

1.1 Motivation

Automotive radar has a distinctive role in the sensor suite of a modern vehicle. Cameras can degrade under poor illumination, glare and reduced visibility. Lidar can be affected by rain, snow and airborne particles. Millimetre-wave radar is comparatively robust under many of these conditions. It also measures radial velocity directly, from the Doppler shift, rather than estimating it by comparing successive frames. For a driver assistance system that must distinguish an object closing at 2 m/s from one closing at 20 m/s, a direct velocity measurement is valuable.

The frequency-modulated continuous-wave (FMCW) waveform makes this practical at low cost. The radar transmits a linear frequency ramp, called a chirp, and mixes the received echo with the transmitted signal. The result is a low-frequency beat tone. Its frequency gives the target range, and its phase change from one chirp to the next gives the radial velocity. Two Fourier transforms turn a block of received samples into a *range–Doppler* (RD) map: a two-dimensional image in which a target appears as a concentration of energy at its range and its radial velocity. Everything downstream — clustering, tracking, classification, and ultimately the driving decision — works from what is extracted from this map.

Extracting targets from the RD map is a detection problem, and it is the first stage of that chain. A tracker can bridge an occasional missed detection, but a target that is missed persistently cannot be tracked, classified or acted upon, because no measurement of it reaches those stages. Excessive false detections cause the opposite problem: they consume tracking resources and can trigger unnecessary responses. Detection therefore sets a limit on how well the later stages can perform.

For about fifty years the standard solution has been the constant false alarm rate (CFAR) family of detectors. Comparing each cell against a fixed threshold would require knowing the noise power, which changes with temperature, terrain, weather and interference. Instead, CFAR estimates the local background from cells surrounding the cell under test (CUT), and scales that estimate by a multiplier chosen to hold the false-alarm rate constant. The detector therefore adapts to conditions it was never told about. CFAR needs no training data, is cheap to compute, and its behaviour can be

predicted mathematically, provided the background behaves as assumed.

Those assumptions are also its weakness, and this was documented early. Gandhi and Kassam [1] showed that cell-averaging CFAR degrades in situations that matter in practice. At a clutter edge, the reference window covers two different background levels and the threshold is set from the wrong one. Under target masking, a second target inside the reference window raises the background estimate and suppresses detection of the first. The main variants — greatest-of, smallest-of and order-statistic — each improve one failure mode and worsen another. Their conclusion was that no single CFAR detector performs well across all backgrounds, and that an adaptive approach may be needed when operating conditions change.

This naturally suggests learning the detector from data instead of designing it by hand, and much recent work has done exactly that. Convolutional networks applied to RD maps have been reported to outperform CFAR, with improvements ranging from a fraction of a decibel to large reductions in false-alarm rate.

1.2 Problem statement

These reported improvements are difficult to compare with one another, and difficult to interpret individually.

A comparison between a classical and a learned detector is informative only if both receive comparable information, are evaluated at the same operating point, and are scored by the same rule. Across the reviewed literature, these conditions are not always simultaneously satisfied. Five differences recur:

- **Input preprocessing.** The network may receive a windowed, matched-filtered or multi-resolution input, while the CFAR baseline receives raw magnitude data.
- **Spatial context.** The network may observe a much larger region than the CFAR reference window, so the two are not solving the same local problem.
- **Retraining.** The network may be retrained for each test condition, while the classical detector keeps a single global parameter set.
- **Operating point.** False-alarm rates are often matched by *design*, using the analytic CFAR expression, rather than by measurement. CFAR detectors do not reach their design P_{fa} in clutter, a point reported by authors who also report learned wins [2].
- **Scoring tolerance.** A detection may count as correct if it falls within some distance of the true position. A detector whose response is spread over a broad region then benefits from that tolerance.

Each of these choices can be reasonable on its own. Together, however, they mean that the reported improvement depends partly on how the experiment was set up, and not only on the detector itself.

A second difficulty compounds the first. A detector trained on simulated data and evaluated on the same simulator has shown that it learned the simulator; whether it learned *radar* is a separate question, and one that much of the literature does not test. That question matters particularly here, for a reason specific to the problem. CFAR estimates its threshold from the data in front of it at inference time, so it keeps adapting in whatever domain it is used. A conventionally frozen feed-forward network has no explicit mechanism to re-estimate its decision rule when the deployment distribution changes: its behaviour is fixed by weights fitted to the training distribution.

This shapes the central question of the project, and it is deliberately not "is a CNN better than CFAR?". A single ranking would hide the thing that turns out to matter most. The question this thesis answers is:

> **When does a learned local radar detector actually provide an advantage over classical CFAR, and what is required for that advantage to survive transfer from controlled simulation to real radar data?**

1.3 Research questions

Five questions follow, each answered by one stage of the work.

RQ1 — The controlled comparison. Under controlled synthetic conditions, with comparable local range–Doppler context and a matched measured operating point, what detection advantage does a small learned local decision rule provide over classical CFAR?

RQ2 — The mechanism. What detector characteristics explain the learned gains and their trade-offs? Three are tested: energy integration over the mainlobe, the role of supervision in localisation, and access to the guard region.

RQ3 — Transfer. Does the learned advantage survive zero-shot transfer from simulated FMCW data to real range–Doppler measurements?

RQ4 — The training domain. If it does not, does reducing the *measured* synthetic-to-real background mismatch improve transfer, without any real radar frame entering neural optimisation?

RQ5 — Real adaptation. With real-domain supervision permitted, can the learned detector beat CA-CFAR at a comparable output burden, on real sequences it has

never been scored on?

RQ1 and RQ2 concern the detector. RQ3, RQ4 and RQ5 concern the distribution it was trained on, and they are the reason this thesis does not end with its synthetic benchmark.

1.4 Objectives

Six objectives operationalise the five research questions above.

O1 — Build a controlled comparison. Construct a classical arm and a learned arm that work from the same range–Doppler data, observe comparable local context, and are scored by one evaluator at a matched *measured* false-alarm rate. Chapter 4 states what the two arms share and what they do not.

O2 — Measure the margin. Quantify the difference across conditions chosen to include both regimes where CFAR is close to optimal and regimes where it is known to fail, with confidence intervals and a correction for multiple comparisons.

O3 — Attribute the margin. Pair each competing explanation with an experiment capable of disproving it, rather than arguing for one of them. Such experiments can support or rule out particular mechanisms; they are not expected to identify a single cause.

O4 — Test external validity. Evaluate both frozen detectors, with no retraining or recalibration of any kind, on real automotive radar data from a public dataset, under a protocol fixed before the evaluation data were scored.

O5 — Test the training domain. If transfer fails, measure the difference between the simulated and real domains, rebuild the synthetic training background from target-domain statistics alone, and re-measure transfer — keeping every real radar frame out of neural optimisation. This separates "the detector cannot transfer" from "it was trained on the wrong distribution".

O6 — Test what real supervision adds. Permit real-domain adaptation and ask whether the same small network can exceed the classical baseline on previously unscored real sequences at a comparable output burden, with the operating point fixed on training data before the evaluation split is opened.

One objective is deliberately excluded. This project does not aim to produce a state-of-the-art detector. A larger network, trained on real data with modern augmentation, would very likely detect more targets than the one built here. The aim is to measure a difference under conditions where the measurement can be trusted, and for that purpose a small, well-understood result is more useful than a large unexplained one.

1.5 Contributions

1. **A controlled like-for-like comparison of CFAR and a learned local detector.** The learned detector's receptive field is matched to the spatial footprint of the CFAR window; both arms are scored by one evaluator through a single code path; operating points are matched at measured rather than design P_{fa}; and results are reported across a tolerance ladder rather than at a single gate. Each choice removes one of the asymmetries listed in Section 1.2. The resulting margin is reported per condition with confidence intervals and a correction for multiple comparisons, and expressed in decibels of cell SNR so that it can be placed beside the comparable literature.

2. **Attribution by intervention.** Three experiments each pair a candidate explanation with a measurement capable of disproving it. One outcome is directly useful: the frozen benchmark's localisation deficit is largely attributable to its supervision, and substantially removable, rather than an inherent property of the learned detector.

3. **A zero-shot synthetic-to-real evaluation.** The frozen detectors are evaluated on the CARRADA automotive radar dataset [3] with no real-data training, fine-tuning or recalibration. Ground truth, matching, metrics and the statistical plan were pre-specified and frozen before any evaluation frame was scored. No directly comparable zero-shot experiment was identified in the reviewed corpus.

4. **A target-domain-informed synthetic intervention.** The simulated background was rebuilt from statistics measured on the CARRADA *training* split alone — smoothness, range dependence and zero-Doppler ridge structure — with no real frame entering neural optimisation or threshold calibration. Exact real-data recall rose from 0.8721 to 0.9449, showing that a substantial part of the transfer failure was a property of the training distribution rather than of the detector. The same experiment shows the limit of that reading: the recovery came with 4.7 times the output density, and did not survive burden matching against CA-CFAR.

5. **A real-domain adaptation result with burden-aware evaluation.** Fine-tuned on CARRADA training sequences with its operating point fixed on training data alone, the same 5,041-parameter network reached exact recall 0.9611 against CA-CFAR's 0.8725 on previously unscored validation sequences, at a slightly *lower* unmatched-detection burden. Within the reviewed corpus, no directly comparable experiment was identified that carries one small detector through controlled synthetic comparison, zero-shot transfer, domain-informed retraining

and real adaptation under a single evaluation protocol.

6. **An empirical argument against single-number detector comparisons.** Across the five stages, detection probability alone would have supported three different and partly contradictory conclusions. Output burden, training domain, tolerance and operating point each changed the answer, and the strongest learned result in the thesis is one where recall and burden had to be read together. A related protocol-sensitivity finding is reported separately: on annotated objects larger than the suppression window, the rule that converts a dense score map into candidates moved the headline result, where on synthetic point targets it did not.

1.6 Report outline

The report follows the order in which the work was carried out, because several conclusions depend on what was known when a given decision was made. Figure 1.1 shows that order as a single roadmap: five stages, one question each, and the conclusion they lead to.

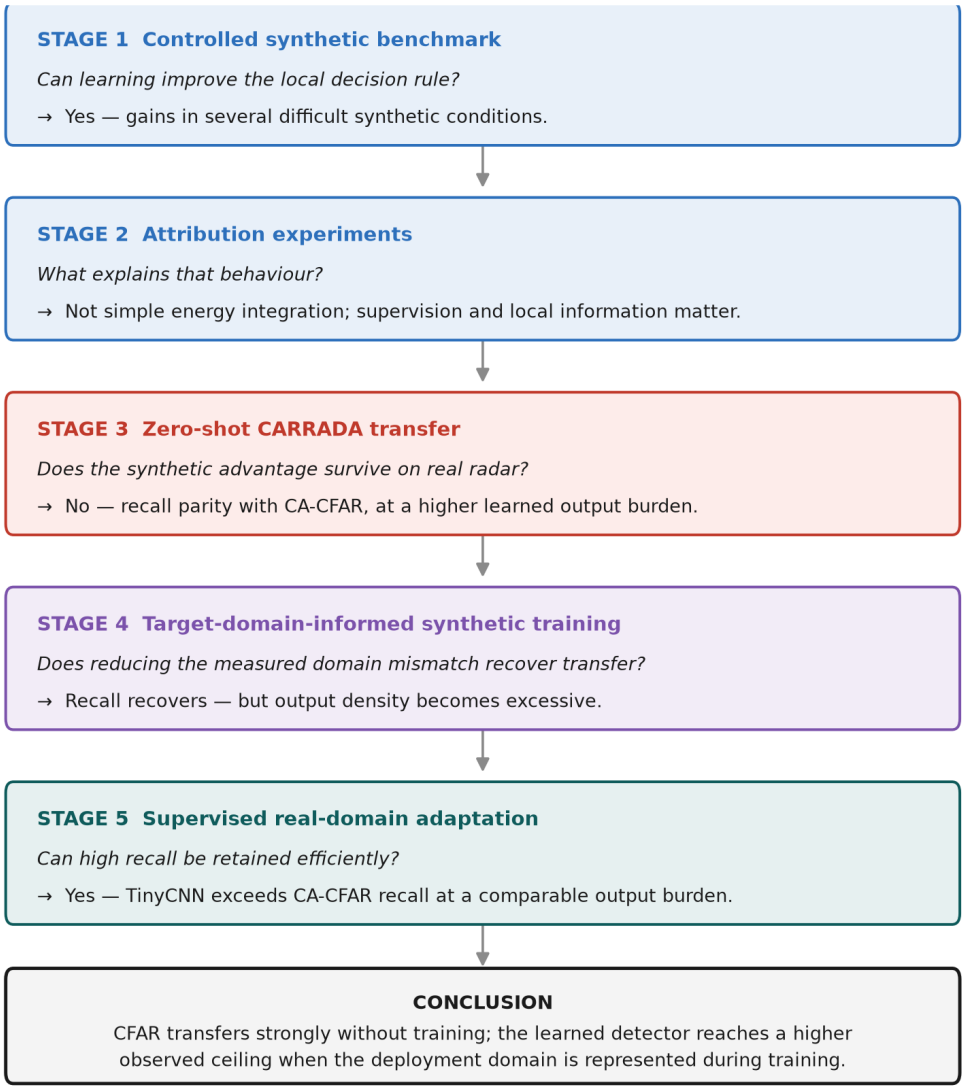


Figure 1.1: The five stages of this thesis. Each answers one question and motivates the next; the later stages are progressively stronger tests of the same hypothesis, not separate experiments. Numbers, intervals and caveats are deliberately omitted — this figure is for orientation, and the chapters carry the evidence.

Chapter 2, *Radar Background and State of the Art.* The FMCW waveform, the RD map, the CFAR family and its documented failure modes, learned detection, and the gap in the published comparisons that this work addresses.

Chapter 3, *Simulator, Radar Specification and Dataset Design.* The simulator, the radar parameters — chosen to reproduce the real dataset's RD bin geometry — and the labelled datasets.

Chapter 4, *Detectors and Evaluation Protocol.* Both detectors and the shared evaluation protocol: candidate extraction, matching, operating point and statistics.

Chapter 5, *Synthetic Benchmark Results.* Stage 1: the controlled comparison

across the seven scored conditions, including detection probability against cell SNR.

Chapter 6, *Attribution Experiments.* Stage 2: three interventions that separate the competing explanations for the measured margin.

Chapter 7, *Real-Data External Validation.* Stage 3: the frozen detectors evaluated zero-shot on real automotive radar data, and a domain audit explaining what changed.

Chapter 8, *Domain Alignment and Real-Data Adaptation.* Stages 4 and 5: rebuilding the synthetic training domain from target-domain statistics, then permitting supervised real-domain fine-tuning.

General Conclusion. The five research questions answered in turn, what the evidence supports, and what follows.

Two appendices support the main text: Appendix A records the protocol revisions and sensitivity checks that a reader needs in order to interpret the final evidence, and Appendix B carries the detailed breakdowns the results chapters summarise. Full provenance — freeze records, file manifests, commit chain and the complete corrections log kept during development — is retained in the accompanying project repository rather than reproduced here.

Chapter 2: Radar Background and State of the Art

2.1 Introduction

This chapter provides the background the rest of the report depends on, and reviews the published work the results will later be compared against.

It has three parts. The first explains how an FMCW radar produces a range–Doppler map, because that map is the input to every detector in this work. The second describes CFAR detection: the principle, the main variants, and the failure modes that were documented decades ago and that motivate learned alternatives. The third reviews learned detection on RD maps and argues that the published literature separates into families whose results are not directly comparable. That argument is used throughout the report, and it is the reason Chapter 7 compares against only a subset of the papers reviewed here.

Two results in this chapter are used directly later. The documented CFAR failure modes (Section 2.3.5) determine which conditions the synthetic benchmark of Chapter 5 must contain. The equivalence between CA-CFAR and a convolution (Section 2.4.2) is what justifies treating a small convolutional network and CFAR as two instances of the same operator, rather than as two unrelated systems placed side by side.

2.2 FMCW radar principles

2.2.1 Chirp generation and the beat frequency

An FMCW radar transmits a signal whose frequency increases linearly with time. One such sweep is called a *chirp*. The treatment here follows the standard accounts [4, 5]. Writing B for the swept bandwidth and T_c for the chirp duration, the instantaneous frequency rises at a constant rate, the *slope*

$$S = \frac{B}{T_c}. \tag{2.1}$$

A target at range R reflects the chirp back after a round-trip delay

$$\tau = \frac{2R}{c}, \tag{2.2}$$

where c is the speed of light. The echo is therefore a copy of the transmitted chirp, shifted in time. Because the frequency is ramping, a shift in time becomes a difference in frequency. Mixing the echo with the transmitted signal and low-pass filtering the result — an operation called *dechirping* — produces a tone at the *beat frequency*

$$f_b = S\,\tau = \frac{2BR}{cT_c}. \tag{2.3}$$

Equation (2.3) is the reason FMCW radar is inexpensive. Range has been converted into a frequency of a few megahertz, so the receiver never has to sample at the radio frequency; it only has to sample the beat tone. Figure 2.1 shows the geometry for the radar used in this work.

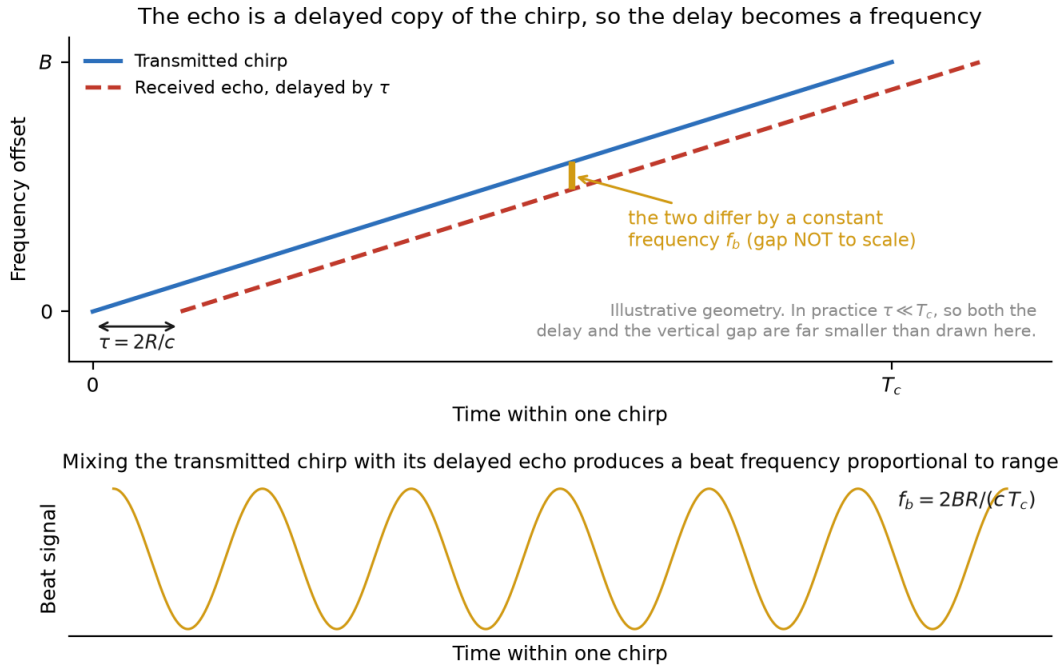


Figure 2.1: Top: the received echo is a delayed copy of the transmitted chirp, so mixing the two produces a beat frequency proportional to range. Axes are normalised, in units of the swept bandwidth B and the chirp duration T_c. The geometry is illustrative: in practice $\tau \ll T_c$, so both the delay and the vertical gap labelled f_b are far smaller than drawn. Bottom: the resulting beat tone. The sensor parameters used in this work are given in Chapter 3.

2.2.2 Range estimation

Sampling the beat signal N times within one chirp and applying a discrete Fourier transform converts the tone into a peak at a known frequency bin. By equation (2.3), that bin corresponds to a range. This first transform is called the *range FFT*, and it is taken along what is known as *fast time*, meaning the samples within a single chirp.

As a nominal resolution criterion, two targets require approximately one range-resolution cell of separation before they appear as separate peaks. This gives

$$\Delta R = \frac{c}{2B}. \tag{2.4}$$

The nominal FMCW range resolution is therefore determined primarily by the swept bandwidth, under the standard assumptions of a linear sweep, an unwindowed transform and sufficient signal-to-noise ratio. In particular it does not improve with a longer chirp, with higher transmitted power, or at shorter target range. Window functions applied before the transform, which suppress sidelobes, broaden the main lobe and so widen the achieved resolution beyond this nominal value.

2.2.3 Doppler estimation and the range–Doppler map

A single chirp gives range but not velocity. Velocity is obtained by transmitting M chirps in quick succession and observing how the *phase* of a target's beat tone changes from one chirp to the next. A target moving radially changes its range slightly between chirps, and although that change is far too small to move the target into a different range bin, it is a substantial fraction of a wavelength and therefore produces a measurable phase rotation.

Taking a second Fourier transform across the M chirps, at each range bin, converts that phase rotation into a frequency: the Doppler shift. This transform is taken along *slow time*, meaning across chirps. The relevant time step here is the *chirp repetition interval* T_r, the spacing between the starts of consecutive chirps, which in general is the chirp duration plus any idle time between chirps. **The radar modelled in this work transmits chirps back to back with no idle time, so $T_r = T_c$ throughout**; the two symbols are used interchangeably from here on, and the simulator samples slow time at exactly this spacing. The velocity resolution then follows the same reasoning as equation (2.4):

$$\Delta v = \frac{\lambda}{2MT_r}, \qquad v_{\max} = \pm\frac{\lambda}{4T_r}, \qquad T_r = T_c \text{ here}, \tag{2.5}$$

where $\lambda = c/f_c$ is the wavelength at carrier frequency f_c. Longer observation, meaning more chirps, gives finer velocity resolution; a shorter repetition interval extends the

unambiguous velocity range.

The output of the two transforms is a two-dimensional array indexed by range bin and Doppler bin: the *range–Doppler map*. Each target appears as a localised concentration of energy at its range and radial velocity. Figure 2.2 summarises the chain.

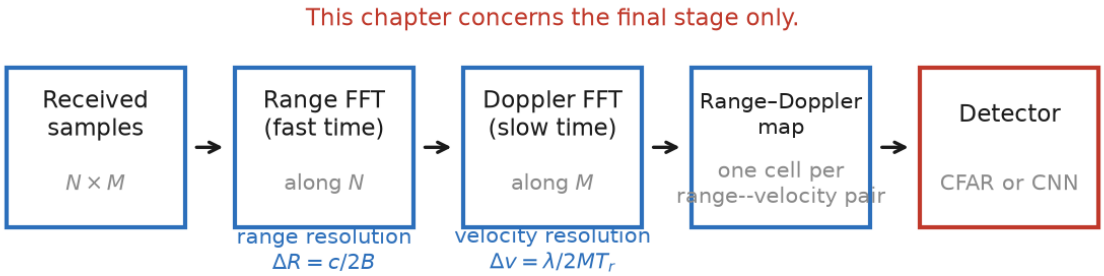


Figure 2.2: From received samples to the range–Doppler map. The detector, which is the subject of this report, operates on the final stage only. The sensor parameters that fix the two resolutions are specified in Chapter 3.

The RD map is an image, but it is not a photograph, and two of its properties matter for detection. First, the background is not smooth: with a single look, the noise in each cell follows an exponential power distribution, so neighbouring cells differ by many decibels purely by chance. Second, a target is not a single bright pixel but a response spread over a few cells, set by the window functions and the transform lengths rather than by the physical size of the object. Figure 2.3 shows both effects on one frame of the benchmark used in this work.

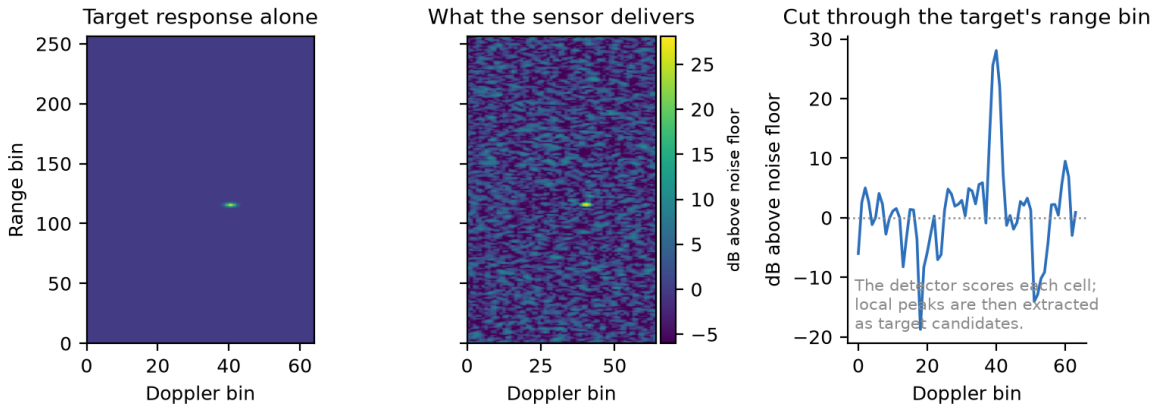


Figure 2.3: Left: the target response on its own. Centre: the same frame as the sensor delivers it. Right: a cut through the target's range bin. The detector assigns scores or decisions to RD cells, after which local peaks are extracted as target candidates; Chapter 4 specifies that extraction step and makes it identical for both detectors.

2.3 Detection in the range–Doppler plane

2.3.1 The detection problem

Detection is a binary hypothesis test applied to every cell of the RD map. Throughout this report x denotes the *measured power* in an RD cell, that is the squared magnitude of the complex range–Doppler sample, $x = |X|^2$. This is the output of a square-law detector, and it is a power quantity rather than an amplitude; every threshold expression below is written in those units. For a cell with measured power x, the two hypotheses are

$$H_0 : \text{background only (noise, clutter, interference)}, \tag{2.6}$$

$$H_1 : \text{background plus a target return.} \tag{2.7}$$

Radar detectors are conventionally designed to the Neyman–Pearson criterion: fix the probability of false alarm P_{fa} at an acceptable value, and maximise the probability of detection P_d subject to that constraint. This is the reason P_d at a fixed P_{fa} is the standard metric in the field, and it is the metric this report uses.

Fixing P_{fa} is straightforward if the background power is known. Under the standard single-look model the background power x is exponentially distributed with mean σ^2, so $\Pr(x > T \mid H_0) = \exp(-T/\sigma^2)$ and the threshold that achieves a chosen P_{fa} is $T = -\sigma^2 \ln P_{fa}$, again in units of power. The difficulty is that σ^2 is not known. It varies with temperature, with the terrain being illuminated, with weather, and with interference from other radars. A fixed threshold set for one condition produces either a flood of false alarms or no detections at all in another.

2.3.2 The constant false alarm rate principle

CFAR detection solves this by estimating the background from the data itself. For each cell under test, a window of surrounding cells is used to form an estimate Z of the local background level, and the threshold is set proportional to it:

$$T = \alpha Z, \qquad \text{decide } H_1 \text{ if } x > T. \tag{2.8}$$

The multiplier α is chosen so that the false-alarm rate takes the desired value. Because the threshold scales with the estimated background, a uniform change in received power scales x and Z together and leaves the decision unchanged. This is the constant false alarm rate property, and it is what allows a single detector to operate across widely varying conditions.

It is worth being precise about what that property guarantees. The rate is held constant *under the assumed background model*: the reference cells are taken to be

independent, identically distributed and free of target energy, and to share the statistics of the cell under test. When those assumptions hold, the achieved false-alarm rate is close to the design value regardless of the absolute power level. When they do not hold — at a clutter edge, where the reference cells no longer share the statistics of the cell under test, or under target masking, where they are not target-free — the achieved rate can depart from the design value in either direction. Section 2.3.5 describes both cases, and this departure is measured rather than assumed in Chapter 4, which is why operating points in this report are matched at the *measured* rather than the design rate.

The window has a specific structure, shown in Figure 2.4. Immediately around the cell under test is a ring of *guard cells*, which are excluded from the estimate. They must be excluded: a target's own response spreads over several cells, so including them would let the target inflate its own threshold and suppress its own detection. Beyond the guard ring are the *reference* or *training cells*, which form the estimate.

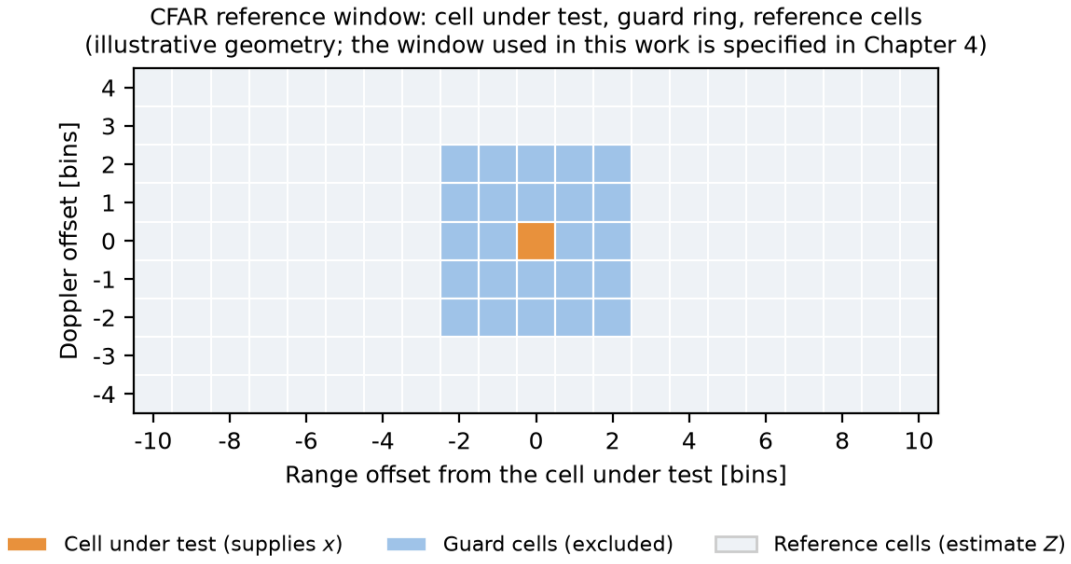


Figure 2.4: The three roles a CFAR window assigns to the cells around each decision. The **cell under test** supplies the test statistic x. The **guard cells** immediately surrounding it are excluded, so that a target cannot raise its own threshold. The remaining **reference cells** form the background estimate Z. The geometry drawn here is illustrative: the window is rectangular because the two axes of the RD map are not alike, and the size actually used in this work, together with the reason for it, is specified in Chapter 4.

2.3.3 Cell-averaging CFAR

The simplest choice, and the original adaptive-threshold scheme of Finn and Johnson [6], is to average all reference cells,

$$Z_{\mathrm{CA}} = \frac{1}{N} \sum_{i=1}^{N} x_i, \tag{2.9}$$

where N is the number of reference cells. For square-law detection in homogeneous exponential noise, the multiplier that achieves a design false-alarm rate is available in closed form:

$$\alpha = N \left(P_{fa}^{-1/N} - 1 \right). \tag{2.10}$$

Under a specific and quite restrictive set of conditions — a homogeneous exponential background, and reference cells that are genuinely free of target energy — CA-CFAR is optimal in the sense of maximising P_d at a given P_{fa}, because the sample mean is then the most efficient estimator of the background power [1]. Outside those conditions the optimality claim does not carry over, and the variants below exist precisely because real backgrounds often violate them. CA-CFAR is also the cheapest variant, requiring only a sum.

2.3.4 Greatest-of, smallest-of and order-statistic variants

Every CFAR variant differs from CA-CFAR in one respect only: how the reference cells are combined into Z. That single choice determines the variant's strengths and weaknesses.

GO-CFAR (greatest-of). The reference cells are split into a leading and a lagging half, and Z is the larger of the two means [7]. Keeping the higher estimate limits the increase in false-alarm rate that CA-CFAR shows at a clutter edge, at the cost of masking targets more readily.

SO-CFAR (smallest-of). The mirror image: Z is the smaller of the two half-means [8]. If an interfering target contaminates one half, the clean half is used, so closely spaced targets are resolved. At a clutter edge the estimate is taken from the quieter side, the threshold is set too low, and the false-alarm rate can increase substantially.

OS-CFAR (order-statistic). The reference cells are sorted and the k-th ranked value is taken as Z, typically with $k \approx 0.75N$ [9]. Interfering targets are high outliers and are excluded by the ranking, which makes the variant robust in multi-target scenes. The cost is a sort per cell and a small additional loss in homogeneous background.

Analytical threshold relationships exist for GO, SO and OS as well, but they are considerably more involved than equation (2.10): they depend on the split of the reference cells or on the distribution of an order statistic, and the published closed forms assume a one-dimensional window and independent reference cells. Neither assumption holds exactly for the two-dimensional window and correlated RD background used here. This work therefore obtains all four multipliers by Monte-Carlo calibration on noise-only data, so that every variant is calibrated under the same conditions in which it is evaluated. Chapter 4 describes that procedure and reports the resulting values.

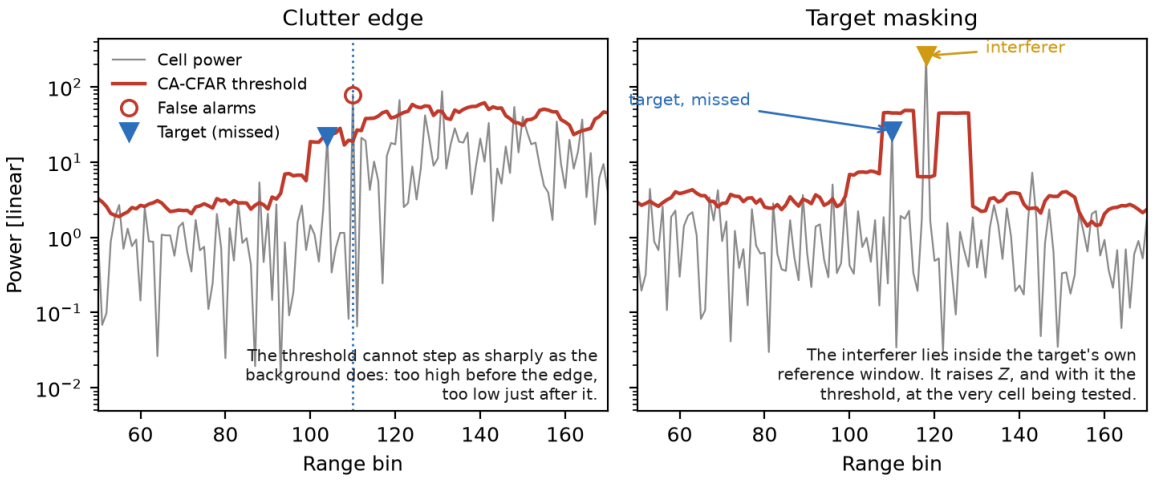
2.3.5 Known failure modes

The limitations of these detectors have been understood since Gandhi and Kassam [1]. Two failure modes dominate, and both follow directly from equation (2.8): the threshold is only as good as the assumption that the reference cells describe the background at the cell under test.

Clutter edges. When the window straddles a boundary between two background levels, the estimate mixes them. On the quiet side of the edge the threshold is set too high and targets are missed; on the strong side it is set too low and false alarms appear. The threshold cannot follow a step that is sharper than the window is wide.

Target masking. When a second target falls inside the reference window of the first, it raises Z and therefore raises the threshold at the very cell being tested. The stronger the interferer, the more effectively it hides its neighbour.

Figure 2.5 illustrates both mechanisms on a one-dimensional cut, using the same guard and reference counts per side as the window used in the rest of this work.



Simulated on the same window as the rest of this work (guard 2, training 8 per side, one-dimensional cut). Exponential background.

Figure 2.5: An illustrative simulation of the two classical failure mechanisms, not a measured benchmark result. A one-dimensional cut through an exponential background is used, with the same guard and reference counts per side as the window of Figure 2.4, so that the mechanisms can be seen directly; the outcome for any single realisation depends on the random draw. Left: at a clutter edge the threshold lags the background step, missing a target before the edge and admitting a false alarm after it. Right: an interfering target inside the reference window raises the threshold at the cell being tested, and the weaker target is missed. Measured performance under these conditions is reported in Chapter 5.

The conclusion Gandhi and Kassam reached is the starting point of this project: no single CFAR variant performs well across all backgrounds, and each variant tends to repair one failure mode while worsening another. GO tends to handle clutter edges better than CA but may worsen masking; SO tends to help in multiple-target cases but can perform poorly at clutter transitions. They suggested that an adaptive approach might be required when operating conditions change.

These failure modes also determine the design of the benchmark. A comparison evaluated only in homogeneous noise would test the one condition under which CA-CFAR is already close to optimal, and would say nothing about the conditions that motivate a learned detector. The conditions used in Chapter 5 are chosen to span both.

2.3.6 CFAR loss

A third limitation applies even when the background is perfectly homogeneous, so it is not a failure mode in the sense above.

CFAR estimates the background power from a finite number of reference cells. That estimate is therefore imperfect, and it varies from one window to the next. To keep the false-alarm rate at its design value despite this variability, the threshold must be set slightly higher than a detector that somehow knew the true background power would need. The resulting penalty in signal-to-noise ratio is called *CFAR loss* [5]. It becomes

smaller as the number of reference cells grows, because the estimate becomes more reliable.

The reason this matters for the present work is that CFAR loss offers a second possible explanation for any advantage a learned detector shows. The obvious explanation is that the network uses its local context differently. But a detector that avoids the estimation penalty would also be expected to do better, even where CFAR has no failure mode at all. Both explanations predict the same observation, so neither can be accepted without testing the other. Chapter 6 treats this as one of the candidate mechanisms to be separated experimentally.

2.4 Learned detection on the range–Doppler map

2.4.1 Convolutional detectors

The RD map is an image-like array, and the detection rule is applied identically at every cell. Both properties suit a convolutional neural network. A convolutional layer applies the same learned kernel at every position, which matches the translation invariance of the problem and means the number of parameters does not grow with the size of the map.

Most learned detectors on RD maps use one of two formulations:

- **Patch.** A small window around each cell is extracted and classified on its own. This mirrors CFAR closely, but it needs one network evaluation per cell.
- **Fully convolutional.** The whole map is passed through the network once, producing a score for every cell in a single pass.

The second is far cheaper and is the formulation used in this work. Importantly, processing the whole map at once does not mean each output depends on the whole map: every output cell still depends only on the input cells within its receptive field, which is a local neighbourhood set by the network's depth and kernel sizes.

2.4.2 CA-CFAR as a convolution

CA-CFAR can be expressed as a fixed convolutional operation followed by thresholding [10, 11]. This provides a useful interpretation for the comparison in this thesis: both CFAR and TinyCNN map a local neighbourhood to a score, but CFAR uses a hand-derived fixed rule whereas TinyCNN learns that rule from data. TinyCNN is a richer nonlinear function rather than a single equivalent CFAR kernel, so the comparison asks whether learning the local decision rule provides an advantage when spatial

context is kept comparable. Figure 2.6 places the two rules side by side over the same footprint.

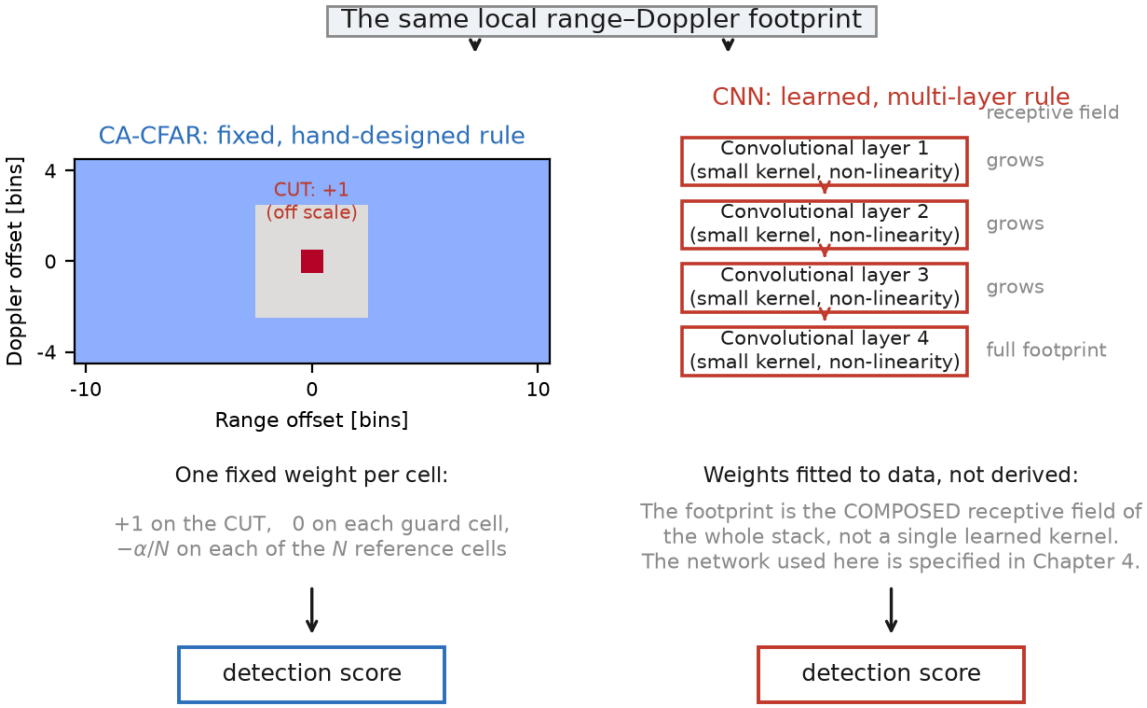


Figure 2.6: Both detectors decide from the same local footprint, but obtain their rule differently. Left: CA-CFAR as one fixed weight per cell — +1 on the cell under test, 0 on the guard cells, $-\alpha/N$ on each reference cell (the colour scale is set by the reference weights, so the cell under test saturates and is labelled instead). Right: a convolutional network reaches the same footprint by composing several small convolutions. **That footprint is the composed receptive field of the whole stack, not a single learned kernel.**

2.4.3 Label representation and supervision

Unlike CFAR, a learned detector must be given a training target: some representation of where the targets are. Three forms are common — a single point, a weighted local region around it, or an object mask covering the target's support.

The choice is not neutral. It affects both what the network learns to detect and how precisely it localises, because it defines what a single output cell is being asked to mean. The representation used in this work, and the reasons for it, are specified in Chapter 4.

2.5 State of the art: learned detectors versus CFAR

2.5.1 Not every result is directly comparable

A substantial body of work reports learned detectors outperforming CFAR, and read together the results appear to show a large and consistent advantage. They do not actually combine, because the works are solving different tasks: some perform per-cell detection, others object detection with bounding boxes, others reconstruction or hardware reformulation. A gain reported for one task is not evidence about another.

Table 2.1 groups the reviewed literature accordingly. Only the first family reports P_d against CFAR at a matched false-alarm rate on a per-cell decision, which is the quantity this report measures.

Table 2.1: Families in the learned-detection literature. Only the first measures the same quantity as this work.

Family	Representative work	Metric	Comparable
Per-cell learned threshold, scored against CFAR	RadCNN [2], Gusland et al. [12], Lallouet et al. [13], Banerjee et al. [11]	P_d at fixed P_{fa}	**Yes**
CFAR-property constrained learning	CFARnet [14]	ROC, plus a test of the CFAR property	Partly
Object detection on RD maps	DAROD [15]	mAP at an IoU threshold	No
Structure-preserving reformulation	Liu et al. [10]	Latency and energy	No

2.5.2 The four directly relevant works

Four studies are particularly relevant to this work, and each for a different reason. RadCNN and Gusland et al. provide direct CNN-versus-CFAR performance comparisons. Lallouet et al. motivate the CFAR-loss hypothesis. Liu et al. and Banerjee et al. supply the theoretical connection between fixed CFAR processing and learned convolutional processing.

RadCNN: a learned detector does not always win

RadCNN [2] evaluates a convolutional detector against CA, GO, SO and OS CFAR on simulated data with Weibull and K-distributed clutter. Its value here is that the reported outcome is mixed rather than uniformly favourable. The network gains over CFAR in some clutter conditions. It ties CFAR in simple homogeneous noise. And it loses in at least one scenario, where the target lies near clutter but its reference window is filled mostly with noise.

Gusland et al.: the closest experimental comparison to this work

Of the reviewed studies, Gusland et al. [12] is the one whose experimental setup most resembles the present work. They use real X-band FMCW measurements, with a UAV carrying GPS for ground truth, rather than simulation. They deliberately restrict the network's receptive field to match the CA-CFAR patch, so that both detectors work from comparable local context.

Their results are consistent with RadCNN. In homogeneous regions the network approximately ties CFAR. Around clutter transitions it gains roughly 0.25 to 0.35 in P_d. That figure is quoted here because Chapter 7 compares it directly against the clutter-edge result measured in this work.

Lallouet et al.: a proposed explanation for the gains

Lallouet et al. [13] report a learned detector outperforming CFAR on simulated exo-clutter data. Their contribution to this report is not the margin but the explanation they offer for it: the finite reference window of Section 2.3.6 makes CFAR's threshold an estimate rather than a known quantity, and a learned detector does not pay that penalty.

This is a testable claim, and it is one of the candidate mechanisms examined in the attribution experiments of Chapter 6.

Liu et al. and Banerjee et al.: the structural connection

These two works are conceptual support rather than a further detector benchmark, and they are cited for that reason. Liu et al. [10] show that CA-CFAR can be represented using convolution-like fixed weights. Banerjee et al. [11] extend the idea towards learned weights.

Together they support the framing used throughout this report: **CFAR is a fixed local rule, and a CNN is a learned local rule**, applied over comparable context.

Two observations run through these comparative studies, and they motivate what follows. Learned detectors often provide little advantage over a well-configured CFAR

in homogeneous conditions but can gain in more difficult clutter; and the reasons for those gains are not fully established.

2.6 Gap addressed by this work

The review supports four observations, which together define what this project adds. They are stated briefly here; the underlying fairness arguments were set out in Section 1.2.

1. **Only a small subset of published work is directly comparable** to the per-cell CFAR-versus-learned detection task addressed here. Most of the corpus answers a different question.
2. **Matching the network's spatial context to the CFAR window is uncommon.** Among the reviewed works only Gusland et al. do so deliberately. This work does the same, while stating explicitly that matching the footprint does not make the two decision rules identical.
3. **The competing explanations for learned gains are rarely separated experimentally.** CFAR estimation loss and a different use of local context both predict the same observation, and no reviewed work distinguishes them. Chapter 6 attempts to.
4. **Zero-shot synthetic-to-real validation is rarely tested.** Simulation-trained detectors are almost always evaluated on the same simulator, and measurement-trained detectors on the same sensor. None of the directly comparable studies identified here evaluates a frozen, simulation-trained detector on real measurements without retraining. Chapter 7 reports that evaluation.

2.7 Conclusion

An FMCW radar turns range into a beat frequency and velocity into a phase rotation across chirps, and two Fourier transforms produce the range–Doppler map on which detection is performed. CFAR detects by estimating the local background from a reference window and applying an adaptive threshold proportional to it; its variants — CA, GO, SO and OS — differ only in how that estimate is formed, and each has different strengths and weaknesses across homogeneous background, clutter edges and multiple targets.

One idea from this chapter carries into the rest of the report: **CFAR is a fixed local operator applied to a neighbourhood of each cell, and a convolutional network is a learned local operator over comparable context**. That framing

is what makes the comparison in this report a controlled one. Two questions are left open by the reviewed work — why the reported learned gains occur, and whether they survive a change of domain — and they motivate the experiments that follow. Chapter 3 begins by specifying the simulator and the datasets, chosen so that the conditions identified in Section 2.3.5 are represented and so that the range–Doppler geometry matches the real dataset used later.

Chapter 3: Simulator, Radar Specification and Dataset Design

3.1 Introduction

Chapter 2 described detection in general terms. This chapter specifies the particular radar, the particular simulator and the particular datasets that every result in this report is measured on. All the numeric values deliberately withheld from Chapter 2 are introduced here.

Three decisions in this chapter shape everything that follows.

The first is the choice to work in simulation. It is what makes a controlled comparison possible at all, because it supplies exact ground truth and independent control over each difficulty. It is also the reason Chapter 7 exists, since a result obtained entirely inside a simulator says nothing on its own about real measurements.

The second is the sensor specification. The simulator reproduces the range–Doppler grid geometry and bin spacing of the real dataset used later, which removes a basic resolution and scale mismatch between the two domains. Without that, any transfer failure in Chapter 7 could be dismissed as a scaling artefact rather than a finding. Section 3.3 records how those parameters were derived, including why the bandwidth published alongside that dataset was not used directly.

The third is the dataset design. The benchmark is divided into predefined conditions rather than generated entirely at random. This ensures that important cases such as clutter edges and target masking occur often enough to evaluate the detectors reliably.

3.2 Why simulate

A comparison of the kind attempted here needs three things that measured radar data does not readily provide.

Exact ground truth. Scoring P_d requires knowing precisely which cells contain targets. Real datasets provide annotations, which are themselves estimates, produced semi-automatically and carrying their own error. In simulation the target position is

an input, not an inference.

Independent control of difficulty. The failure modes of Section 2.3.5 — clutter edges, target masking — must be produced deliberately and in isolation to be measured separately. In recorded data they co-occur uncontrollably.

Sufficient quantity in rare conditions. The conditions that discriminate between detectors are exactly the ones that are rare in naturally sampled data.

The cost is external validity, and it is a real cost. A detector tuned inside one simulator may have learned that simulator rather than radar. This is why the synthetic benchmark of Chapter 5 is treated in this report as an internally valid measurement whose generalisation is a separate question, tested in Chapter 7 and not assumed here.

3.3 Radar specification

3.3.1 Matching the real dataset's bin geometry

The sensor parameters were chosen so that the simulator reproduces the range–Doppler grid geometry and bin spacing of CARRADA [3], the real dataset used in Chapter 7. The reason is that detectors operate in bins: a CFAR window is a fixed number of cells wide, and a network's receptive field is a fixed number of cells across. If one metre of range corresponded to five cells in one domain and two in the other, any difference measured later would be partly a unit mismatch.

What this matching does and does not achieve should be stated clearly. It removes the geometric confound, so that a given physical separation means the same number of cells in both domains. It does *not* make target responses identical. CARRADA contains extended road users whose energy spreads over a region set by the object's physical size and aspect, whereas the simulator produces point-target responses whose extent is set by the transform and its windows. Those two supports differ, and Chapter 7 treats that difference as one of its central issues rather than assuming it away.

Table 3.1 shows the match achieved.

Table 3.1: Range–Doppler bin geometry: the real dataset and the simulator used here.

	CARRADA	**Simulator**
Map size (range × Doppler)	256×64	256×64
Range span	0 m–50 m	0 m–49.97 m
Range per bin	0.195 m	0.1952 m
Velocity span	± 13.43 m/s	± 13.44 m/s
Velocity per bin	0.42 m/s	0.4201 m/s
Frame time	4.635 ms	4.6336 ms

3.3.2 Deriving a consistent effective bandwidth

The bandwidth could not simply be copied from the dataset's published parameters, because the published range figures do not agree with one another.

The published table gives 256 range bins over a maximum range of about 50 m, which corresponds to 0.195 m per bin. The same table also lists a sweep bandwidth of 4 GHz. Using the standard relation $\Delta R = c/2B$, that bandwidth would imply a nominal range resolution of about 0.0375 m, which is more than five times finer than the published grid spacing. The two cannot both describe the same range axis.

Since the objective here is to reproduce the grid the dataset actually uses, the published grid geometry was taken as the reference. Matching 0.195 m per bin over 50 m requires an effective bandwidth of

$$B_{\text{eff}} = \frac{c\,N_{\text{bins}}}{2R_{\text{max}}} \approx 767\,\text{MHz}, \tag{3.1}$$

which is the value adopted, rounded to 768 MHz.

Two cross-checks support reading the published table this way. The same table quotes an angular resolution of 0.70°, while 180° divided by 256 angle bins is 0.703°: the entries labelled "FFT resolution" therefore appear to be bin spacings rather than physical resolutions, and reading the range entry the same way makes it self-consistent. The velocity axis, meanwhile, is consistent throughout — the quoted velocity resolution implies a frame time of 4.635 ms, which divided by 64 chirps gives a chirp interval of 72.4 µs and reproduces the quoted maximum velocity to within 0.01 m/s. Only the range side requires interpretation, and nothing in this report depends on why: what matters is that the simulator reproduces the grid geometry the dataset actually exhibits.

3.3.3 Resulting parameters

Table 3.2 lists the specification adopted. These values are read directly from the simulator's configuration when this report is built, so the two cannot drift apart.

Table 3.2: Sensor specification and the quantities derived from it.

Quantity	**Symbol or formula**	**Value**
Carrier frequency	f_c	77 GHz
Sweep bandwidth	B	768 MHz
Chirp duration	T_c	72.4 µs
Samples per chirp	N	512
Chirps per frame	M	64
Wavelength	c/f_c	3.8934 mm
Chirp slope	B/T_c	10.6077 MHz/µs
Sampling frequency	N/T_c	7.0718 MHz
Range resolution	$c/2B$	0.1952 m
Maximum range	$cf_s/4S$	49.97 m
Velocity resolution	$\lambda/2MT_r$	0.4201 m/s
Maximum velocity	$\pm\lambda/4T_r$	13.44 m/s
Frame time	MT_r	4.6336 ms
Ideal coherent FFT gain	$10\log_{10}(NM)$	45.15 dB
Effective gain after windowing	from window coefficients	40.33 dB

Three points about this table are worth drawing out.

Maximum range. This is set by the Nyquist limit of the sampled beat band: a target at 49.97 m produces a beat frequency of exactly $f_s/2$.

Maximum velocity. A target moving faster than 13.44 m/s radially does not disappear. Its Doppler shift falls outside the unambiguous interval and wraps around into it, so the target still appears on the map but at an incorrect velocity. This is Doppler ambiguity, and it is why the benchmark of Section 3.6 draws velocities well inside the unambiguous interval.

Processing gain. Combining 512 fast-time samples and 64 chirps concentrates a target's coherent energy into a small region of the RD map while noise remains spread across it. In the ideal, unwindowed case this concentration is worth 45.15 dB. The range and Doppler window functions applied before each transform suppress spectral leakage [16] but cost some of that gain, leaving an effective 40.33 dB. The second figure is computed directly from the window coefficients rather than measured from data, so it tracks any change to the configuration automatically. It is the one used when setting dataset difficulty, because it is the gain the simulated detector input actually receives.

3.4 The simulator

3.4.1 Signal generation and processing

The most important thing to understand about the simulator is what it does *not* do: it never draws a bright spot onto a range–Doppler map. Targets are specified in physical units, and the map is produced by the same operations a real receiver performs.

1. Each target is assigned a range R, a radial velocity v and an amplitude representing its reflectivity.
2. The transmitted chirp is synthesised as a linear frequency ramp.
3. The target's range evolves as $r(t) = R + v\,t$, with t running continuously across the whole frame, giving a time-varying round-trip delay $\tau(t) = 2r(t)/c$.
4. The echo is formed as a copy of the transmitted signal evaluated at $t-\tau(t)$, scaled by the target's amplitude. **No Doppler term is inserted by hand.** Because the delay changes as the target moves, the carrier phase $2\pi f_c\,(t - \tau(t))$ advances slightly from one chirp to the next, and that advance *is* the Doppler shift.
5. The echoes of all targets are summed, with a delayed and attenuated echo added per target where multipath is enabled.
6. Thermal noise of a calibrated variance is added, together with the clutter field if the stratum requires one.
7. The sum is mixed with the transmitted signal, giving the beat signal.
8. The range transform is applied along fast time and the Doppler transform along slow time, each with its window function.

Only at that last step does the 256×64 range–Doppler map exist. The consequence is that the shape of a target's response — its mainlobe width, its sidelobes, the scalloping loss it suffers when it falls between bin centres, and the leakage it produces along each axis — emerges from the signal processing rather than being modelled or drawn.

Figure 3.1 shows the additions that turn this chain into a benchmark generator.

3.4.2 Propagation, clutter and multipath

Propagation. Received power follows the monostatic radar equation, falling as $1/R^4$. Since the simulator synthesises signal amplitudes rather than powers, this is applied as an amplitude scaling of $1/R^2$, which is the same law expressed in amplitude. In decibels both readings coincide at a loss of $40\log_{10}(R/R_{\text{ref}})$, and this is how it is implemented, referenced to $R_{\text{ref}} = 25\,\text{m}$ so that a target at that range keeps its nominal amplitude.

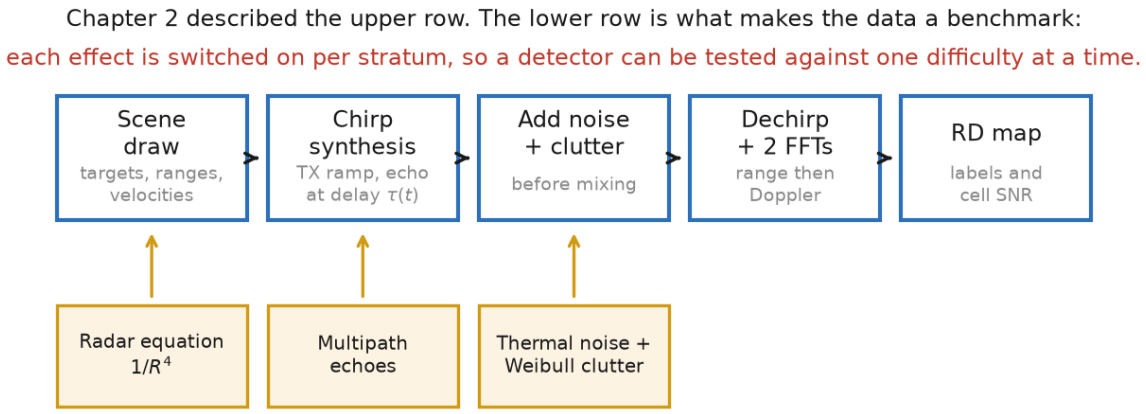


Figure 3.1: The upper row is the processing chain of Chapter 2. The lower row is what makes the output a benchmark: each effect can be enabled independently per stratum, so a detector can be tested against one difficulty at a time.

$R_{\text{ref}} = 25\,\text{m}$ was chosen as a neutral mid-range reference within the simulated operating region: targets nearer than it are strengthened by propagation and targets farther away are attenuated, rather than referencing the model to the 50 m maximum range and amplifying almost every generated target. Propagation is enabled in every stratum, so the physics stays consistent across the benchmark. The consequence is that range and difficulty are coupled: a distant target is genuinely harder, as it is in practice.

Clutter. The zero-Doppler ridge is modelled as a band of elevated return around zero velocity, with a Weibull texture rather than a Gaussian one. The shape parameter is swept, lower values giving spikier clutter, so that the rate at which each detector degrades as clutter becomes heavier-tailed is measurable rather than assumed. Its strength is specified as a clutter-to-noise ratio, which keeps it meaningful when the noise level changes.

Multipath. A delayed, attenuated echo is optionally added, producing a secondary response offset in range from the true target.

3.5 Difficulty calibration

3.5.1 What signal-to-noise ratio means here

Difficulty must be specified in a way that means the same thing to every detector, and the obvious choice turns out to be the wrong one.

The detector never sees the raw samples. An SNR defined on the received time-domain samples describes a signal that no detector in this report operates on. Between those samples and the detector lie the two Fourier transforms, which concentrate

a target's coherent energy into a small region of the map while leaving noise spread across all of it. For this configuration that concentration is worth 40.33 dB. A target specified at −10 dB on the raw samples therefore does *not* arrive 10 dB below the noise floor of the map; it arrives roughly 30 dB *above* it, and every detector finds it trivially.

Difficulty is therefore specified at the detector's input. The quantity used throughout this report is the *cell SNR*: the ratio of a target's peak cell power to the background noise power in one cell of the finished range–Doppler map,

$$\mathrm{SNR}_{\mathrm{cell}} = \frac{P_{\text{peak cell}}}{P_{\text{noise per cell}}}. \tag{3.2}$$

Given a requested cell SNR, the input noise variance needed to realise it is computed from the window functions and the transform lengths.

The achieved value is then measured, not assumed. The realised cell SNR of a given target generally differs a little from the requested one, for several reasons at once: propagation attenuates it according to its range; it rarely falls exactly on a bin centre, so it loses some peak power to scalloping; and the window functions redistribute energy between the peak cell and its neighbours. Each target's achieved cell SNR is therefore measured on the generated frame and stored beside its label, and it is that stored value — never the requested one — that every analysis in this report uses.

3.5.2 Where the difficulty sits

Figure 3.2 shows the resulting distribution across the test split.

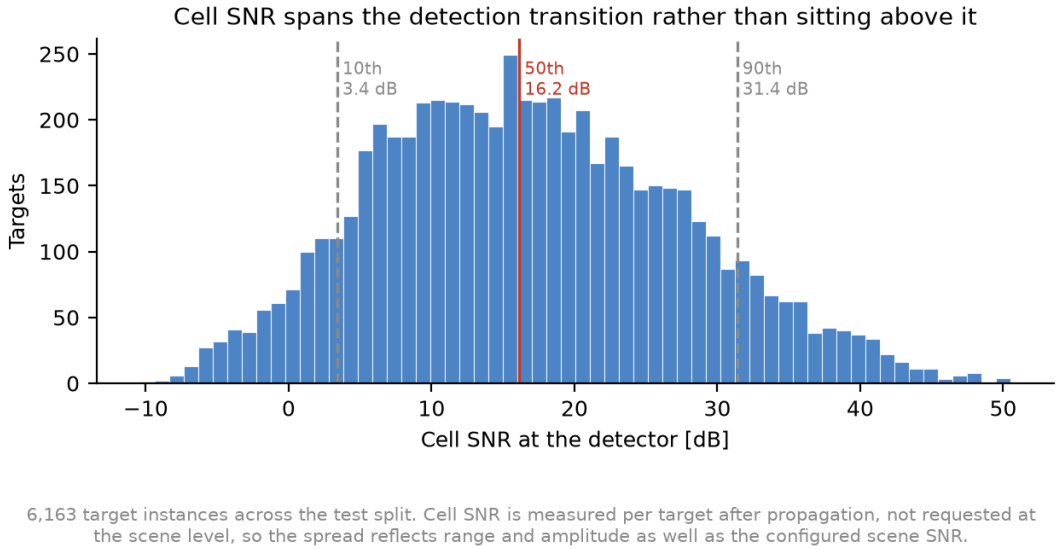


Figure 3.2: Measured cell SNR for every target in the reported holdout of Table 3.4. The spread reflects range and target amplitude as well as the configured scene SNR, because propagation is applied before the measurement.

The spread is deliberate, and the reasoning is simple. A useful benchmark needs many targets near the region where detection changes from difficult to reliable. If almost every target were strong, every detector would succeed and there would be little to compare. If almost every target were very weak, every detector would fail and again there would be little to compare. The distribution was therefore placed across the detection transition rather than above or below it.

An earlier configuration sat about 2 dB lower. At that level roughly a fifth of all targets fell below 5 dB cell SNR, a region in which all the detectors evaluated here had very low detection probability, and the clutter strata bunched near $P_d \approx 0.15$, where methods cannot be told apart. The scene level was raised by 2 dB in response.

That adjustment belongs to dataset development and not to the reported comparison. It was made while the benchmark was being designed, before the final splits were generated and well before the test split was scored. It should be noted, though, that the observation prompting it came partly from preliminary detector behaviour on earlier development data, so the calibration was not blind to how detectors behave in general — only to the results reported in Chapters 5 and 6, which did not exist at the time.

3.6 Dataset design

3.6.1 Why stratified rather than randomly sampled

When scenes were generated entirely at random, the difficult conditions occurred very rarely. Target masking appeared in only 1.1 % of frames and unresolved pairs in 0.4 %. An overall score computed on such a dataset is dominated by the much more common easy cases, and changing the random composition would change the headline number even if the detector had not changed at all.

The final benchmark instead generates each condition in equal quantity and reports results condition by condition. This gives a reliable estimate within every scenario of interest, and it makes the composition of the benchmark an explicit design choice rather than an accident of sampling.

One point avoids a misreading. Equal strata are *not* a claim about how often these conditions occur in real driving; a road scene does not contain unresolved target pairs a seventh of the time. This is a controlled benchmark, built so that each condition can be measured reliably, and the per-condition results should be read that way rather than as a prediction of average field performance.

3.6.2 The eight strata

Table 3.3 lists them, and Figure 3.3 shows one frame from each.

Table 3.3: The eight strata. Each is generated in equal quantity. Seven contain targets and are scored; `noise` contains none and supports false-alarm measurement.

Stratum	Content	Purpose
`isolated`	One target, homogeneous noise	Control: the condition where CA-CFAR is close to optimal
`masking`	Strong and weak pair, separation swept across the resolution limit	The masking failure mode of Section 2.3.5
`clutter`	Target inside the zero-Doppler ridge	Detection against heavy-tailed clutter
`clutter_edge`	Target in clean noise, CFAR window reaching into the ridge	The clutter-edge failure mode. Also the condition in which a published CNN *loses* [2]
`multi`	Four to five targets, all pairs beyond the window reach	Dense scenes without masking
`noise`	Empty frames	False-alarm measurement and threshold calibration
`mixed`	Variable target count, clutter and multipath together	A realistic combination rather than one isolated effect
`unresolved`	Pair inside a single mainlobe	Physics cap: no detector can separate them, so it acts as a sanity check

Two design points in that table are worth drawing out.

The `clutter_edge` stratum exists specifically because the literature review of Section 2.5 identified it as the condition where a published learned detector loses. A benchmark that omitted it would be systematically flattering.

Its velocity band needed care. The two clutter strata are meant to be different: in `clutter` the target sits *inside* the strong zero-Doppler ridge, whereas in `clutter_edge` the target should sit just *outside* the ridge while the CFAR reference window still overlaps it. The difficulty is that the ridge does not end where its nominal width says it does — the Doppler window spreads its energy outwards, so the region beside it is still elevated (still 13.6 dB above the noise floor at 1.68 m/s). The usable edge was therefore measured on generated frames. Had the nominal edge been used, a substantial fraction of `clutter_edge` targets would have landed inside the residual clutter skirt and the stratum would simply have duplicated `clutter`.

The `unresolved` stratum places target pairs closer together than the measured resolution of the range–Doppler representation used here, so the detectors evaluated in this report are not expected to separate them reliably. It is retained as a sanity check: a method reporting high P_d on this stratum would point to a fault in the evaluation rather than to a capable detector, and this is what exposed one such fault during development.

The frame shown for each stratum is chosen by a fixed rule: the frame whose target count is closest to that stratum's median target count, with ties broken by frame index. The rule depends only on the scene composition, never on appearance and never on how any detector performed on the frame. At full map scale most targets are not obvious by eye, which is by design — the benchmark places them near the detection transition.

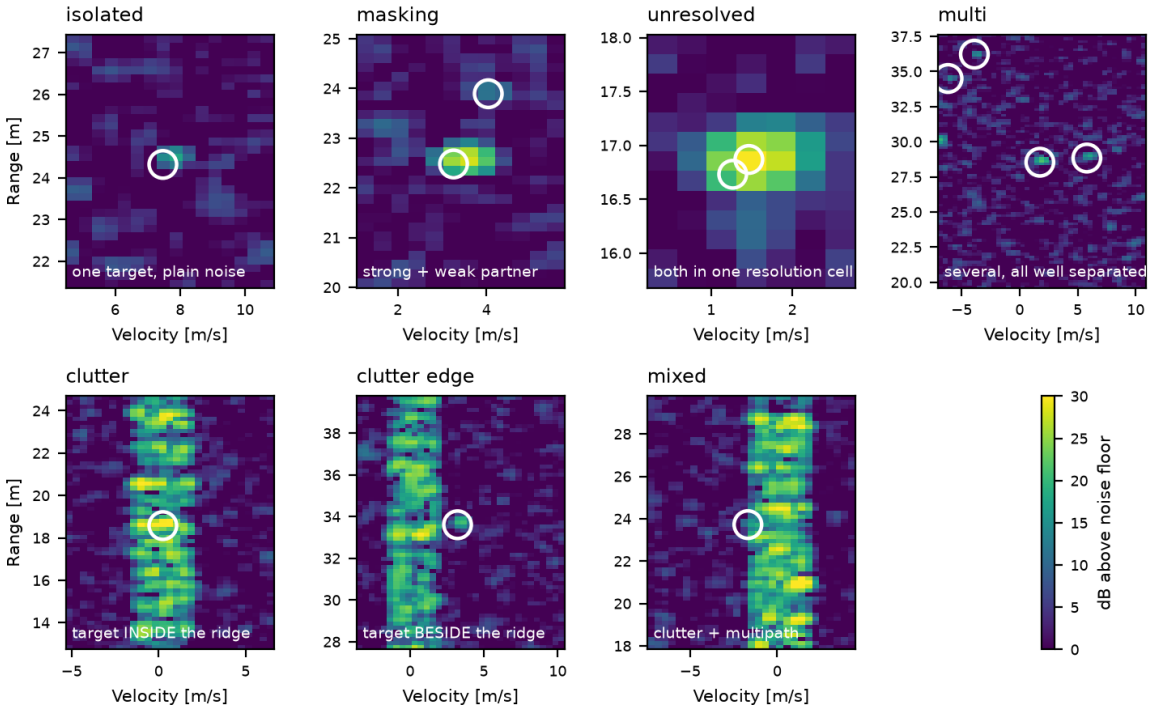


Figure 3.3: What each stratum actually contains. One frame per stratum, centred on the strongest target, with white circles marking the annotated targets: the weak partner beside a strong one in `masking`; two targets sharing a single mainlobe in `unresolved`; well-separated returns in `multi`; a target inside the ridge in `clutter` against one beside it in `clutter_edge`; and clutter with multipath in `mixed`. Window sizes differ per stratum so that the relevant structure is visible, and the `noise` stratum is omitted because it contains no target to centre on. Figure B.1 in Appendix B shows the same frames at full map scale.

3.6.3 Splits

Six datasets are generated from the same configuration with different random seeds.

Table 3.4: Dataset splits. Frame counts are read from the generated data when this report is built.

Split	Role	Frames
`bench2_train`	Network training	8,400
`bench2_val`	Threshold selection, epoch selection	1,600
`bench2_tune`	CFAR window geometry and suppression radius	400
`bench2_snr_sweep`	Detection probability against cell SNR	1,300
`holdout`	**The reported numbers. Read once**	3,200
`bench2_test`	Superseded benchmark read; retained, not reported	3,200

The separation of `tune` from `val` is deliberate. The CFAR window geometry, the suppression radius and the network's operating point are all selected quantities, and

selecting them on the same data that sets the other arm's threshold would introduce exactly the asymmetry Section 1.2 identified. Chapter 4 describes those selections.

The final row needs explaining. **The final holdout is a separate test set kept completely out of model training, parameter selection and threshold tuning, and used only for the final synthetic evaluation.** Separate random seeds ensure that it contains no generated frame shared with the training or development splits. The benchmark of Chapter 5 is scored on `holdout`, which was generated after the evaluation protocol was finalised and verified to share no frame content with any earlier split. `bench2_test` was the corresponding read under the superseded post-processing protocol; it is kept on disk as part of the record but no number in this report is taken from it. Appendix A states why a new split was generated rather than the old one re-scored: a split whose answer is already known is no longer a one-shot read.

3.7 Verification

A simulator is an instrument, and an instrument that has not been checked cannot support a measurement. Three checks were run before any detector result was produced.

Point spread function. An isolated target was placed and its response measured along both axes, on-bin and deliberately off-bin. Figure 3.4 shows the result. The mainlobe spans approximately two bins, which is what fixes the guard-cell count of the CFAR window in Chapter 4: the guard region must cover a target's own spread, or a target would inflate its own threshold.

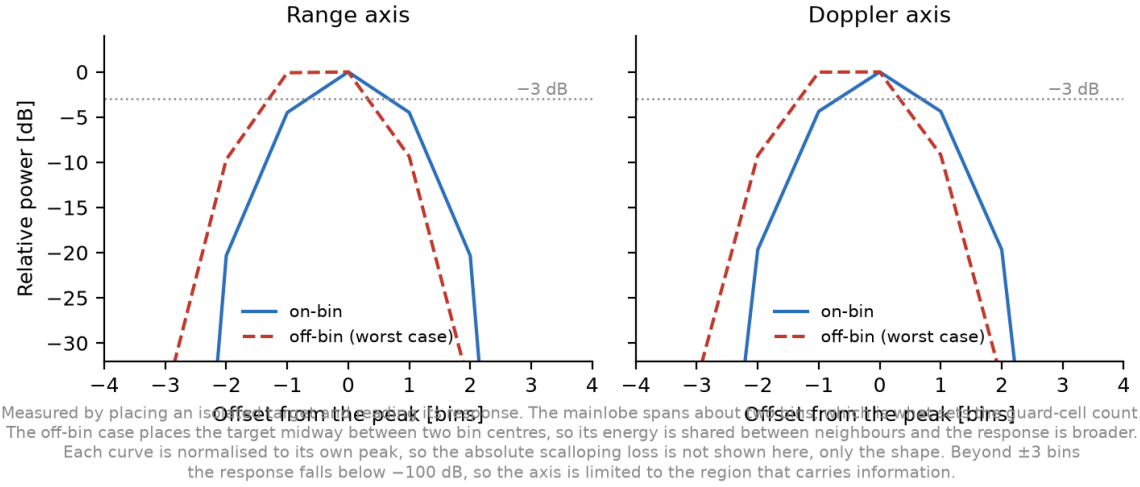


Figure 3.4: Response of an isolated target along each axis, measured rather than assumed. The off-bin case places the target midway between two bin centres, so its energy is shared between neighbouring cells and the response is broader. Each curve is normalised to its own peak, so this figure shows the *shape* of the response and not the absolute scalloping loss.

Resolution limits. Measured on the same data, two equal-strength targets require about 0.39 m of range separation to appear as separate peaks, rising to 0.59 m when one partner is weak, and 1.16 m/s in Doppler. These measurements set the separation ranges swept in the `masking` and `unresolved` strata, so that both sit across the real limit rather than at an assumed one.

Label alignment. Every generated target was checked against the peak actually present in the map. Alignment is exact above about 22 dB cell SNR. Below that a small fraction of labels do not coincide with the strongest local peak; this was investigated and traced to noise spikes exceeding a weak target, which is a property of the physics rather than a labelling fault, and it disappears at high SNR as a genuine bug would not.

3.8 Conclusion

This chapter specified the instrument that produces every synthetic result in this report.

Simulation was chosen because it supplies exact ground truth, independent control over each difficulty, and enough examples of the rare conditions that actually separate detectors. Its cost is external validity, which is why Chapter 7 exists.

The sensor reproduces the range–Doppler grid geometry and bin spacing of the real dataset used later, so that a physical separation corresponds to the same number of cells in both domains. The effective bandwidth of 768 MHz was derived from that dataset's own published grid rather than from its quoted bandwidth figure, which is inconsistent with it. Matching the grid removes a scaling confound; it does not make target responses identical, and Chapter 7 treats that remaining difference explicitly.

Targets are not drawn onto the map. They are specified physically and passed through the full signal chain, so the shape of a response, its sidelobes and its scalloping loss emerge from the processing rather than from a model of it. Difficulty is measured at the detector as cell SNR and recorded per target, so the value used in analysis is the one realised rather than the one requested.

The benchmark is divided into eight equal conditions spanning both the regime where CA-CFAR is close to optimal and the regimes where it is documented to fail — including one chosen specifically because a published learned detector loses there. Five independent splits keep network training, threshold selection, CFAR window selection, the SNR sweep and the single scored test set apart, so no detector is tuned on the data used to score it. The simulator's point spread function, resolution limits and label alignment were all verified before any detector was run.

Chapter 4 now specifies the two detectors that operate on these frames, and fixes the evaluation protocol before any comparative result is produced.

Chapter 4: Detectors and Evaluation Protocol

4.1 Introduction

This chapter specifies the two detectors compared in this report and fixes the rules by which they are scored. Choosing an evaluation protocol after seeing comparative results risks introducing bias, so the protocol was fixed before the benchmark result was examined.

The chapter is organised around the fairness problem of Section 1.2. Each asymmetry listed there is addressed by a specific choice recorded here: the network's receptive field is matched to the CFAR window; both detectors are reduced to the same output type and scored by one code path; operating points are matched at measured rather than design false-alarm rate; and results are reported at more than one matching tolerance.

What is held in common, and what is not

Both arms turn the same range–Doppler map into a per-cell score map, and both score maps are then handled identically. What they share is the **same underlying data** — the identical frames from the identical processing chain of Chapter 3; a **comparable local spatial footprint** of 21×9 cells, by construction; and the **same downstream evaluation** — one peak extractor, one matcher, one set of metrics, one code path.

What they do *not* share is the transform applied between the map and the score. CFAR operates on range–Doppler power and forms a local ratio statistic; TinyCNN applies a fixed logarithmic normalisation relative to the frame median (Section 4.3.3). **Their input processing is therefore not mathematically identical**, and this report does not claim it is. Each arm has a fixed, detector-specific transform that is part of what that detector *is*: a CFAR detector without its ratio is not CFAR, and a network evaluated on a different normalisation from the one it was trained on is not the trained network. The controlled variable is the local decision rule, with as many other factors held common as is practical.

4.2 The classical arm

4.2.1 Window geometry

The reference window used throughout this report spans 21 bins in range by 9 in Doppler. Of its 189 cells, **1** is the cell under test, **24** are guard cells excluded from the estimate, and **164** are reference cells forming the background estimate Z (Figure 4.1).

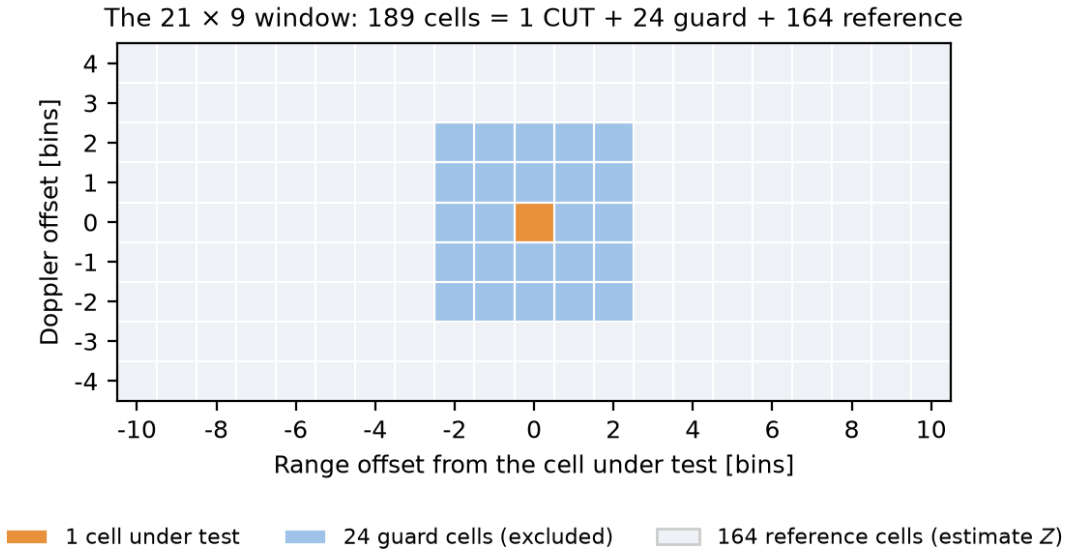


Figure 4.1: The reference window used by every CFAR arm in this report. Counting outward from the centre, the range axis has 2 guard cells then 8 reference cells on each side; the Doppler axis has 2 and 2.

Two aspects of that geometry were decided by measurement rather than convention.

The guard count follows the measured point spread function. Chapter 3 measured a mainlobe spanning about two bins. The guard ring must cover a target's own spread, or the target inflates its own threshold and suppresses itself, so 2 guard cells per side is the smallest value that does the job.

The window is rectangular because the two axes are not alike. The range axis has 256 bins and no large-scale structure; the Doppler axis has only 64 and carries the zero-Doppler clutter ridge. A window sized for range would span a third of the Doppler axis and drag ridge energy across a wide band of thresholds.

Candidate geometries were compared on `bench2_tune`, a split used for nothing else, and 21×9 was selected there before any read of the test split (Figure B.2, Appendix B). The separation matters: the network's operating point is selected on `bench2_val`, so using one split for both selections would tune one arm on the other's selection set. What this geometry establishes for the comparison is that both detectors observe the same local extent of the map. One qualification is recorded rather than repaired: **SO-CFAR did not take part in the selection**, having been added to the project after

the geometry was chosen, so it inherits a window selected on the evidence of the other three variants. If SO is disadvantaged by that, the disadvantage is a property of this evaluation and not of the algorithm.

4.2.2 Variants

Four variants are evaluated, differing only in how the reference cells become Z: CA, GO, SO and OS, as described in Section 2.3.5. OS uses a rank fraction of 0.75, which was never swept — a stated limitation, since a better-tuned OS might do better than the one reported here.

Implementation uses vectorised two-dimensional filtering rather than explicit Python loops over every cell under test. CA-CFAR uses box-filtered reference sums, GO/SO use separate half-window sums split along the Doppler axis, and OS-CFAR uses a rank filter. The vectorised outputs were checked against direct brute-force window calculations before evaluation.

4.2.3 Threshold calibration

Two distinct calibrations occur in this report and are easy to confuse. **(A) Threshold-multiplier calibration**, described here, fixes a baseline multiplier α per CFAR variant using *noise-only* data; it concerns the CFAR statistic alone, before any comparison with the network. **(B) Operating-point matching**, described in Section 4.4.3, is the headline comparison: each detector's *score threshold* is swept and the arms are compared at the same *measured* false-alarm rate. The P_d values reported in Chapter 5 come from (B), not from the table below.

The multiplier achieving a chosen false-alarm rate has a closed form for CA-CFAR, equation (2.10). It is not used. All four multipliers are instead obtained by Monte-Carlo calibration: pure-noise frames are passed through the identical processing chain, the CFAR ratio map is computed for each, the ratio values are pooled over all validly tested cells, and the multiplier is taken as the $(1 - P_{fa})$ empirical quantile of that pooled distribution. The result is the level exceeded by exactly a fraction P_{fa} of noise cells, by construction, and it requires no distributional assumption — which matters, because only CA has a tractable one.

The comparison multiplier in the right-hand panel is the CA expression of equation (2.10) applied with the cell count appropriate to each statistic: the full 164 reference cells for CA and OS, and the half-ring count for GO and SO. That makes it *exact for CA*, a reasonable approximation for GO and SO, and for OS only a first approximation, since an order statistic has a different distribution entirely. It is therefore labelled an *analytic reference*, not a closed-form solution per variant.

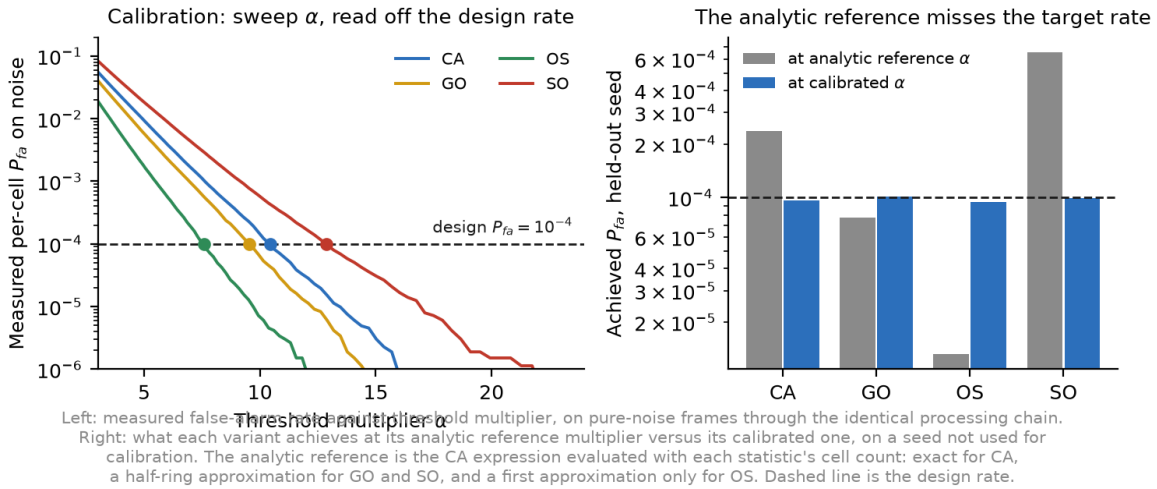


Figure 4.2: Analytic multipliers do not deliver the design false-alarm rate for any variant; calibrated ones do. Left: measured false-alarm rate against threshold multiplier on pure-noise frames, with the calibrated value marked. Right: the rate each variant achieves at its *analytic reference* multiplier against its calibrated one, on a verification seed not used for calibration.

With that definition, the right-hand panel is the argument. At the analytic reference the achieved rates are not the design rate: CA overshoots by roughly a factor of two and SO by more than six, while OS undershoots by nearly an order of magnitude — the largest error, as expected, for the variant whose statistic the expression fits worst. After calibration all four sit at the design rate on data not used to set them; Table 4.1 lists the multipliers actually used.

Table 4.1: Calibrated threshold multipliers at a design P_{fa} of 10^{-4} for the 21×9 window.

Variant	Calibrated α	Analytic reference	Status of the reference
CA	10.4586	9.4739	exact for this statistic
GO	9.5458	9.8081	half-ring approximation
SO	12.8494	9.8081	half-ring approximation
OS	7.5786	9.4739	first approximation only

For CA-CFAR, the empirically calibrated multiplier is slightly higher than the ideal closed-form value because neighbouring RD cells are correlated rather than independent. This calibration difference should not be confused with CFAR loss, which is the detection penalty caused by estimating the background from finite reference data and is measured separately in Chapter 5.

4.3 The learned arm

4.3.1 Architecture

The learned detector, referred to throughout as TinyCNN, is a fully convolutional network that maps a range–Doppler map to a per-cell score at the same resolution. It has 5,041 parameters, and its layer stack is drawn in Figure 4.3.

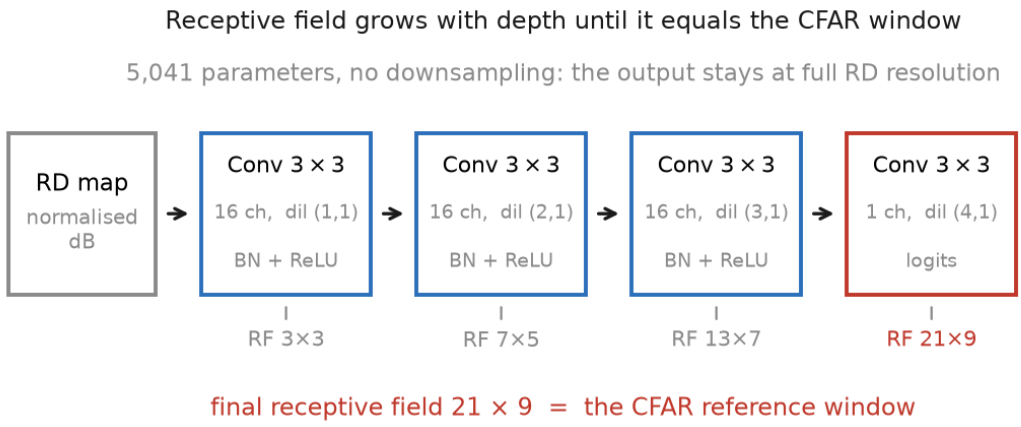


Figure 4.3: Four dilated convolutional layers with the receptive field growing to match the CFAR window. There is no downsampling, so the output remains at full range–Doppler resolution and no peak is displaced.

Three choices define it.

The receptive field is matched to the CFAR window. This is the central design decision of the project. The layers use dilations $(1, 2, 3, 4)$ along range and $(1, 1, 1, 1)$ along Doppler, composing to a receptive field of 21×9 — exactly the window of Figure 4.1. Dilation [17] enlarges the receptive field without adding parameters and without downsampling. The value was verified empirically, by back-propagating from a single output cell and recording which input cells receive non-zero gradient, rather than trusting the arithmetic.

There is no downsampling. Padding equals dilation at every layer, so spatial size is preserved exactly through the stack. Avoiding a pooling and upsampling stage removes one source of localisation error; it does not by itself guarantee that peaks are placed correctly, which is an empirical question addressed in Chapter 5.

The network is deliberately small. At 5,041 parameters and 16 channels it is tiny by modern standards. The question is what a small learned local rule adds over a fixed one, not how well a large network can be made to perform.

4.3.2 What "matched" does and does not mean

The receptive-field match is the fairness argument of this report, so its limits should be stated plainly.

Matching the footprint does not equalise the decision rule. Both detectors see the same 21×9 neighbourhood, but they use it differently. CFAR takes the cell under test as its statistic, excludes the guard ring entirely, and estimates the background from the 164 reference cells. The network has no such partition: all 189 cells in its footprint are available to it. This is a structural difference between the two decision rules, not a difference in spatial footprint: CFAR excludes the guard cells out of necessity, because a simple mean cannot distinguish a bright neighbour caused by a target's own mainlobe from one caused by high noise, while a learned function is not subject to that constraint. The accurate statement is therefore *the same footprint, but CFAR is constrained to a hand-derived statistic that requires discarding the inner cells.* Whether that difference contributes to any measured margin is tested in Chapter 6 rather than assumed here.

Nominal and effective receptive fields are not the same thing. It is a general property of stacked convolutions that influence decays from the centre outward, whereas CFAR weights its reference cells uniformly. The effective receptive field of this particular network was *not* measured, so this is noted as an architectural consideration rather than a measured property, and no part of the fairness argument rests on it. The 21×9 figure quoted above is the nominal receptive field, which *was* measured.

4.3.3 Input representation

Raw range–Doppler power spans a very large dynamic range and its absolute scale can vary between frames. TinyCNN therefore receives power in decibels relative to the frame median. This compresses the numerical range and removes irrelevant global scale changes while preserving the local target-to-background contrast used for detection. CFAR obtains a related scale invariance naturally through its CUT-to-background ratio.

The transform is a fixed part of the detector, applied identically during training and inference:

$$X = \frac{1}{20}\,\mathrm{clip}\left(10\log_{10}\frac{P}{\mathrm{median}(P)},\ -10,\ +60\right), \tag{4.1}$$

where P is the range–Doppler power map.

The median is used rather than the frame maximum because peak normalisation would rescale every frame by its brightest target and destroy the absolute signal-to-noise information the detector needs. It is nonetheless a statistic of the frame, not a measurement of physical noise power, and the two need not coincide in cluttered scenes

or in real data; Chapter 7 returns to this, since the same fixed transform must then be applied to frames whose background statistics differ from those it was fitted on.

4.3.4 Labels

Section 2.5 noted that a learned detector must be given a training target and that the choice is not neutral. The label used here is a 3×3 block centred on each target's cell. A block label asks the network to respond over a small region rather than at one exact cell, which relaxes the localisation demand relative to a single-cell label. It is retained because it belongs to the frozen benchmark: changing it would mean the reported results no longer describe the system that produced them. Its effect is not left open — Chapter 6 trains under alternative supervision while leaving the frozen benchmark intact, so the influence of this choice on localisation is measured rather than assumed.

4.3.5 Training

Table 4.2 gives the training configuration. Every entry is a recorded choice rather than a default, and the whole configuration was frozen before any benchmark split was scored. It was not revisited afterwards: the one later protocol change described in Appendix A touches candidate extraction, not the network or its training.

Table 4.2: Frozen training configuration, read from the frozen contract when this report is built.

Setting	Value	Note
Loss	`BCEWithLogits`	with positive-class weighting
Positive weight	100	chosen by ablation, not class balancing
Optimiser	Adam [18]	learning rate 0.001, batch 32
Epochs	21	selection rule below
Early stopping	off	deliberately
Seeds	8	all reported; single-seed results not trusted
Threads	2	fixed for bit-level reproducibility
Training split	`bench2_train`	8,400 frames
Selection split	`bench2_val`	1,600 frames

Two entries need explanation.

Epoch selection. The reported epoch was not chosen by early stopping, because validation loss plateaus later than detection probability does and would select the wrong epoch. Instead, every epoch from 1 to 25 was evaluated on the validation split, detection probability was pooled over the 8 seeds at each epoch, and the selected epoch

is the *earliest* whose pooled value lies within one standard error of the plateau. The rule was written down before the sweep was examined.

Positive weighting. The positive class is vastly outnumbered, and exact class balancing would imply a weight near 974. The weight of 100 won a three-way ablation on the validation split.

Eight seeds are reported, always. A single training run is one sample from a distribution, and the spread between seeds is not negligible. Every network result in this report is a mean over 8 seeds with the individual seeds shown.

4.4 The evaluation protocol

4.4.1 One rulebook

Each detector produces its own score map: CFAR works from linear power and produces a ratio map, the network applies the normalisation of Section 4.3.3 and produces a logit map. Everything after that point is shared. Both maps go through the same code, in the same order:

$$\text{score map} \rightarrow \text{peak extraction (NMS)} \rightarrow \text{operating threshold}$$
$$\rightarrow \text{one-to-one matching} \rightarrow \text{metrics}$$

Figure 4.4 shows the same chain with the stage boundaries marked.

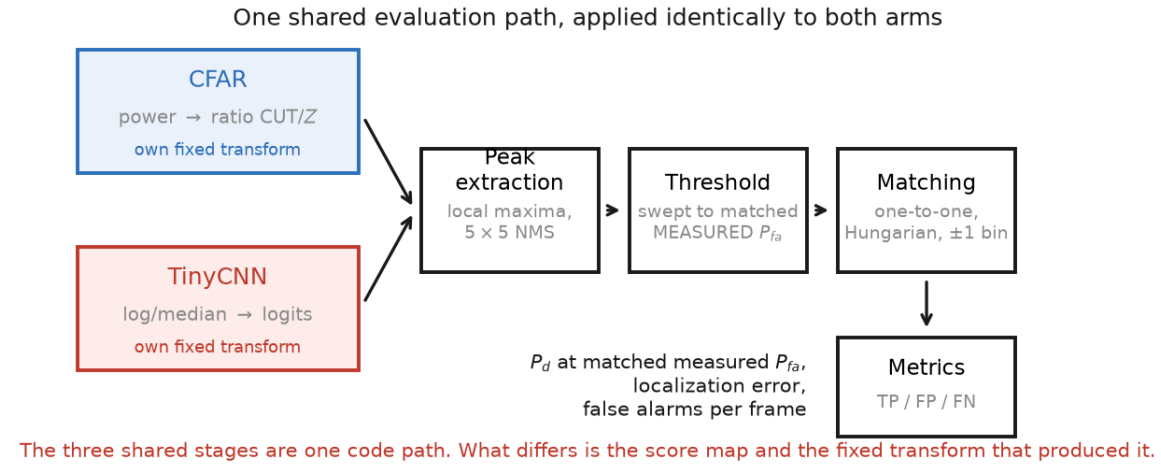


Figure 4.4: Both arms are reduced to point lists and scored through a single code path. What differs between the two routes is the score map and the fixed detector-specific transform that produces it; everything downstream is shared.

Peak extraction

A strong target does not produce one bright cell. It produces a small cluster of bright cells, several of which may be local maxima. Non-maximum suppression (NMS) reduces

each cluster to its strongest cell: a cell is kept as a candidate if it is the largest value in a square neighbourhood of side $2r + 1$, set by a suppression radius r. This protocol uses $r = 2$, a 5×5 window. Candidates are extracted *before* any threshold is applied, so the candidate set is computed once and reused at every threshold the score sweep visits.

The radius is a post-processing constant, not a property of either detector, and it was chosen on development data only — the tuning split for selection, the validation split for confirmation. Neither the held-out benchmark of Chapter 5 nor any real data of Chapter 7 contributed to the choice. Radii 1, 2, 3, 5 were compared at matched measured P_{fa}, with the threshold re-derived at each radius so that a drifting false-alarm rate could not confound the comparison.

A radius of 1 left duplicate peaks around individual targets, while radii 2, 3 and 5 removed them. Larger radii increasingly suppress nearby genuine targets, so radius 2 was selected as the smallest clean setting. This corresponds to a 5×5 local-maximum neighbourhood. The full sensitivity sweep is reported in Appendix B.

Because CARRADA objects are spatially extended, this point-target suppression rule is revisited in Chapter 7.

The valid region and the candidate floor

CFAR cannot form its full 21×9 window near the border, so the outer margin is excluded: $2 + 8 = 10$ bins on the range axis and $2 + 2 = 4$ on Doppler. The same mask is applied to the network, so both detectors are scored on exactly the same cells and the false-alarm denominator counts the same cells for both.

The shared extractor also keeps only cells with a strictly positive score. CFAR scores are ratios of powers and are always positive, so nothing is removed; network scores are logits, which are negative wherever the predicted probability is below one half, so those cells can never become candidates. This does not affect any reported result, since the operating points used here lie well above zero, but it truncates the extreme high-P_{fa} tail of the network's ROC curve, which cannot extend below a logit of zero.

4.4.2 Matching

A detection matches a truth if it lies inside the tolerance gate, $|\Delta r| \leq G_r$ and $|\Delta v| \leq G_v$. Matching is one-to-one, computed by Hungarian assignment [19] on the distance between candidate and truth — the pairing that minimises total distance, with pairs outside the gate made unaffordable. A matched detection is a true positive; an unmatched truth is a false negative.

Two consequences follow for detections landing near a real target.

- **One truth gives at most one true positive.** If several detections fall inside one isolated target's gate, exactly one is matched and the rest are ignored — neither credited nor charged as false alarms. Cells inside a truth's gate also leave the false-alarm denominator, so the detector is not credited for correctly rejecting them either.
- **Two nearby truths are different.** When two truths are very close, their matching regions can overlap. Several peaks from one broad physical response may then be assigned to different truths, allowing one unresolved response to be credited more than once. NMS limits this effect by reducing each local response to a single candidate peak.

That second case is why the suppression radius was not chosen on detection probability alone. The rule is applied identically to both arms.

Two shaded regions appear in this chapter and they are easy to confuse. The **CFAR guard cells** of Figure 4.1 belong to the detector: they keep a target's own energy out of the background estimate. The **near-target exclusion region** of Figure 4.5 belongs to the evaluator: it decides which unmatched detections count as near misses rather than false alarms. One is built from the CFAR window geometry, the other from the matching tolerance, and they need not have the same extent.

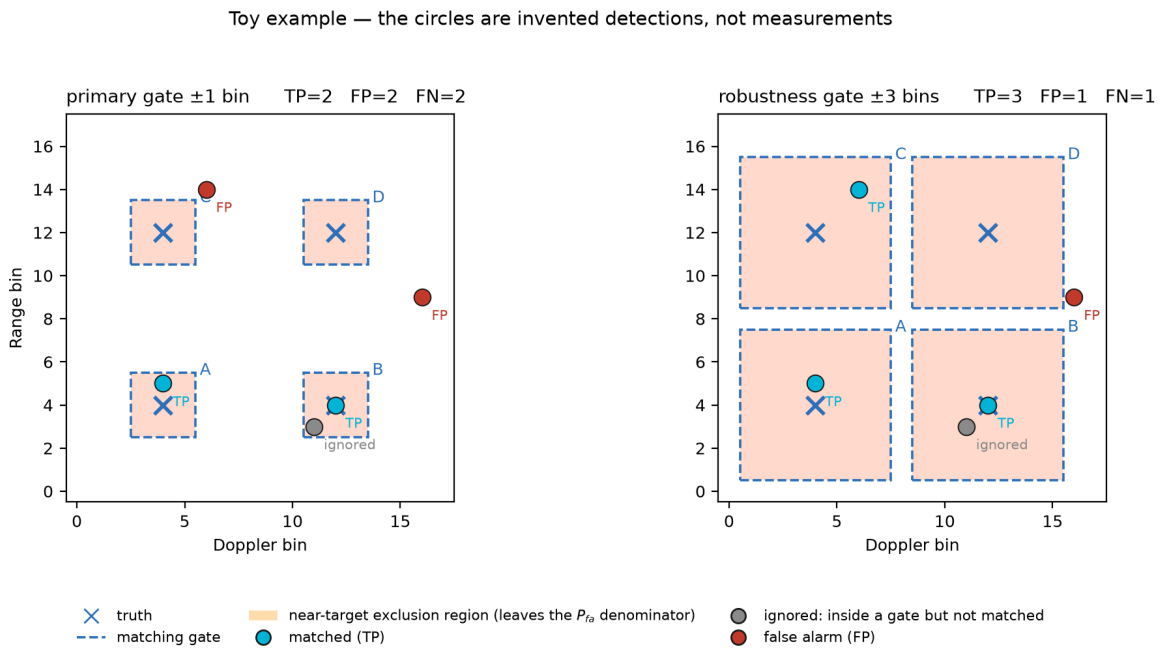


Figure 4.5: How the gate, one-to-one matching and the two tolerances interact. A toy frame with four truths and five detections, scored by the same function used throughout this report; **the detections are invented to exercise every outcome, and are not measurements**. Left, the primary ±1-bin gate: A and B are matched, the second detection near B is ignored because B is already taken, the detection two bins from C falls outside the gate and so is both a miss and a false alarm, and D is a plain miss. Right, the same frame at ±3 bins: the detection near C becomes a match. The shaded area is the near-target exclusion region, drawn from the matching gate — **not** the CFAR guard region of Figure 4.1.

4.4.3 Operating point

The headline synthetic comparison is P_d at matched *measured* P_{fa}. Each detector's score threshold is swept, the achieved false-alarm rate is measured at each setting, and the two arms are compared where those achieved rates agree. This is stricter than matching at design P_{fa}: Figure 4.2 showed that achieved and design rates differ, and by different amounts for different variants, so matching on the achieved rate removes that confound.

The sweep is performed *per condition*. Each stratum gets its own sweep, and the arms are compared at the same measured false-alarm level within that stratum. This is an **evaluation operating point, not a deployable threshold**: it requires labels on the data being scored, and one global setting would not achieve the same rate in every condition. Its purpose is to remove threshold choice as a confound from the comparison.

Chapter 7 necessarily works differently. There the network must run at a single fixed threshold derived from synthetic validation data alone, without access to the real data it is scored on, because real data supplies no false-alarm axis to sweep

against. **Within each experiment all detector arms share the same evaluation rule and operating-point definition; the synthetic and real-data experiments necessarily use different operating-point protocols, and are compared only within domain.**

4.4.4 Tolerance

The primary gate is ±1 bin, fixed in advance. A gate of ±3 bins is reported beside it as a robustness check, never instead of it. Reporting both separates detection from localisation: a detector that finds a target but places it two bins away is credited at the wider gate and not at the narrower one — the detection near truth C in Figure 4.5. Comparing the two numbers therefore shows how far a conclusion depends on localisation tolerance. Both are reported for every condition, whichever way they point.

4.4.5 Statistics

Differences are assessed by a paired bootstrap [20] with 2,000 resamples. **The resampling unit is the frame.** Frames are drawn with replacement, and within each resample both arms are re-scored on exactly the same frames, so frame difficulty cancels instead of adding noise. Resampling whole frames also preserves the dependence among targets belonging to the same frame.

Seven conditions are tested, so using $p < 0.05$ separately for each would increase the chance of a false positive across the full set. A Bonferroni correction therefore uses $0.05/7 = 0.0071$ as the significance threshold. A condition is called statistically significant only when $p < 0.0071$. This is why the unresolved result ($p = 0.017$) is not considered significant despite being below 0.05.

One limitation should be noted. The operating threshold is computed once per condition and held fixed across resamples, so the intervals capture sampling error in P_d at a fixed operating point. They do not include the uncertainty in choosing that threshold, which would require redoing the sweep inside every resample. The reported intervals are therefore slightly optimistic.

4.5 Conclusion

Both detectors are now fully specified, and so is the instrument that will measure them.

The classical arm uses a 21×9 window whose guard count follows the measured point spread function and whose shape was selected on a split reserved for that purpose, with SO the one variant that did not take part in that selection. Its four multipliers

are calibrated on noise-only data rather than taken from an analytic expression which, as measured here, does not deliver the design false-alarm rate for any of them.

The learned arm is a 5,041-parameter fully convolutional network whose receptive field composes to the CFAR window. The limits of that match are set out in Section 4.3.2: the arms share a spatial footprint, not a decision rule, and the network is not obliged to discard the guard cells that CFAR must exclude. Whether that difference matters is left to Chapter 6 to test.

The protocol reduces both arms to point lists and scores them through a single code path at matched measured false-alarm rate, at two tolerances, with paired frame bootstrap intervals and a correction for multiple comparisons. Its two known asymmetries — the positive-score candidate floor, and the fact that the operating point is an evaluation construct rather than a deployable threshold — are recorded above rather than left for a reader to find. The protocol was fixed, and the network configuration frozen, before the test split was read.

Chapter 5 now reports what that measurement returned.

Chapter 5: Synthetic Benchmark Results

5.1 Introduction

This chapter reports what the protocol of Chapter 4 returned when applied to the benchmark of Chapter 3. It is a results chapter: it states what was measured, and leaves the question of *why* to Chapter 6 and the question of whether it generalises to Chapter 7.

Four measurements are reported, in the order that makes them interpretable.

1. **Was the operating point actually matched?** A comparison at a supposedly common false-alarm rate means nothing until that is verified, so it is checked first.
2. **The headline result**: detection probability per condition, with confidence intervals and a correction for multiple comparisons.
3. **The same result in decibels**, which is the currency the comparable literature of Section 2.5 reports, and which makes the margin externally interpretable.
4. **What the headline does not show**: localisation, tolerance sensitivity and computational cost, each of which qualifies the result in a different direction.

Every number below comes from a single read of one held-out split. That split (3,200 frames, 400 in each of its 8 strata, of which 2,800 contain targets and are scored for detection) was generated after the configuration and the evaluation protocol were both fixed, from a seed never previously used, and was verified to share no frame content with the training, validation, tuning or SNR-sweep data. It had not been scored by anything before the results in this chapter were produced. Appendix A records why a new split was generated rather than an earlier one re-scored.

5.2 The operating point was matched

Chapter 4 argued that matching at design P_{fa} is not sufficient, because the achieved rate and the design rate differ, and differ by a different amount for each variant. That argument is only useful if the matching then worked.

Figure 5.1 shows the achieved per-cell false-alarm rate for every arm and every condition, as a ratio to the design rate of 10^{-4}.

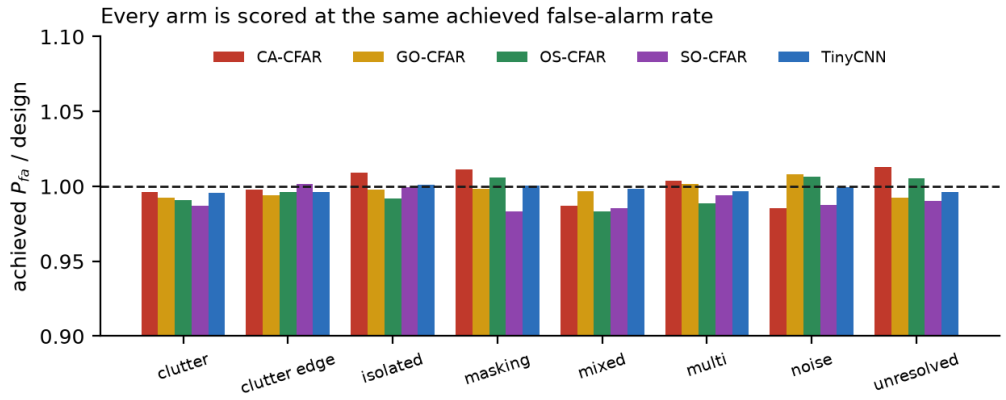


Figure 5.1: Achieved per-cell false-alarm rate divided by the design rate, for all five arms across every condition of the held-out split, including the target-free noise stratum.

Every value lies between 0.983 and 1.013 — within two per cent of unity. The comparison that follows is therefore at a genuinely common measured operating point, not at an assumed one.

Figure 5.2 puts that operating point in context by sweeping the score threshold across its whole usable range, pooled over the seven scored conditions.

5.3 Detection probability by condition

Figure 5.3 gives the headline measurement: detection probability for the frozen network against the best CFAR variant for each condition, at the primary ±1-bin gate.

Two features of the comparison deserve emphasis before the numbers.

The classical arm is the best variant per condition. Rather than fixing one variant throughout, each condition is scored against whichever of CA, GO, SO or OS performs best there. This is a stronger baseline than any single detector could achieve in deployment, because it amounts to an oracle choosing the right variant for a condition it has already seen. The margin reported is therefore a conservative one.

Eight seeds, not one. Every network value is the mean over 8 independently trained seeds.

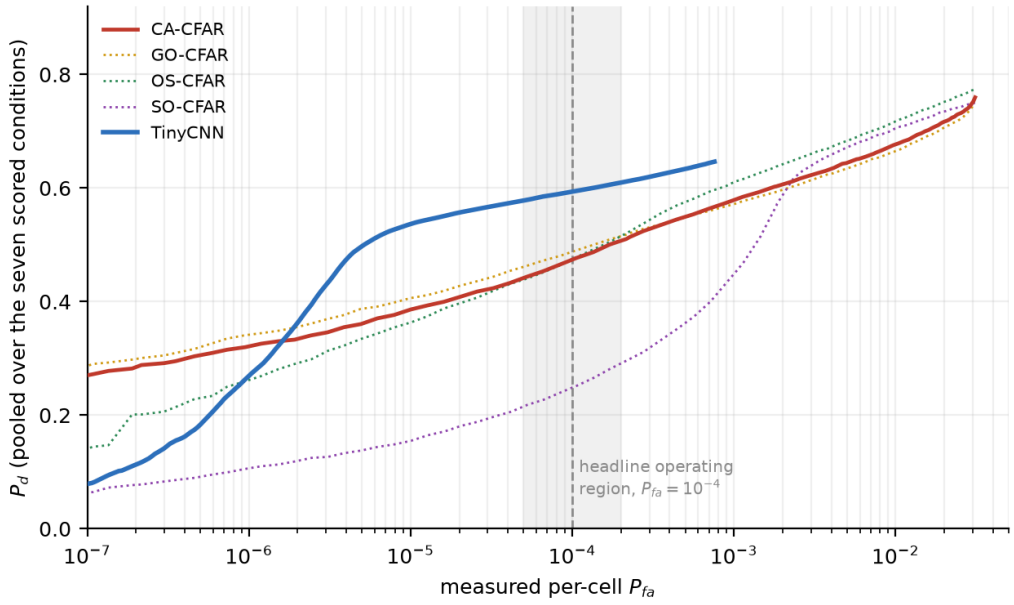
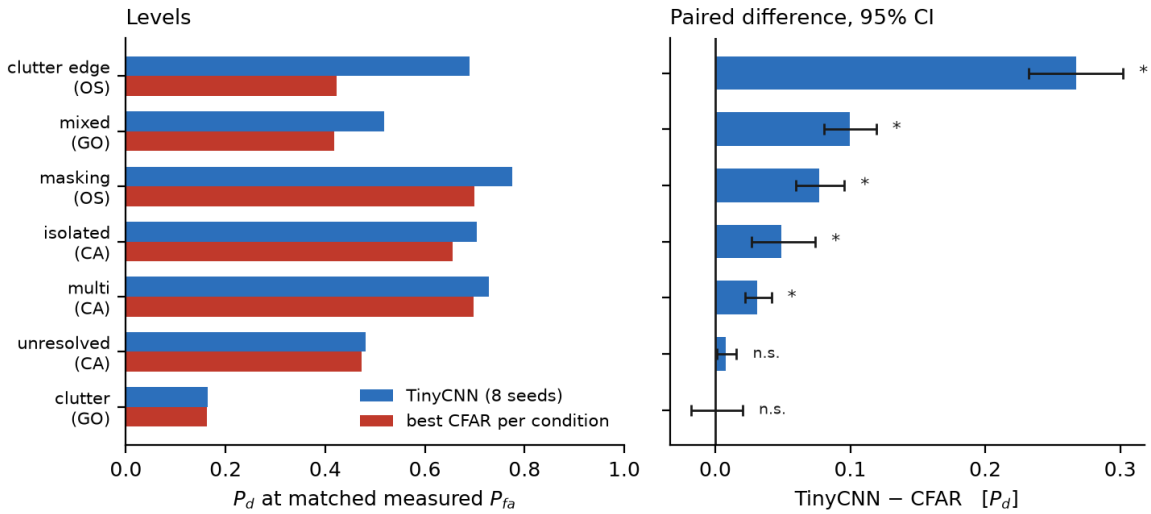


Figure 5.2: Each point corresponds to a different score threshold; higher P_d at the same measured P_{fa} is better. Pooled over the seven scored conditions of the held-out split under the final radius-2 protocol, with TinyCNN the mean over 8 seeds. The shaded band marks the 10^{-4} operating region used for the headline comparison — which sweeps each condition separately, so a pooled value here need not equal the per-condition mean of Table 5.1. The network's curve stops near 8×10^{-4} because the shared extractor keeps only positive logits (Section 4.4.3), which truncates its high-P_{fa} tail.



Held-out split, 400 frames per condition, read once. The classical arm is the best CFAR variant for that condition, named in the label.
* marks significance after Bonferroni correction across the seven conditions; n.s. marks conditions that do not reach it.

Figure 5.3: Left: detection probability at matched measured P_{fa}. Right: the paired difference with its 95 % bootstrap confidence interval. The classical arm is the best-performing CFAR variant for that condition, which is a deliberately demanding comparator.

Table 5.1 gives the values.

Table 5.1: Detection probability at matched measured P_{fa} on the held-out split, primary gate. The CFAR column is the best variant for that condition, named in brackets; it was re-selected on this data rather than inherited.

Condition	TinyCNN	Best CFAR	Difference	95 % CI
Clutter edge	0.6904	0.4232 (OS)	**+0.2672**	[+0.2324, +0.3020]
Mixed	0.5175	0.4178 (GO)	+0.0997	[+0.0807, +0.1192]
Masking	0.7758	0.6987 (OS)	+0.0770	[+0.0598, +0.0956]
Isolated	0.7041	0.6550 (CA)	+0.0491	[+0.0269, +0.0741]
Multi-target	0.7290	0.6979 (CA)	+0.0311	[+0.0219, +0.0415]
Not significant after Bonferroni correction ($\alpha = 0.0071$)				
Unresolved	0.4805	0.4725 (CA)	+0.0080	[+0.0011, +0.0158]
Clutter	0.1635	0.1631 (GO)	+0.0005	[−0.0176, +0.0202]

Read practically rather than by sign count, the result is: **5 established gains, one small difference on the unresolved stratum that does not survive correction, and a practical tie on homogeneous clutter.** Every difference happens to carry a positive sign, but two of the seven are too small to be claimed, and reporting the result as "ahead in all seven" would rest on a +0.0005 margin that means nothing. Among the 5 established differences the margin ranges from +0.0311 on multi-target scenes to +0.2672 at clutter edges, a factor of more than eight between the smallest and the largest.

That pattern is itself informative, and it is consistent with the failure modes of Section 2.3.5.

The largest margin is where CFAR is documented to fail. The clutter-edge condition — target in clean noise with the reference window reaching into the ridge — is where the classical threshold is set from the wrong background. The network gains +0.2672 there, roughly three times its next largest margin.

The two unestablished conditions are unestablished for different reasons.

Unresolved. The unresolved-pair condition is primarily a protocol and physics sanity check: it places two truths inside a single resolvable response, so the intended detection ceiling is near $P_d = 0.5$ for either arm. Its small TinyCNN–CFAR difference (+0.0080, $p = 0.017$) does not survive the Bonferroni correction, so no detector advantage is claimed.

Homogeneous clutter. In homogeneous clutter the two approaches are effectively tied at the primary operating point (+0.0005, $p = 0.963$).

5.4 The result in decibels

Per-condition probabilities are hard to compare against published work, which reports margins in decibels of signal-to-noise ratio. Figure 5.4 expresses the same frozen result in that currency, using the dedicated SNR-sweep split.

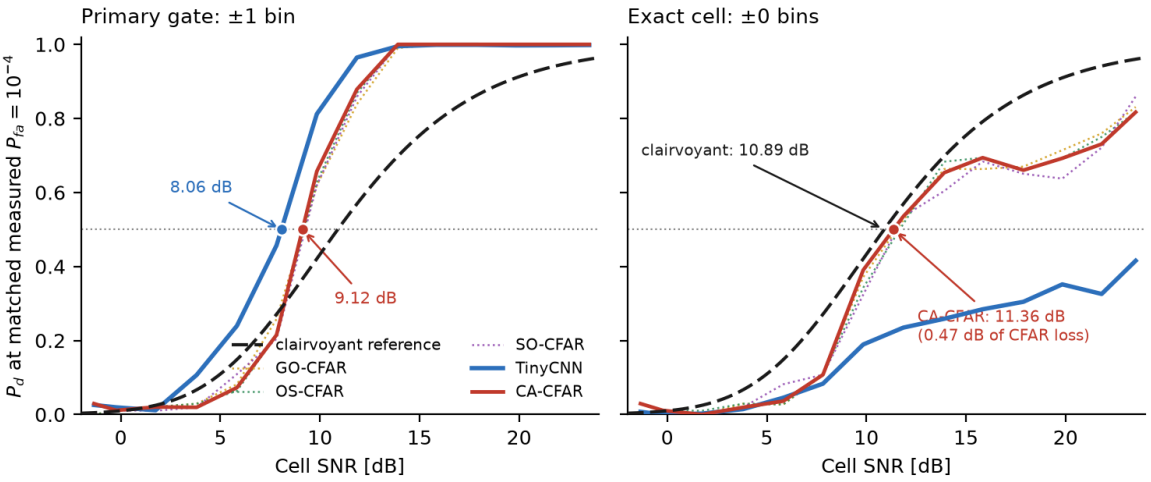


Figure 5.4: Both empirical detectors against the clairvoyant reference. Detection probability as a function of measured cell SNR at matched measured P_{fa}; TinyCNN and CA-CFAR are drawn solid, the secondary CFAR variants dotted, and the dashed curve is a clairvoyant detector that knows the true noise power. Left, the primary gate: TinyCNN reaches $P_d = 0.5$ at 8.06 dB against CA-CFAR's 9.12 dB, and both sit above the clairvoyant curve — the ±1-bin gate is more permissive than the exact-cell problem that curve describes, so it is not a bound there. Right, the exact cell, where the comparison is like for like: CA-CFAR crosses at 11.36 dB, 0.47 dB above the clairvoyant reference, while TinyCNN never reaches $P_d = 0.5$.

The same result can be expressed as an SNR margin. TinyCNN reaches $P_d = 0.5$ at 8.06 dB, compared with 9.12 dB for CA-CFAR, giving a 1.06 dB advantage at this detection level. The corresponding CA-CFAR exact-cell crossing lies 0.47 dB above the clairvoyant reference, indicating that CFAR loss explains only part of the observed margin.

The right-hand panel carries one qualification. Scored on the exact cell rather than within one bin, the network never reaches $P_d = 0.5$, peaking at 0.415, while every CFAR variant crosses it. **The network's advantage exists at the ±1-bin gate and reverses at the exact cell**, which points to localisation without isolating it. Section 5.6 measures placement directly, and Chapter 6 tests the mechanism.

5.5 Sensitivity to matching tolerance

Chapter 4 committed to reporting both tolerances for every condition. Figure 5.5 does so.

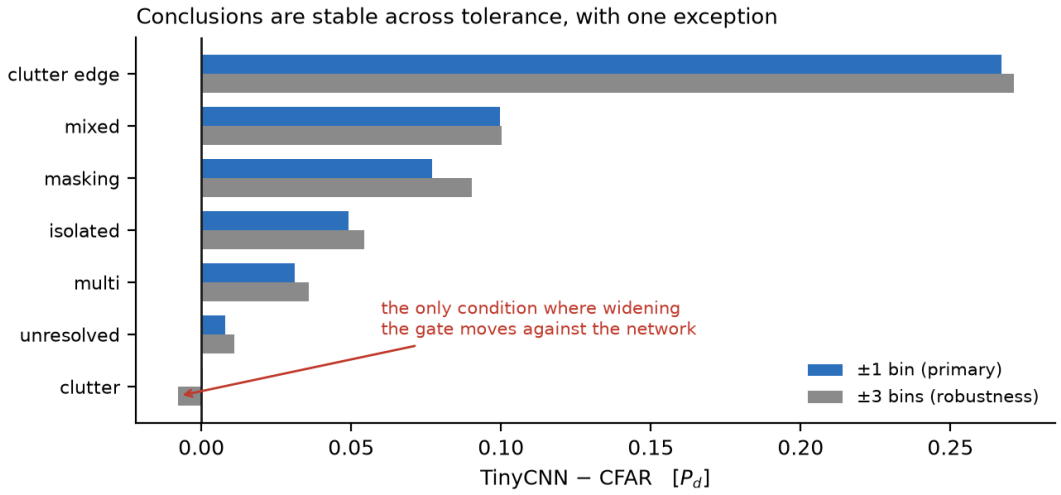


Figure 5.5: The paired difference at the primary ±1-bin gate and at the ±3-bin robustness gate, for every condition.

In six of the seven conditions the ordering is unchanged and the magnitude moves little; widening the gate slightly *increases* the network's margin on most of them, and the unresolved stratum, marginal at the primary gate, becomes significant at the wider one. The conclusion that the network leads on those conditions does not depend on the tolerance chosen, which is what the robustness gate was included to establish.

The exception is homogeneous clutter, where the difference moves from +0.0005 at the primary gate to -0.0078 at the wider one, changing sign. The bootstrap p-value there is 0.476, nowhere near the corrected significance threshold, so this report does not claim the classical arm wins in that condition either. It is a second indication that localisation contributes to the observed difference — not an independent one, since both indications come from the same detections on the same frames. Section 5.6 measures placement directly.

5.6 Localisation

Both previous sections pointed to the same place, so it is measured directly. Localisation error is computed over true positives only — it asks, of the targets a detector did find, how precisely it placed them.

That definition carries a caveat which should be read before the numbers. **Each detector's error is averaged over its own true positives, so the two averages are not taken over the same set of targets.** The network detects more targets, and the extra ones it finds are on average the harder, weaker targets that CFAR missed; those are exactly the targets whose peaks are least well localised. Part of the difference reported below is therefore a difference in which targets are being averaged, not only

in how well each arm places them. The metric is also bounded by the gate: a detection displaced further than the gate is scored as a miss rather than as a large error, so severe displacement appears as lost P_d instead. Both gates are reported for that reason, and Figure 5.6 gives the measured errors.

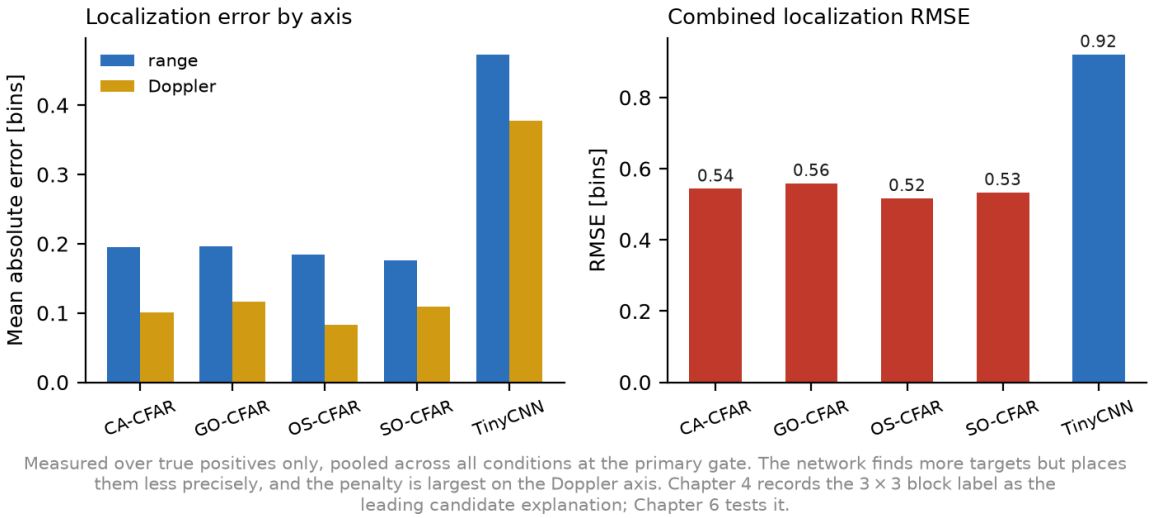


Figure 5.6: Mean absolute localisation error by axis, and combined RMSE, pooled over all conditions at the primary gate. Measured over true positives only.

Pooled over all conditions, the frozen network's combined RMSE is 0.921 bins against 0.544 for CA-CFAR — 1.69 times larger. Broken down by axis, the range error is 0.474 bins against 0.195, and the Doppler error is 0.378 bins against 0.101, a factor of 3.8. In physical units the network places detections to 0.092 m and 0.159 m/s against CA-CFAR's 0.038 m and 0.042 m/s.

All four CFAR variants localise similarly well, between 0.517 and 0.559 bins. That the four classical arms agree with one another and the learned arm sits outside their range indicates that this is a property of the learned arm as configured here, rather than of the evaluation.

This is a measured cost of the benchmark configuration, and it should be scoped carefully. It is a property of *this* frozen arm, trained with the 3×3 block label of Section 4.3.4 — not a general property of learned detection. A block label asks the network to respond over a region rather than at one cell, which is precisely the behaviour measured here. Chapter 6 varies the label directly and finds that most of this gap is attributable to that choice and is largely removable: supervision concentrated on the target cell reduces the error substantially at essentially no cost in detection. The figure above therefore records what the frozen benchmark does, not a limit on what a small convolutional detector can do.

5.7 Computational cost

Detection probability is not the only axis on which these detectors differ. Table 5.2 gives the measured cost per frame.

Table 5.2: Measured detector cost per range–Doppler frame. All five arms were timed back to back in one process at the same thread count. **The figures cover score-map computation only**: candidate extraction, thresholding and matching are shared by every arm through the same code and would cancel, so they are excluded — which also means these values do not depend on the suppression radius.

Detector	Time per frame	Note
CA-CFAR	0.86 ms	cheapest; a box sum per cell
TinyCNN	6.04 ms	5,041 parameters, 80.2 MMAC/frame
GO-CFAR	7.79 ms	two half-window sums, then a maximum
SO-CFAR	7.82 ms	the same, then a minimum
OS-CFAR	69.95 ms	a rank selection per cell

The ordering is worth noting because it cuts both ways, and because the intuition it violates is a common one. The network performs roughly three orders of magnitude more arithmetic per cell than CA-CFAR, and it is 7.0 times slower — but it is faster than three of the four classical variants, and 11.6 times faster than OS-CFAR. The reason is not that the network is cheap; it is that CFAR here is vectorised, so a box mean costs the same regardless of window size, whereas a per-cell rank selection does not vectorise at all. **A learned detector is therefore not automatically the expensive option**: it is expensive relative to the simplest classical rule and cheap relative to the more robust ones.

5.8 Conclusion

On the synthetic benchmark, measured at a verified common operating point:

- **The network establishes a gain on the five resolvable challenge conditions** — isolated, masking, multi-target, mixed and clutter edge — after correction for multiple comparisons, against a per-condition best-CFAR comparator stronger than any deployable single variant. The margin is largest where CFAR is documented to fail, reaching +0.2672 at clutter edges.
- **Homogeneous clutter is a practical tie** (+0.0005), and the difference moves against the network when the matching tolerance is widened, without approaching significance.

- **The unresolved stratum is a resolution and protocol stress test rather than an ordinary competitive condition.** Both arms sit near the ceiling its construction imposes, and the small difference does not survive correction.
- In the currency the comparable literature uses, the network reaches half detection at a cell SNR 1.06 dB lower than CA-CFAR.
- **This frozen arm places its detections less precisely than CFAR** (1.69 times the combined RMSE, 3.8 times on the Doppler axis), and at the exact-cell gate it does not reach half detection. That is a measured property of the 3×3 block label this arm was trained with, not of learned detection in general: Chapter 6 varies the label and removes most of the gap.

This establishes a controlled in-domain advantage, not real-world superiority. A learned local decision rule *can* improve on CFAR under controlled conditions — that is a statement about potential, measured on the simulator that produced the training data, and nothing in this chapter licenses more than it.

The question the benchmark leaves open is whether that improvement is a property of the detector or of the distribution it was trained on. The next chapters take one piece each: Chapter 6 asks where the margin comes from and how much of the placement behaviour is set by supervision; Chapter 7 freezes the detectors and asks whether any of it survives contact with real radar measurements; and Chapter 8 asks what would have to change for it to.

Chapter 6: Attribution Experiments: Explaining the Learned Advantage

6.1 Introduction

Chapter 5 established that the learned detector holds a measured advantage on the synthetic benchmark, and that the frozen arm places its detections less precisely than CFAR. Neither result says *why*. A margin reported as a single unexplained number cannot be transferred, defended or designed against: without a mechanism there is no way to predict whether the same advantage will appear under different conditions, which is exactly the question Chapter 7 goes on to ask.

This chapter therefore states the candidate explanations and pairs each with an experiment capable of disproving it. Three were run. Each was specified in full before any result existed, each is isolated from the frozen artefacts, and none of them touched the held-out split.

A note on scope that applies to the whole chapter. All three attribution experiments are reported under the final radius-2 post-processing protocol of Appendix A. They use the validation split only, and they do not alter the frozen benchmark of Chapter 5. Section 6.7 states what that does and does not permit.

6.2 Motivation

Three explanations for a learned advantage appear in the literature reviewed in Section 2.5, and they are not equivalent.

Energy integration. A single-cell CFAR test discards the target energy that spills into neighbouring cells. A network with a 21×9 receptive field can combine it. If this is the mechanism, giving CFAR the same integrated energy should close most of the gap.

Supervision. The 3×3 block label tells the network that nine cells are equally correct. This is the documented suspect behind the localisation deficit of Section 5.6, and it predicts a response broader than one cell.

Information access. CFAR must discard its guard cells to estimate a background; the network is under no such constraint and may use them. If that access is what matters, denying it should cost detection.

The first is the cheap explanation and is tested first. The second and third are tested by intervention rather than by argument.

6.3 Integrated CFAR: is the advantage mainlobe energy?

What was tested. CFAR's noise estimate was left exactly as it is — the estimators are imported from the project's own code and called unchanged — and only the cell under test was replaced by an integrated statistic over a 3×3 region. Two kernels were used: a uniform sum, and a kernel weighted by the measured point spread function, which is the matched filter for this response. The region was derived rather than assumed: a 1×1 cell captures 27.1% of a target's energy, 3×3 captures 94.2% and 5×5 captures 99.7%, so 3×3 takes almost all the available energy while still sitting strictly inside the guard region, where it cannot contaminate the reference ring.

The measured result, in Table 6.1. Integration does not close the gap. It makes the classical detector *worse*, on every condition and for both kernels.

Table 6.1: Effect of replacing CFAR's single-cell test statistic with an integrated one, at matched measured P_{fa} on the validation split. Negative values mean integration lost.

Kernel	Mean ΔP_d	Negative on	Significant
Uniform 3×3	-0.0376	7 of 7	4 of 7
PSF-weighted 3×3	-0.0237	7 of 7	4 of 7

The ordering is informative. Performance is monotone in how sharply the kernel concentrates on the centre: single-cell CFAR is best, the matched filter second, the uniform sum worst. Localisation follows the same order — averaged over the seven conditions, Doppler error rises to 1.63 times CA-CFAR's under the PSF kernel and 2.42 times under the uniform one.

What it supports. The response is concentrated in very few cells, so summing nine of them adds little signal and a great deal of noise; because the window functions correlate adjacent cells, the noise sum grows faster than the square root of the cell count

and the deflection falls. The learned advantage is therefore **not** access to mainlobe energy that CFAR discards. The cheap explanation is eliminated, which promotes the learned decision rule itself to the live one.

Caveat. This shows that *this* form of integration, with the noise estimate held fixed, does not reproduce the advantage. It does not show that the network gains nothing from its receptive field: a network can weight that field in ways a fixed 3×3 kernel cannot, including adaptively.

6.4 Label ablation: what causes the localisation deficit?

What was tested. Three label representations, eight seeds each, trained identically and otherwise matching the frozen configuration: a single-cell *point* label, the incumbent *flat* 3×3 block, and a *Gaussian* label over the same 3×3 footprint with weights taken from the measured point spread function. The positive-class loss weight was recomputed per arm so that total positive gradient mass is held constant, which is what makes the arms comparable.

The control is verified directly: retraining the flat arm reproduces the frozen checkpoints bit-for-bit at epoch 21 on all eight seeds. **The control is the frozen arm**, so any difference between arms is attributable to the label alone.

Table 6.2: Concentrating the supervision cuts localisation error sharply at negligible detection cost. Pooled over conditions on the validation split at epoch 21; P_d and Doppler error are at the primary gate, and the last column is exact-cell P_d.

Label	**Doppler error**	P_d **(±1)**	ΔP_d	P_d **(exact cell)**
Flat 3×3 (incumbent)	0.3946 bins	0.5949	—	0.1932
Gaussian 3×3	0.1719 bins	0.5982	+0.0033	0.3376
Point 1×1	0.1116 bins	0.5899	-0.0050	0.4018

The measured result, in Table 6.2. The incumbent label costs 3.54 times the Doppler error of a point label. The sharpest test was specified in advance as the growth of localisation error with training, and it lands as the hypothesis predicted: between epoch 5 and epoch 21 Doppler error grows by a factor of 1.97 under flat supervision, 1.26 under the Gaussian and 1.04 under the point label — monotone in how flat the supervision is. At the exact-cell gate the incumbent actually *degrades* with training while the other two hold, so flat supervision does not merely cap exact-cell accuracy; more training destroys it.

What it supports. Most of the frozen arm's localisation deficit is attributable to the supervision, and is largely removable. Concentrating the label reduces Doppler error from 0.3946 bins to 0.1719 (weighted) and 0.1116 (point), while the weighted label moves detection by +0.0033 — within seed noise. That converts a limitation of the benchmark configuration into a design choice with a measured alternative, and it retires two competing explanations: the deficit is not the network's capacity, and not the matching gate.

Caveat. The point label was predicted in advance to train poorly at its extreme class imbalance, which would have vindicated the 3×3 choice. It trained stably on all 8 seeds and localised best, so that prediction failed. The 3×3 label's only advantage is 0.0050 of P_d at the primary gate, and even that is not uniform: the point label is significantly worse on 3 of 7 conditions, all of them clutter-bearing, where sparser supervision gives less gradient signal — but significantly *better* on 2, the two resolution-limited conditions, where a response concentrated on one cell is what separates two targets that a broader one merges. The weighted label avoids the cost without incurring either, and is the better choice on this evidence.

6.5 Guard-masked patch network: is it information access?

What was tested. Whether the network's freedom to use its guard cells — the cells CFAR must discard — is what buys its detection advantage. Denying them turned out to be impossible on a dense fully-convolutional network, and the reason is structural: with a guard of ± 2, the cell one bin from a given cell under test lies inside that cell's guard region while being its own cell under test, and one input map cannot hold it simultaneously hidden and visible. The masked arm must therefore be a patch model centred on each cell under test. The primary contrast is consequently **masked patch against unmasked patch**, so that the patch-training differences are common-mode and cancel.

The measured result, in Table 6.3. The two gates disagree, and that disagreement is the finding.

Table 6.3: Effect of denying the network its guard cells, on the validation split. ΔP_d is unmasked minus masked, so a positive value means guard access helped.

Gate	Mean ΔP_d	Sign	Significant
± 1 bin (primary)	+0.0173	positive 6 of 7	3 of 7
Exact cell	-0.1597	negative 7 of 7	7 of 7

At the primary gate guard access contributes, but the effect is small and not uniform: removing those cells costs under two percentage points of P_d on average, the sign is positive on 6 of 7 conditions rather than all of them, and only 3 reach significance. At the exact-cell gate the sign reverses on every condition and the effect is far larger — on isolated targets the masked model reaches 0.511 against the unmasked model's 0.260, nearly double. Mainlobe energy in the guard cells makes the network's response *broad*: broad enough to match within one bin, not sharp enough to identify the peak. Deny those cells and the response sharpens.

What it supports. The guard cells carry information the learned detector genuinely uses for detection, but that information is not simply energy — E2 integrated the same energy into CFAR and it hurt. The network's use of those cells is also part of what makes it place targets imprecisely.

Caveat, and it is a substantial one. Both patch arms are undertrained relative to the dense frozen model, because uniform sampling of cells under test gives far fewer positive examples per epoch. This is common-mode, so the *difference* between the arms is unbiased, but the absolute P_d of both arms sits below the dense model's and should not be compared with Chapter 5.

6.6 Synthesis

Taken together the three experiments narrow the field without closing it.

- **Energy integration is ruled out** as the explanation for the detection margin. Given CFAR the same integrated energy, it performs worse, not better (E2).
- **The frozen arm's placement behaviour is largely set by its supervision, and is substantially fixable** (E3). Guard-region access broadens the response as well (E4), so two different design choices push in the same direction.
- **Guard-cell access buys detection, but little of it**, and it is paid for in placement precision (E4).

Figure 6.1 puts the three results side by side.

The explanation these leave standing for the *detection* margin is the learned decision rule itself — the network's ability to weight its receptive field in a way no fixed kernel reproduces — rather than any simple property of what it is given. That is a narrowing, not an identification, and this chapter does not claim to have isolated a single cause.

One consequence carries into Chapter 7. Two different interventions produced the same signature: a response broader than one cell, harmless at a tolerant gate and costly at a strict one. The real-data tolerance ladder shows the same shape, which is why

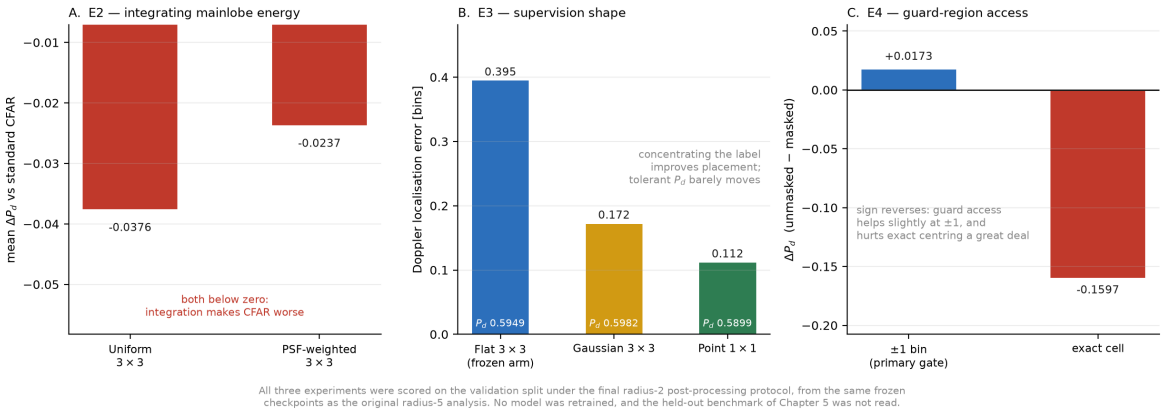


Figure 6.1: Summary of the three attribution experiments on the validation split under the final radius-2 post-processing protocol. **A**: giving CFAR the same integrated mainlobe energy moves P_d below zero for both kernels, so integration does not explain the learned advantage. **B**: concentrating the supervision reduces Doppler localisation error sharply, while detection probability at the primary gate (annotated in each bar) barely changes. **C**: denying the network its guard region helps slightly at the ±1-bin gate and costs a great deal at the exact cell — the sign reverses between the two gates, which is the finding.

Section 7.6 treats a broad response as *consistent with* that behaviour rather than as its established cause.

6.7 Conclusion

Three interventions were run, each fixed before its result existed, at zero cost to the frozen benchmark. The cheap explanation for the detection margin was eliminated; the frozen arm's placement behaviour was shown to be largely attributable to its supervision, and substantially reducible; and the value of guard-region access was measured.

Two limits bound everything in this chapter.

Validation split only. No experiment here touched the held-out data, so none of these results changes any number in Chapter 5. **This chapter explains that benchmark; it does not replace it.**

Interventions, not a general mechanism. Each experiment removes or substitutes one thing and reports what that costs. That is enough to eliminate a candidate explanation and enough to locate a design choice, but it does not add up to a complete account of why the learned detector behaves as it does, and no such account is claimed.

All three experiments are reported at the final suppression radius of 2. They were first analysed at the superseded radius 5 and re-scored from the same frozen checkpoints, with no model retrained; Appendix A.1 records what that involved.

Chapter 7 now asks whether any of this survives real radar measurements.

Chapter 7: Real-Data External Validation

7.1 The question, and how the transfer test was set up

Every result in Chapters 5 and 6 came from a single simulator. Training, validation and held-out test splits were generated by the same code from the same signal model, differing only in random seed and in the scenarios they sample. A held-out split of that kind shows the detector generalises to *new samples* of its own domain. It cannot show that the detector generalises to a *different* domain, and no quantity of additional synthetic data can close that gap.

This chapter therefore asks a single question:

> Do the already-frozen TinyCNN and classical CFAR detectors transfer zero-shot from the synthetic development domain to real automotive radar data?

Zero-shot here means that neither detector is retrained, fine-tuned or recalibrated against real data. The detector weights, the input preprocessing, the CFAR window geometry and the suppression radius all remain exactly as frozen in Chapter 4.

The operating point is the one thing that cannot carry over. Chapter 5 sets each detector's threshold by sweeping it per condition and comparing the arms where their *measured* false-alarm rates agree. That rule needs labels on the data being scored, and CARRADA provides no false-alarm axis to sweep against. For real data the threshold is therefore **fixed in advance: one value per frozen seed, derived from synthetic validation data alone**. No CARRADA data of any split enters that derivation. The two chapters use different operating-point protocols, and their headline numbers are never pooled — a distinction Section 7.2 makes precise.

What was fixed before the evaluation, and what was revised afterwards. This evaluation was run at the end of the project by the person who produced the

synthetic result, knowing what that result was. The complete protocol — ground truth, eligibility, matching, metrics, the statistical plan, and the wording permitted for unmatched detections — was therefore written down and frozen *before any real detector output existed*, and the real-data work ran in a directory that reads the frozen benchmark and writes nothing back to it. Implementation and debugging used the CARRADA *training* sequences only; the validation sequences were not used at all in this chapter.

One qualification belongs with that. The results below come from a **re-run**. The suppression radius was later reselected on synthetic development data, and the same test sequences were scored once more under the revised rule. No detector changed and no real measurement informed the revision, so the result remains zero-shot as defined above — but it is not the original untouched read. Appendix A states what changed and what did not.

Scope. This is an **external validation**, not a replacement for the synthetic benchmark: a poor real-data outcome bounds the external validity of the synthetic findings rather than retracting them. The two experiments also do not share a headline metric — the synthetic benchmark scores a point target within a ± 1-bin gate at a matched measured false-alarm rate, while the real dataset annotates extended regions and supports no false-alarm measurement at all. Numbers from the two domains are never pooled anywhere in this report.

7.2 CARRADA and the real-data evaluation protocol

7.2.1 The dataset

CARRADA [3] is a public dataset of synchronised camera and automotive radar recordings, collected on a test track with a 77 GHz FMCW sensor and released with annotations on both range–angle and range–Doppler representations. It was selected for three reasons: it publishes range–Doppler data directly, so no re-processing from raw ADC samples is required; its annotations are given in range–Doppler coordinates, so no projection step is needed; and its published grid geometry matches the geometry this project's simulator was configured to reproduce.

That last point requires care. The simulator was specified in Chapter 3 to reproduce CARRADA's *published range–Doppler grid* — map dimensions and bin spacings — so that a target occupies a comparable number of cells in both domains. It does not reproduce CARRADA's waveform, its scenes, its clutter statistics or its sensor

hardware. **Matching a grid is not matching a radar**: the geometric correspondence removes one confound, and Section 7.4 measures how far apart the domains remain.

The dataset holds 30 sequences and 12,666 range–Doppler frames, of which 7,193 contain at least one annotated object. Two counts must be kept apart, because they act as denominators for everything reported later. A single physical vehicle tracked across two hundred frames contributes *one* track but *two hundred* frame-level object instances: the dataset holds 8,750 frame-level object instances drawn from only 77 distinct tracks. Detection statistics are computed per frame-level instance, but 77 objects across 30 sequences is a small amount of genuinely independent content, which is why the statistical treatment takes the *sequence* — not the frame or the object — as its unit of analysis.

Objects belong to three classes — pedestrian, cyclist and car — and no frame contains more than two annotated objects. The dataset therefore offers no counterpart to the multi-target, mutual-masking and unresolved-pair conditions that Chapter 5 used to separate detector behaviour; that limitation is revisited in Section 7.7.

CARRADA publishes a sequence-level partition, reproduced in Table 7.1, and this work adopts it unchanged. Splitting instead by frame would be indefensible here: with only 77 tracks, consecutive frames of the same object are near-duplicates, and a random frame-level split would place the same physical vehicle on both sides of the partition. The reported test performance would then be closer to a memorisation check than to a generalisation measurement. Only the test split is scored in this chapter.

Table 7.1: Official CARRADA sequence-level splits, used unchanged.

Split	Sequences	Frames	Annotated	Object instances
Train	18	8,088	4,319	5,318
Validation	6	2,448	1,483	1,804
Test	6	2,130	1,391	1,628

7.2.2 Feeding the frozen detectors real data

Two checks were needed before either detector could be run, and both were passed without modifying any detector.

The representation transfers. CARRADA publishes more than one numerical representation of each range–Doppler frame. A cross-check over 240 frames confirmed that the published decibel array is consistent with a logarithmic representation of the radar amplitude/power data.[1] This is sufficient for both detectors: CFAR uses power

[1]Regressing the logarithm of one representation against another gives a slope of 2.0046 ± 0.0183 with a residual standard deviation of 0.330 dB and no dependence on signal level or map position — the signature of a $20\log_{10}$ transform.

ratios, so an unknown constant scale cancels, while TinyCNN subtracts the frame-median level during preprocessing, which removes the same constant. Absolute sensor calibration is therefore not required, and **no detector-specific preprocessing was redesigned for real data**: both arms run the code they were frozen with.

The grid transfers exactly. Every one of the 12,666 frames is a 256×64 array matching the simulator's map dimensions, and the annotation coordinates index that array directly. No resampling, interpolation or coordinate transformation is required anywhere in the pipeline.

7.2.3 Why the synthetic metric cannot be reused

One difference forces a new metric. A synthetic target occupies a single true cell, and Chapter 4 scores detections against it with a ± 1-bin gate. CARRADA annotates *extended object support*: a segmentation region covering tens of cells, whose median extent is 18×13 bins for a car, 11×11 for a cyclist and 9×9 for a pedestrian (Figure B.4 in Appendix B). There is no point target to match within ± 1 bin, and inventing one — by taking a centroid, a box centre or the brightest cell inside the mask — would mean scoring the detectors against a construct of this project rather than an annotation the dataset provides.

7.2.4 The protocol

Redesigning a metric with results in view is exactly where unconscious bias operates, so the protocol was written in full and frozen before any real detector output existed. Table 7.2 summarises what it fixes.

Table 7.2: The real-data evaluation protocol, fixed before any real detector output existed.

Element	Decision
Ground truth	The dense segmentation support. No point target is invented; centroid, box centre, brightest cell and sparse centroid are all explicitly rejected.
Eligible region	The frozen 21×9 detector-valid interior, unchanged from Chapter 4.
Eligible object	One whose support intersects that region in at least one cell.
Matching	One-to-one, maximum-cardinality, identical code for every detector arm.
Primary metric	Exact dense-mask object recall: the fraction of eligible objects for which an assigned peak lies inside the annotated support.
Tolerance analysis	The same recall with the mask dilated by ± 1 and ± 3 bins.
Localisation	**Not reported.** CARRADA supplies no point truth.
False alarms	**Not reported as** P_{fa}, in any form.
Burden metric	Unmatched detections per frame.
Statistics	Paired bootstrap over the 6 test sequences, 10,000 replicates. Seed variability reported separately and never merged with sequence variability.
Primary classical arm	CA-CFAR.

Two entries deserve explanation, because they are where this evaluation departs most sharply from Chapter 5.

Why no false-alarm rate is reported. A false-alarm measurement requires knowing that a detection corresponds to nothing real. CARRADA annotates only its 77 tracked road users; it does not annotate the buildings, parked vehicles, poles and roadside structure that a real scene contains, and bright unannotated returns occur in a measurable fraction of frames. A peak outside every annotated mask may therefore be infrastructure, static clutter, multipath, an unannotated object — or a genuine false alarm, with no way to distinguish them. Reporting such counts as P_{fa} would assert knowledge the dataset does not contain. They are reported instead as *unmatched detections per frame*, an output-density measure, and that term is used consistently throughout.

Why the operating point is a different protocol. Within each experiment in this thesis all detector arms share the same evaluation rule and operating-point definition. Across domains they cannot. Chapter 5 sweeps each detector's score threshold *per condition* and compares the arms where their measured false-alarm rates agree — an **evaluation operating point**, not a deployable setting, whose purpose is to remove threshold choice as a confound. This chapter has no false-alarm axis to sweep against, so a **single global threshold** was derived per frozen seed from *synthetic validation data alone*, pooled across all eight scenario conditions at a measured per-cell false-alarm rate of 10^{-4}, and applied unchanged to every real frame. That is the harder of the two requirements: one fixed threshold, chosen without sight of the data it will run on. The consequence, fixed in advance, is that the two chapters' headline numbers are **not poolable**.

7.3 Zero-shot transfer: the result

The official test sequences were scored under the protocol above — once in the original read, and once again in the disclosed re-run at the revised suppression radius, which is what is reported here. All eight frozen network seeds and all four CFAR variants were evaluated on the same 2,130 frames. Of the 1,628 object instances in the test split, 1,627 remain eligible once the detector-valid interior is applied; the difference is a single instance whose support lies entirely in the excluded border.

Table 7.3: Exact dense-mask object recall on the held-out CARRADA test split, over 1,627 eligible object instances.

Detector	Exact dense-mask object recall
Frozen CA-CFAR (primary classical arm)	**0.8838**
Frozen TinyCNN (mean over 8 seeds)	**0.8721**
Pooled difference, TinyCNN − CA-CFAR	-0.0118
Sequence-balanced paired difference	**-0.0136**
95 % confidence interval on that difference	[−0.0269, +0.0045] — includes zero

At exact annotated support the two arms are statistically indistinguishable. The sequence-balanced difference of -0.0136 carries a 95 % confidence interval of [−0.0269, +0.0045], which includes zero. The point estimate favours the classical detector and CA-CFAR is ahead in 5 of the 6 test sequences, but the margins are small and the network leads in the remaining one. The interval does not license a claim that either arm wins. **The synthetic ordering is not reproduced on real data, and it is not cleanly reversed either: it disappears.**

The raw pooled recall difference is -0.0118 when all eligible objects are counted together. For statistical inference, the sequence-balanced estimate is -0.0136, which gives each of the 6 test sequences equal weight so that a long sequence cannot dominate the result. Confidence intervals and conclusions therefore use the sequence-balanced estimate.

The second half of the result is on a different axis. **At their frozen operating points the network emits 1.9 times CA-CFAR's unmatched output** for statistically indistinguishable exact recall: 23.48 unmatched detections per frame against 12.28 (Table 7.4).

Table 7.4: Unmatched detections per frame beside exact recall. This is an output-density measure and **not** a false-alarm rate.

Detector	Unmatched detections / frame	Exact recall
CA-CFAR	12.28	0.8838
TinyCNN	23.48	0.8721
GO-CFAR	30.29	0.9219
SO-CFAR	30.36	0.9687
OS-CFAR	31.87	0.9662

The secondary CFAR variants are important to the interpretation. GO-CFAR reaches 0.9219 recall, SO-CFAR 0.9687 and OS-CFAR 0.9662, all above the frozen TinyCNN's 0.8721. Their higher recall comes with substantially higher unmatched output, around 30.29–31.87 detections per frame rather than TinyCNN's 23.48 and CA-CFAR's 12.28. Thus the zero-shot TinyCNN does not dominate the classical family: several CFAR variants achieve substantially higher raw recall, albeit at greater output burden. **On this dataset recall and output density move together for every detector tested**, which is precisely why recall alone cannot rank them. CA remains the primary comparator as fixed in advance; no variant was promoted after the fact. Figure 7.1 plots every arm, and Figure 7.2 gives the primary comparison together with its per-sequence breakdown.

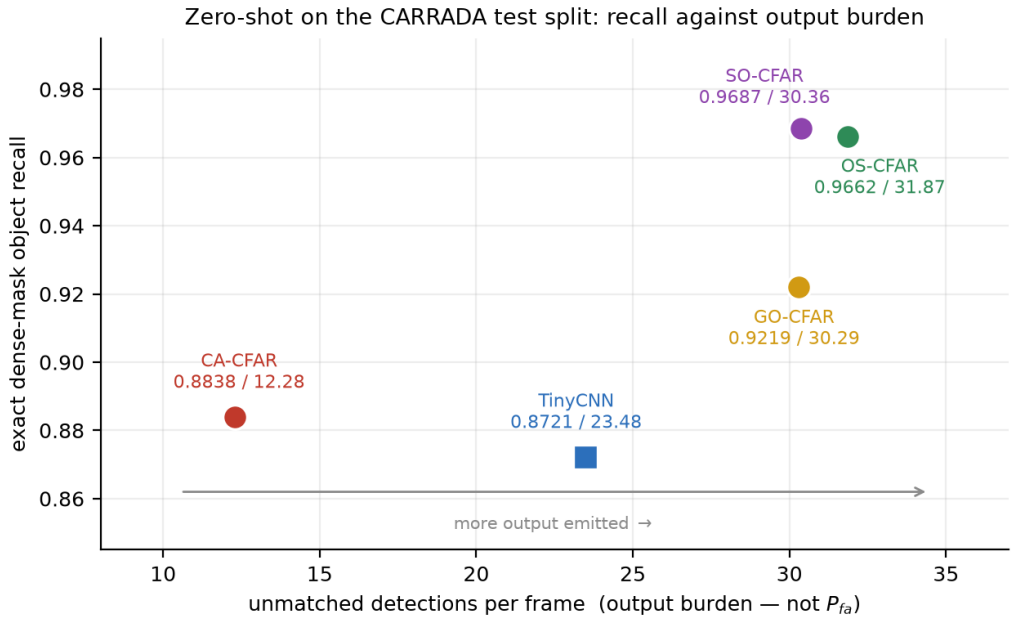


Figure 7.1: The robust CFAR variants achieve the highest raw recall, while CA-CFAR gives the lowest output burden. TinyCNN lies between these regimes and does not dominate the classical family under zero-shot transfer. Each point is one detector at its frozen operating point; labels give recall and unmatched detections per frame.

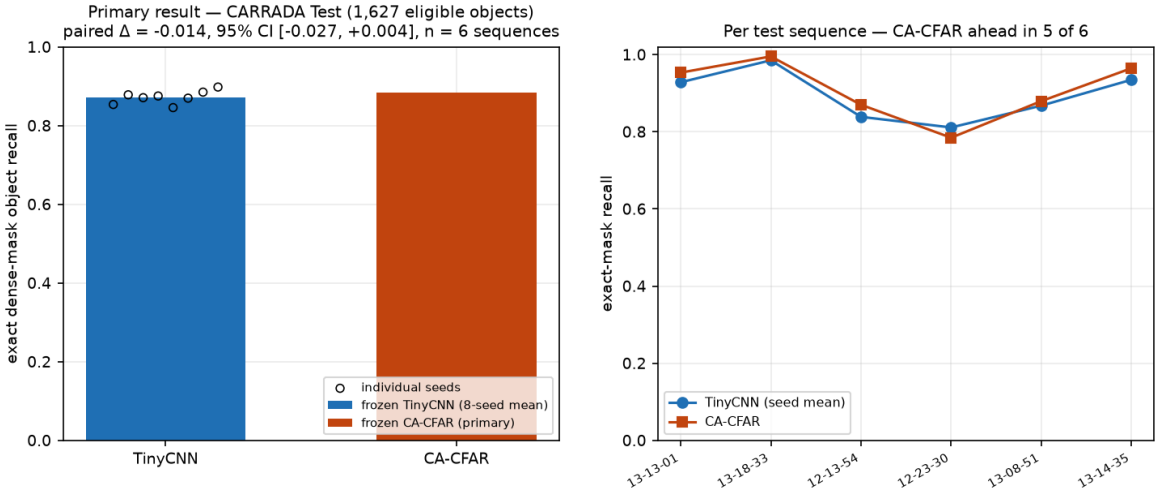


Figure 7.2: The synthetic ordering does not carry to real data. Left: exact dense-mask object recall, with the eight individual network seeds overlaid. Right: the same comparison per test sequence — CA-CFAR ahead in 5 of 6 sequences, the network in the remaining one, all six margins small.

Three cautions belong with the burden column, and they are the reason it must not be read as a false-alarm rate. The annotation covers only tracked road users. The suppression window is sized for point targets: a 5×5 window is far smaller than a median car's 18×13 support, so one real object can legitimately produce several surviving peaks — which occurs for 14.1% of objects for the network and 8.2% for CA-CFAR, of which one-to-one matching credits exactly one. And the extra output is not

generated by the annotated objects: the network averages 23.49 unmatched detections on annotated frames against 23.45 on annotation-empty ones, and CA-CFAR 12.41 against 12.04. Output density is a property of the scene, not of the annotation — which is exactly why CARRADA cannot price it.

7.4 Why the synthetic advantage disappeared

Before either detector was run on real data, the two domains were compared directly, so that whatever the detectors then did could be read against a measured baseline rather than an assumption. No detector was involved in any measurement in this section. Its purpose here is to explain the result above and to motivate the interventions of Chapter 8.

To compare background statistics fairly, the same background mask was used in both domains. Cells were excluded if they were close to an annotated target, inside the strong zero-Doppler ridge, or inside the border where the 21×9 detector window cannot operate ($\pm$3-bin target dilation and $\pm$4-bin zero-Doppler exclusion, leaving the same $236 \times 56 = 13{,}216$ cells the detectors themselves see). Statistics were then computed only from the remaining background cells, with no percentile trimming at any point. Uncertainty was measured across independent groups — CARRADA sequences and simulator conditions — rather than treating neighbouring frames or millions of correlated cells as independent samples.

Table 7.5: Background statistics, measured on the same eligible population in both domains with no trimming. Uncertainty is across groups, not frames.

Statistic	**CARRADA**	**Simulator**	**Ideal**
Coefficient of variation, $CV(P)$	0.473 ± 0.016	1.005 ± 0.003	1.000
Effective Gamma shape, k	4.771 ± 0.223	0.990 ± 0.006	1.00
Background std. deviation [dB]	2.211 ± 0.011	5.573 ± 0.005	5.57
Lag-1 correlation, range axis	0.563 ± 0.003	0.573 ± 0.001	—
Lag-1 correlation, Doppler axis	0.226 ± 0.006	0.574 ± 0.003	—

Four readings follow, and they are the four differences that Chapter 8 acts on.

The simulator behaves exactly as designed. Its background has $k = 0.990$ against a theoretical value of 1.0 for single-look exponential speckle, and a decibel spread of 5.573 dB against a theoretical 5.57 dB, uniformly across range. This is an independent validation of the clutter model of Chapter 3, obtained as a by-product of building the comparison.

CARRADA is markedly smoother. Its effective Gamma shape is $k = 4.771$ and its decibel spread 2.211 dB, less than half the simulator's. The wording is deliberate: the measurement licenses the statement that CARRADA background statistics are consistent with an *effective Gamma shape* near 4.771. It does not license a claim that the sensor performs some number of independent looks, which the dataset authors document nowhere.

Smoother does not mean cleaner. Lower cell-to-cell fluctuation is not less clutter. The same audit finds CARRADA carrying far stronger *localised* structure — a zero-Doppler ridge of 24.0 dB with a half-width of only ±1 bin, against the simulator's broader and weaker 14.8 dB spread over ±4 bins (Figure 7.3). The real domain is *different*, not simply easier or harder: a flatter field interrupted by strong narrow structure, rather than a uniformly grainy one.

CARRADA is not stationary in range. Measured over the same fixed blocks used throughout this chapter, its effective Gamma shape varies by roughly a factor of two and a half between the nearest and the middle blocks. The simulator's is flat across every block. A detector tuned on a stationary background therefore meets, in the real domain, a field whose statistics change with range.

One residual mismatch is recorded alongside these because Chapter 8 does *not* correct it: the lag-1 Doppler correlations differ sharply between the domains (0.226 against 0.574), while the range correlations are statistically indistinguishable.

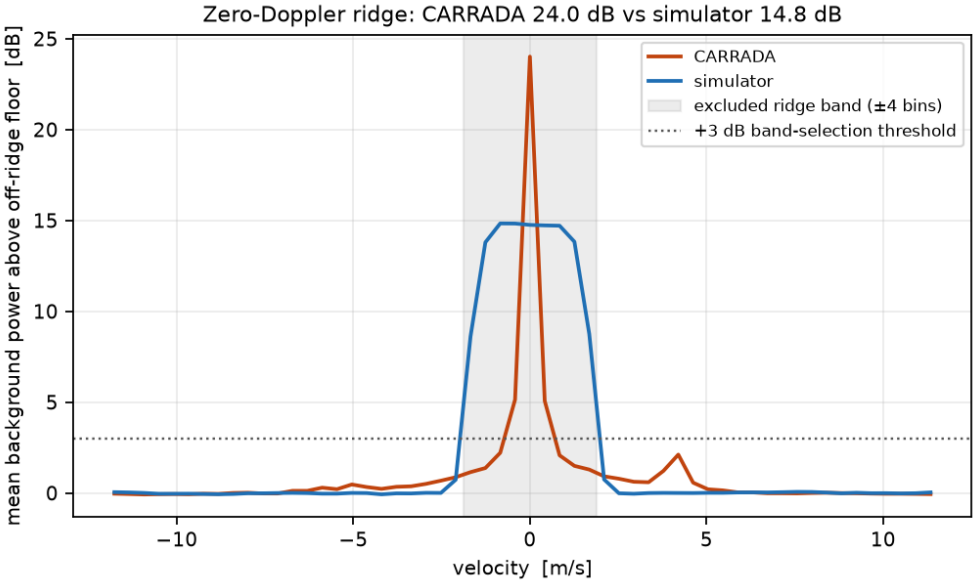


Figure 7.3: The largest single structural difference between the domains: CARRADA's zero-Doppler ridge is narrow and intense, the simulator's broad and weak. The shaded band is the region excluded from the background statistics of Table 7.5.

Figure 7.4 shows one frame from each domain on identical axes, to make the measured differences visible. It is included for intuition and is explicitly not evidence: it shows two hand-picked frames, whereas Table 7.5 measures every frame in both domains.

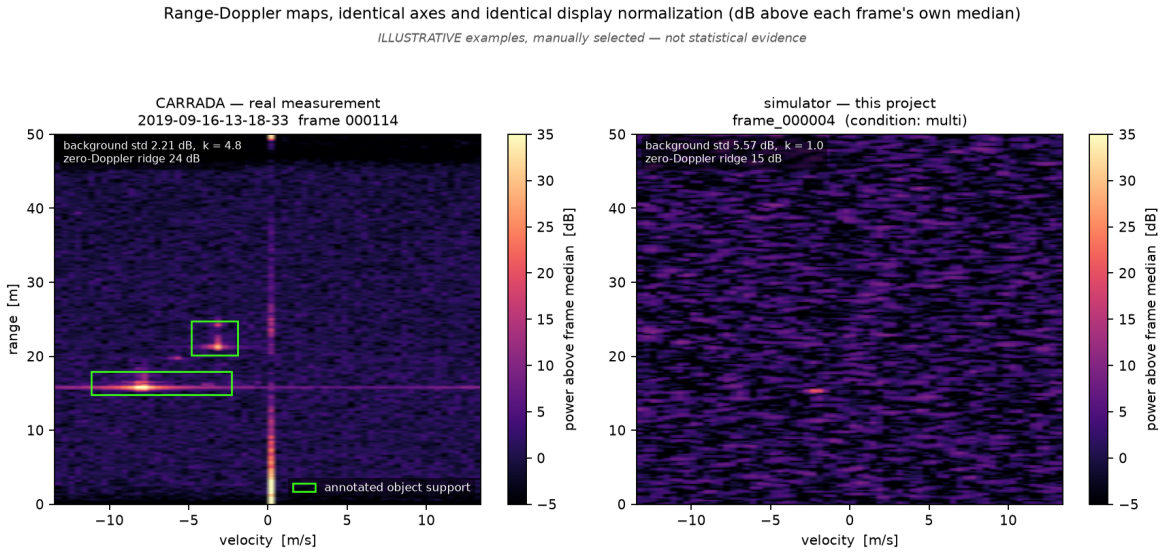


Figure 7.4: One range–Doppler frame from each domain, on identical physical axes and with identical display normalisation (decibels above each frame's own median). Left: CARRADA, with the annotated object support outlined. Right: this project's simulator. **Illustrative examples, manually selected — not statistical evidence.**

7.5 Secondary findings

7.5.1 Near range is where the two arms genuinely diverge

Stratifying by range block is the one stratification in which the arms separate clearly, and the pattern is not monotone. The network is behind in the nearest block — 0.8455 against CA-CFAR's 0.9475, a deficit of -0.1020 — and *ahead* in the three middle blocks, by as much as +0.0807 at 160–210 bins. The near-range deficit is the largest single stratified difference in the chapter, and it is consistent with the range non-stationarity recorded in Section 7.4: the simulator's background statistics are stationary in range, CARRADA's are not, and the near field is where they differ most. A single pooled number would have hidden both halves of this. Chapter 8 returns to the near-range block directly, because it is the pre-specified test of whether the domain intervention worked.

Performance near and away from the zero-Doppler ridge is similar between Tiny-CNN and CA-CFAR, so the transfer difference is not concentrated there. Class-wise recall is difficult to interpret because CARRADA classes have different annotation-

mask sizes, which directly changes the area in which a detection can count as correct. Detailed stratified results are therefore left to Appendix B.

> **A large structural difference between domains does not translate mechanically into a detector difference.** The audit's biggest finding — the ridge — shows no detector separation; the clearest separation appears where the audit measured range non-stationarity instead.

7.5.2 Tolerance around the annotated support

Since the truth is a region rather than a point, it is worth asking whether the misses fall just outside it. Dilating the mask by ±1 and ±3 bins, identically for both arms, gives the ladder in Table 7.6.

Table 7.6: Recall with the annotated support dilated. This measures tolerance around the annotation, *not* localisation accuracy.

Detector	**Exact**	**±1 bin**	**±3 bins**
TinyCNN	0.8721	0.9127	0.9521
CA-CFAR	0.8838	0.9047	0.9213
Difference	-0.0118	+0.0080	+0.0307

The network gains substantially more from dilation than the classical detector does, and the ordering changes as the tolerance widens. A larger share of the network's misses are therefore near misses lying just outside the annotated region. Two cautions apply. This is **not** a localisation measurement: with extended truth there is no point against which localisation could be measured, and dilating a mask asks whether a peak fell *near* the annotated support, which is a different question from how precisely it was placed. And a detector that emits more candidates has more chances to land one inside a dilated region; the network emits 1.9 times CA-CFAR's output, so part of this gain is bought rather than earned.

The same shape appears in the synthetic domain, where the frozen network gains from a wider gate and loses at the exact cell (Section 5.6). A broader detector response is consistent with both observations and may contribute to both, but it is not established as the cause of either, and separating response breadth from output density would need an intervention that was not performed. The breadth itself belongs to this frozen arm's 3×3 block label, which Chapter 6 shows is largely removable, not to learned detection as such.

7.5.3 Seed variability

The eight frozen seeds span 0.8457 to 0.8980, a standard deviation of 0.0166; 2 of the eight exceed CA-CFAR's 0.8838 and the rest fall below it, which is what a genuinely close comparison looks like at the seed level. No seed was selected, dropped or ensembled at any point. The spread is small in absolute terms — comparable to the paired difference being measured — so the pooled result is not an average over wildly different networks. It is not negligible either: eight networks with near-identical synthetic development performance still differ measurably under domain shift, and a single-seed result would have been reported as a win or a loss depending only on which seed happened to be trained. That is why every network value in this report is a mean over 8 seeds. This is a property of this detector and this training configuration, not evidence about neural networks in general. Figure 7.5 collects the tolerance ladder, the output burden and the seed spread.

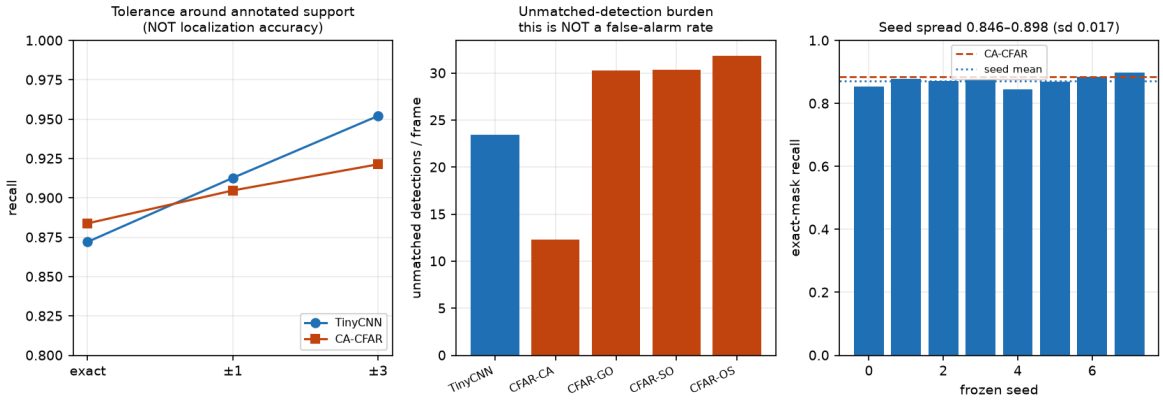


Figure 7.5: The three secondary findings in one view. Left: recall against mask dilation — the network gains more from a wider tolerance. Centre: unmatched detections per frame, the axis on which the arms clearly differ. Right: the eight frozen seeds against CA-CFAR's recall (dashed line).

7.5.4 Post-processing matters more on extended objects

A methodological finding emerged that the chapter did not set out to make. This comparison is sensitive not only to the two score maps but to the fixed rule converting a dense score map into object candidates. The synthetic benchmark never exercises that sensitivity: its targets are points, and any reasonable suppression window holds one maximum per target. Real annotated objects are *larger* than the suppression window, so the same rule operates *inside* a single object, and which local maximum survives is close to arbitrary. The ranking of two detectors on real extended-object data therefore depends on that post-processing choice as well as on the detectors. Appendix A records how this was found and what it moved.

7.6 Placing the result against published work

Section 2.5 separated the reviewed corpus into families and established that only one of them — per-cell learned thresholding scored against CFAR at a matched false-alarm rate — measures the same quantity as this work. Only that family is used for numerical comparison, and the purpose is calibration, not advocacy. Table 7.7 sets out both what each work measured and the training regime it measured it under.

Table 7.7: The comparable literature: reported margin over a classical baseline, and whether the evaluation domain was represented in training.

Work	Data and training	Reported margin	Test domain seen in training?
RadCNN [2]	Simulated pulse-Doppler, Weibull and K clutter; same simulator for test	+1 dB best case, −0.8 dB worst; P_{fa} matched by design	Yes
Lallouet et al. [13]	Simulated exoclutter; same simulator for test	All network variants ahead of CFAR	Yes
Gusland et al. [12]	Real X-band FMCW, trained and tested at different locations	+0.25 to +0.35 P_d at clutter transitions; parity below 450 m	Yes — same sensor
Banerjee et al. [11]	Simulation, plus on-air captures of a single parked reflector	Comparable P_d at lower P_{fa}; below 80 % under interference	Partly
This work	**Simulation only; CARRADA test split**	**+1.06 dB** synthetic, at measured P_{fa}; **-0.0136** zero-shot	**No — zero-shot**

Three statements are supported, and they answer the three questions this comparison can settle.

The synthetic margin is of a normal size. At 1.06 dB it sits alongside RadCNN's average of about +0.5 dB, and was measured under a stricter condition: RadCNN matches P_{fa} by design and reports its CFAR arms drifting to between 1.2 and 1.7×10^{-3} against a design value of 10^{-3}, whereas the arms here are matched at measured P_{fa}. The clutter-edge margin, +0.2672 in P_d, is similar in magnitude to the +0.25 to +0.35 band Gusland et al. report for clutter transitions on real X-band data with a different architecture, waveform and target class. The homogeneous-clutter tie is likewise consistent with RadCNN's noise-only scenario and Gusland's sub-450 m

regime.

This transfer regime is not directly represented in the reviewed corpus. No work in the comparable family evaluates a detector on real data drawn from a domain it never saw in training. The nearest is Banerjee et al., whose on-air data is a single parked reflector at high SNR — a controlled capture, not a moving street scene. Gusland et al. come closest in spirit, training and testing at different physical locations, but both sets are real measurements from the same sensor: the gap there is one of scene, not of data-generating process. **No directly comparable zero-shot synthetic-to-real experiment was identified in the reviewed corpus**, so the transfer result has no external benchmark against which to calibrate it, and a single measurement is not a norm.

The nearest supporting observation points the same way. Tosi et al. [21] train a CNN peak detector on a linearised signal model, evaluate it on data containing hardware nonlinearities, and find it comparable to CFAR within its training regime and degrading outside it — attributing that to the coverage of the training distribution rather than to the architecture. That is the same failure mode, observed independently on a different problem.

Two further points bound the reading. Every comparable work reports a condition in which its network loses; the synthetic suite of Chapter 5 contains no such condition, which is more plausibly a gap in the suite than a property of this network. RadCNN's losing scenario — target in noise with clutter intruding into the reference window — has no counterpart among the seven conditions evaluated here, and constructing one would be a direct extension of this work. And none of these comparisons establishes that the detector built here outperforms any published detector: the systems differ in architecture, data, waveform, metric and operating point, and the table compares *margins over a classical baseline*, not detectors.

One adjacent calibration is admissible for a single purpose. DAROD [15], a 3.4-million-parameter detector trained on the CARRADA training split, reports a recall of 46.03 % at an IoU threshold of 0.5 on the same test split used here. That number is **not** comparable with the 0.8721 of Table 7.3: DAROD's metric requires a predicted box to overlap the annotation by a fixed fraction *and* carry the correct class label, whereas the metric here asks only whether a detected peak falls inside the annotated support, and is by a wide margin the more permissive of the two. No ranking between the systems is claimed. What it does establish is that CARRADA is a demanding benchmark, on which a purpose-built supervised detector trained on the dataset itself still misses roughly half of the annotated objects under its own criterion — against which a simulator-trained network reaching recall comparable to a classical detector reads as a reasonable zero-shot outcome rather than a defective one.

7.7 Limitations

1. **Dataset scope.** CARRADA is one sensor on one test track, and the effective sample size for inference is 6 test sequences. The confidence interval is wide because six sequences is not many, and a second dataset could reinforce or overturn this result.

2. **Ground truth and metric mismatch.** The primary metric asks whether a peak lands inside an extended annotated region. It is not the synthetic point-target P_d, and the two are never pooled. Class sizes and mask extents are also unbalanced — the most numerous class has the smallest masks — so per-class recall cannot be read as class difficulty alone.

3. **No false-alarm measurement exists.** The comparison is one-sided: it measures what each detector finds, not what it invents. Unmatched detections are an output-density measure and not a false-alarm rate. This bounds the chapter's central trade-off rather than resolving it, since the network's larger output cannot be priced against its recall on this dataset.

4. **Post-processing and data imbalance.** The post-processing was designed for synthetic point targets, whereas a real CARRADA object can cover many range–Doppler cells. A single object can therefore produce several surviving peaks; after one is matched, the others contribute to the unmatched-output count. The burden metric is therefore partly sensitive to the chosen NMS radius. In addition, 84% of test objects overlap the zero-Doppler region, so the headline result mainly reflects performance in that part of the map.

5. **Transfer scope.** No real-data training was performed *in this chapter*, so its result is a zero-shot bound and not a ceiling. CFAR is also zero-shot here: its multipliers were calibrated for single-look statistics, so its operating point on real data is not the designed one, and that it transferred well is an observation rather than evidence its calibration was correct. Finally, this chapter performs no causal intervention, so none of the measured domain differences can be said, on its evidence alone, to have caused the result. Chapter 8 tests exactly what this bound leaves open, as follow-up work carried out after this evaluation.

7.8 Conclusion

This chapter asked whether the frozen detectors transfer zero-shot from the simulator to real automotive radar data, and answered it under a protocol fixed in advance of the answer.

Both detectors transfer in the sense that both work: neither collapses on real measurements, and both recall the large majority of annotated objects from a purely synthetic development process. But the ordering established in Chapter 5 does not carry over. Frozen CA-CFAR reaches 0.8838 exact dense-mask object recall against the frozen network's 0.8721, a sequence-balanced paired difference of -0.0136 whose confidence interval $[-0.0269, +0.0045]$ includes zero. **At exact annotated support the two arms are statistically indistinguishable; the synthetic advantage is neither retained nor reversed, but absent.** And comparable recall is bought with substantially more output: 23.48 unmatched detections per frame against CA-CFAR's 12.28. **The two arms are separable on output density even where they are not separable on recall.**

The classical family as a whole deserves its due here. **On real zero-shot transfer, the learned detector's synthetic advantage disappears. CA-CFAR provides the cleaner low-burden baseline, while SO- and OS-CFAR achieve the highest raw recall at the cost of denser output.** The learned arm occupies neither position: it emits roughly twice CA-CFAR's output without matching the recall of the robust variants.

Two qualifications complete that statement. The tolerance ladder favours the network — more of its misses are near misses — but that is tolerance around an annotated region rather than localisation accuracy, and a denser output has more chances to fall inside a dilated one. And the difference is not concentrated at zero Doppler, which the audit had identified as the largest structural difference between the domains; the clearest stratified separation is in the nearest range block, where the network trails by -0.1020.

None of this retracts the synthetic findings of Chapters 5 and 6. Those were controlled experiments and they remain valid within the domain that produced them. What this chapter establishes is the boundary of that validity: the learned detector's measured advantage was specific to its training domain and did not transfer, while a classical detector whose threshold adapts to local statistics by construction reached comparable recall from a far sparser output.

That shifts the question from detector capacity to domain alignment. The audit of Section 7.4 was run to decide whether this comparison was fair, but it also measured something else: the simulator and the real sensor differ in background smoothness, spread, ridge structure and range behaviour by margins far larger than any constant tuned anywhere in this project. A detector whose decision rule is fixed by weights fitted to one of those distributions was never guaranteed to carry to the other. That turns a negative result into a testable hypothesis — that the training distribution, not the architecture, is what failed to transfer — and it is the hypothesis Chapter 8 sets out to test.

Chapter 8: Domain Alignment and Real-Data Adaptation

8.1 Introduction

Chapter 7 ended on a negative result and a measurement. The negative result is that the synthetic margin did not survive transfer: frozen on real data, the learned detector and CA-CFAR became statistically indistinguishable in exact-support recall while the learned arm emitted roughly twice the output. The measurement is the domain audit, which found the simulator and the real sensor to differ in background smoothness, spread, ridge structure and range behaviour by margins far larger than any tuning constant in the project.

Those two things leave exactly two questions open, and this chapter answers one each.

Question 1 — the training distribution. Was the transfer loss caused partly by the training distribution? If the synthetic *training* domain is made statistically closer to the real one, does transfer improve — *without any real radar frame entering neural optimisation*? **Stage 4** addresses this.

Question 2 — real supervision. Can real supervision turn recovered recall into an efficient operating point? If real supervision is finally permitted, can the same $5,041$-parameter network exceed CA-CFAR on real sequences at a comparable output burden? **Stage 5** addresses this, and is the first and only point in the thesis where a real radar frame reaches a gradient.

The status of these two experiments, stated plainly. Neither is a fresh zero-shot external validation, and neither is presented as one. Chapter 7's evaluation was the single planned read of the CARRADA test split; everything here comes afterwards and is reported as follow-up work. The chronology is recorded in Appendix A.

8.2 Stage 4: domain-informed synthetic training

8.2.1 What is being isolated

The domain audit of Section 7.4 measured differences, not causes. Stage 4 turns those measurements into an intervention: the synthetic background is rebuilt so that its *measured* statistics move toward the real ones, and everything else is held fixed. If transfer improves, the training distribution mattered. If it does not, the loss of the margin lies elsewhere.

The generator's parameters were derived from the **18 official CARRADA training sequences and nothing else**. The validation split was not used, and the test split was not opened by any code that influenced the design. Three background properties were changed, all of them set from train-derived measurements:

- **Multi-look-like smoothness.** The original chain produces a single-look background: its measured spread is 5.569 dB against the closed-form single-look value of 5.570 dB. The real background is consistent with an effective Gamma shape near seven to eight, so frames are formed as a power average of several independent background realisations sharing one deterministic target field.
- **Range-dependent mean background.** The measured train range profile, unit-mean in power, is applied to the background alone — never to the targets, whose amplitudes come from the frozen radar equation.
- **A narrow, persistent zero-Doppler ridge.** Strong, about one bin wide at half power, and present in essentially every frame rather than in the minority of frames the original strata produced.

The target model, the point spread function, the range and Doppler windows, the labels, the target placement, the SNR sampling and the dataset sizes are untouched. Only the background changed.

8.2.2 What could not be matched, and why

This is a **partial domain-gap reduction**, not a reproduction of CARRADA. Two mismatches survive deliberately, and both trace to one frozen constraint. The lag-1 Doppler correlation of the real background is far lower than the simulator's. That correlation is set by the slow-time window, and the same window defines the target point spread function, the measured mainlobe width and hence the guard geometry that makes the CFAR comparison fair. Changing it would change the detector contract established in Chapter 4, so it was left alone and declared. The same constraint widens the synthetic ridge's skirt relative to the real one, whose fall-off is sharper than any

window of this family permits. Both residuals were recorded before any model was trained.

8.2.3 Result on the CARRADA test split

Eight seeds were trained on the revised synthetic domain under the frozen training contract — same architecture, optimiser, learning rate, batch size, epoch count, positive weighting and flat 3×3 label — and thresholds were derived from the new synthetic validation split alone.

Table 8.1: Stage 4 on the CARRADA test split: exact dense-mask object recall over 1,627 eligible objects, and output burden.

Arm	Exact recall	Unmatched det./frame
Original synthetic TinyCNN	0.8721	23.48
CARRADA-informed TinyCNN	**0.9449**	**109.28**
CA-CFAR	0.8838	12.28
Informed − original: +0.0629, 95 % CI [+0.0353, +0.0879]; positive on 6 of 6 sequences		

Recall rose by +0.0629, with an interval clear of zero and the same sign on every test sequence. The pre-specified range question was answered too: in the nearest block, where Chapter 7 measured the largest deficit, recall moved from 0.8455 to 0.9468 against CA-CFAR's 0.9475 — the deficit closed.

8.2.4 Recall–output trade-off

> **The recall gain was bought with output density.** Unmatched detections rose from 23.48 to 109.28 per frame, a factor of 4.7, against CA-CFAR's 12.28.

A detector that fires more often recovers more objects, so a burden-matched diagnostic was run to ask whether anything survived once output was equalised. Two things did, and they point in opposite directions. The informed arm beats the original arm at *equal* output, so the gain is not merely density. But CA-CFAR leads *both* learned arms at every matched burden level.

Supported conclusion. Reducing the measured synthetic-to-real background mismatch substantially changed transfer behaviour and recovered much of the learned detector's real-data recall, without any real frame entering neural optimisation. It did **not** establish superiority over CA-CFAR once detector output burden is controlled.

8.2.5 What this stage can and cannot attribute

Several background properties changed together. The experiment supports a causal statement about the *combined* intervention — changing the synthetic training domain toward the measured target statistics changed transfer performance — and nothing narrower. It does not establish that the smoothness, the ridge or the range profile was individually responsible. Separating them would need one ablation per property, which the General Conclusion lists as future work and this thesis does not claim to have run.

8.3 Stage 5: supervised real-domain adaptation

Stage 4 recovered recall and left an efficiency problem. Stage 5 asks the question the whole thesis has been circling: with real supervision finally permitted, can the same small network exceed the classical baseline at a comparable output burden?

8.3.1 Protocol

Fine-tuning starts from each of the eight Stage-4 checkpoints, preserving seed identity, and uses **CARRADA training sequences only**. The 18 training sequences were divided at **sequence level** into 15 for fitting and 3 for internal development; frames from one recording never straddle the two sides, because consecutive frames of a sequence are near-duplicates and a frame-level split would leak one into the other almost perfectly.

The supervision changes here, and that is the point. The synthetic arms were trained on flat 3×3 blocks around point targets. Fine-tuning uses the **provided dense segmentation masks**, restricted to the detector-valid interior, with annotation-empty frames kept as genuine all-background examples. Nothing is invented: no centroids, no brightest cells, no bounding-box centres. This is therefore real-domain *adaptation*, not a continuation of the original point-target task.

All layers train, including the normalisation statistics — a $5,041$-parameter network has no sub-network worth freezing. The architecture is untouched. The configuration is deliberately small: 1×10^{-4} learning rate, ten epochs at most, early stopping on internal-development loss, positive weight 100 fixed by a rule declared before the class balance was measured.

The operating point was fixed on training data, before the evaluation split was opened, and it is part of what Stage 5 changes. CARRADA cannot support a false-alarm measurement, so the threshold cannot be set by a measured P_{fa} as it is on synthetic data. Instead, CA-CFAR was run on the three internal-development sequences and its output burden measured (9.00 unmatched detections per frame); each seed's threshold is the one maximising exact recall on those sequences subject to

not exceeding that burden. This is a **burden-constrained operating-point rule** using training data alone. It is stated explicitly because the frozen operating points of Stage 4 and Stage 5 were chosen by different rules, and Section 8.3.3 depends on that distinction.

8.3.2 Result on previously unscored validation sequences

The official CARRADA validation split had never been scored by any detector, nor used for any model or threshold decision. It was read once, after the configuration above was frozen, giving Table 8.2.

Table 8.2: Stage 5 on the CARRADA validation split: 2,448 frames, 1,804 eligible objects. All four arms were scored together on the same objects.

Arm	Exact recall	Unmatched det./frame
Original synthetic TinyCNN	0.9033	22.50
CARRADA-informed TinyCNN	0.9577	109.59
Real-fine-tuned TinyCNN	**0.9611**	**11.17**
CA-CFAR	0.8725	11.64
Fine-tuned − CA-CFAR: +0.0554, 95 % CI [+0.0024, +0.1210]; ahead on 4 of 6 sequences		

The fine-tuned detector recalls more objects than CA-CFAR *and* emits slightly fewer unmatched detections. This is the one place in the thesis where the learned detector is ahead on both axes at once, and it is the only place where it has been given real supervision. The advantage holds across the tolerance ladder — 0.9611 against 0.8725 at exact support, 0.9806 against 0.8997 at ±1, 0.9860 against 0.9285 at ±3.

The interval only narrowly excludes zero. Its lower bound is +0.0024 on 6 sequences: the fine-tuned arm is ahead on 4 of 6 and marginally behind on the other two, and the aggregate leans on two sequences where the classical detector does poorly. Substantial sequence-level uncertainty remains, and a different six sequences could give a different interval.

8.3.3 Fine-tuning improves selectivity rather than recall

Stage 5 is not a further recall improvement, and reading it as one would misstate what changed.

> **Against the Stage-4 arm, fine-tuning established no recall gain:** +0.0022, 95 % CI [-0.0090, +0.0138], an interval that includes zero.

What changed is the *shape* of the recall–output trade-off, and the evidence for that is the burden curve of Figure 8.2 rather than the frozen operating points alone. At CA-CFAR's own 11.64 unmatched detections per frame, the fine-tuned arm reads 0.9613 against the CARRADA-informed synthetic arm's 0.8569, the original synthetic arm's 0.6552 and CA-CFAR's 0.8725. That comparison holds output constant, so it isolates the trade-off from the threshold choice: **real-domain adaptation substantially improved the learned detector's recall–output trade-off.**

The change at the *frozen* operating points — 109.59 to 11.17 unmatched detections per frame at equal recall — is larger than the weight update alone accounts for, because Stage 4 and Stage 5 thresholds were chosen by different rules. The burden-constrained operating-point selection of Section 8.3 is what converted the improved trade-off into a deployment-relevant operating point; the sweep in Figure 8.2 is what shows the trade-off itself improved. Both steps are part of Stage 5, and neither on its own produces the result.

Two further observations, both pre-specified. The near-range block continues to improve, from 0.8929 through 0.9656 to 0.9745 against CA-CFAR's 0.9268. And pedestrians, the most numerous and hardest class, move from 0.8541 through 0.9385 to 0.9496 against CA-CFAR's 0.8007. Class recall is confounded by mask size — larger objects offer more support cells to hit — so these are reported as strata, not as class-wise capability claims.

Seed spread also collapses along the sequence: 0.0262 for the original arm, 0.0082 after domain-informed training, 0.0025 after fine-tuning. Real supervision made the eight independently seeded networks behave far more alike.

8.3.4 Limitations of the fine-tuning result

The comparison gives the learned detector something CA-CFAR does not get. The fine-tuned network is supervised with the same dense segmentation support that the primary metric scores against. That is legitimate supervised adaptation — it is exactly what the experiment set out to test — but it aligns the network's output geometry with the truth geometry, and the classical detector has no equivalent opportunity. The defensible statement is therefore *under supervised target-domain adaptation, the learned detector outperformed CA-CFAR on this evaluation*, not that it is inherently the better detector.

The fine-tune is deliberately small, and was still improving when it stopped. Every seed reached the ten-epoch ceiling with its development loss still falling; early stopping never triggered. The result is a lower bound on what this configuration can do, not a converged one.

Six sequences bound the inference. The statistical caveat recorded with the

primary result applies to every comparison in this section.

8.4 Cross-stage synthesis

Table 8.3 places the three training domains and the classical baseline side by side on the CARRADA *validation* split, where all four arms were scored together on the same 1,804 objects and are therefore directly comparable. The Stage-4 result on the test split is Table 8.1; the two splits are different sequences and are never compared across.

Table 8.3: The training-domain progression, all four arms scored together on the CARRADA validation split.

Training regime	Exact recall	Unmatched det./frame
Original synthetic	0.9033	22.50
Target-domain-informed synthetic	0.9577	109.59
+ real fine-tuning	**0.9611**	**11.17**
CA-CFAR (untrained)	0.8725	11.64

Read down the columns, the progression is: a strong in-domain synthetic result that does not transfer; a domain-informed intervention that recovers recall at a heavy output cost; and a real adaptation step that keeps the recall and gives the output back. Figure 8.1 shows the same three domains graphically.

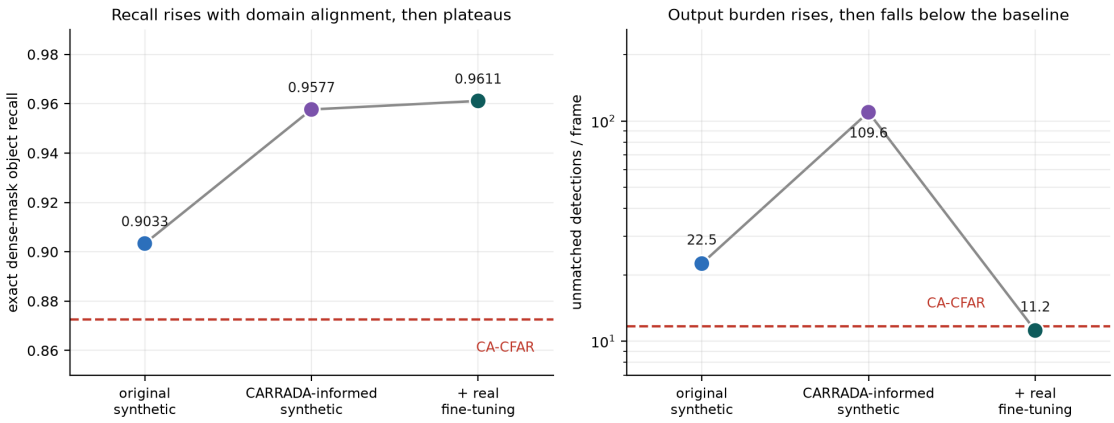


Figure 8.1: Recall alone tells less than half of this story. Left: exact dense-mask recall rises with domain alignment and then plateaus — real supervision adds no established recall over target-informed synthetic training. Right: output burden, on a logarithmic axis, rises sharply under target-informed synthetic training and falls below the classical baseline under real fine-tuning. All four arms are scored on the CARRADA validation split; the dashed line is CA-CFAR.

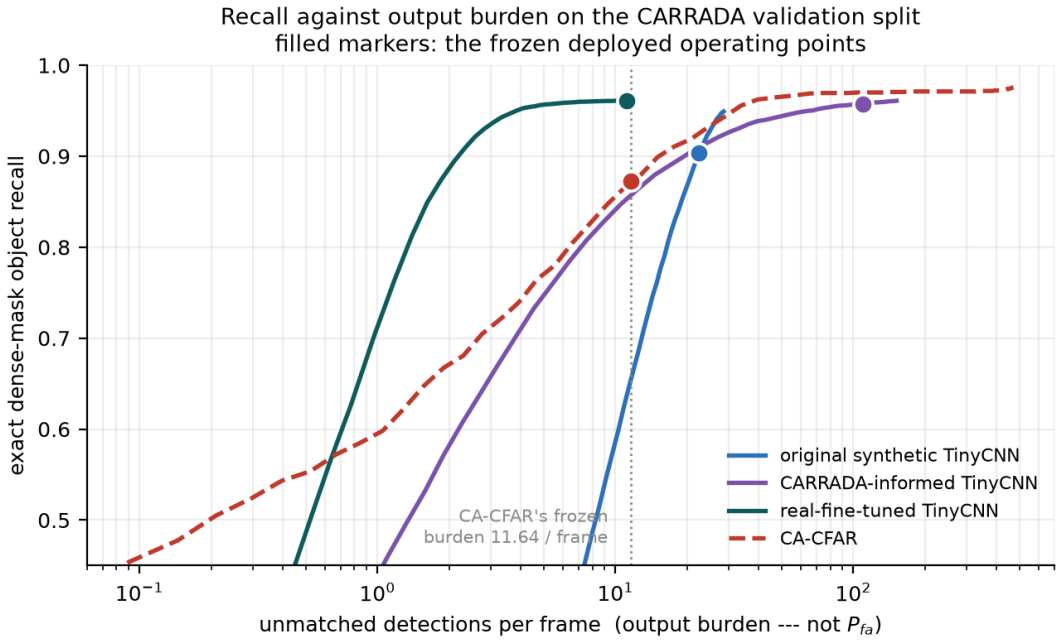


Figure 8.2: The evidence that adaptation improved the trade-off rather than only the threshold: at any fixed output burden the fine-tuned detector recalls more than the other learned arms and than CA-CFAR. From an evaluation-only threshold sweep that selected nothing; filled markers are the frozen deployed operating points. **Unmatched detections are an output-burden measure, not a false-alarm rate.**

8.5 Conclusion

Two follow-up experiments, each answering the question the previous stage left open.

Target-domain-informed synthetic training showed that the synthetic training distribution was a material part of the transfer failure: recalibrating the background toward measured target-domain statistics recovered a large part of the lost real-data recall *without any real frame entering neural optimisation.* It also showed that recall is not a sufficient summary, because the recovery came with roughly 4.7 times the output density and did not survive burden matching against the classical baseline.

Supervised real-domain adaptation then produced the strongest learned-versus-classical result in this thesis: higher exact recall than CA-CFAR at a slightly lower output burden, on sequences no detector in this project had previously scored. Its contribution was not recall — that had already been recovered — but the recall–output trade-off, which improved enough for a burden-constrained operating point chosen on training data to land above the classical baseline on both axes at once.

Taken together these two stages convert Chapter 7's negative result into a conditional one. The learned detector's advantage was never a fixed property of the architecture. It was a property of how well its training distribution matched deployment, and it became an advantage over the classical baseline only once the deployment domain was represented during training — first statistically, then with real supervision.

General Conclusion

This project asked when a learned local radar detector actually provides an advantage over classical CFAR, and what is required for that advantage to survive the move from controlled simulation to real measurements. It answered that question in five stages, each of which changed the answer the previous one would have given.

To do so it built: a simulator producing labelled FMCW range–Doppler maps with a calibrated cell-SNR definition; four CFAR detectors on a measured window geometry; a 5,041-parameter convolutional detector whose receptive field equals the CFAR window by construction; and one shared evaluation path — the same candidate extraction, the same one-to-one matching, the same statistics. *Within each experiment, all detector arms share the same evaluation rule and operating-point definition.* The synthetic and real-data experiments necessarily use different operating-point protocols, because real data supplies no false-alarm axis to match on, and they are therefore compared only within domain.

Answering the research questions

RQ1 — what a learned local rule adds, in-domain. On a fresh held-out synthetic test set, at matched *measured* false-alarm rate, the learned detector establishes gains in five challenge conditions, ties in homogeneous clutter, and shows no significant difference on the resolution-limited stress case after correction. It reaches $P_d = 0.5$ at a cell SNR 1.06 dB below CA-CFAR. This establishes potential under controlled conditions; it establishes nothing about deployment.

RQ2 — where the advantage comes from. Attribution by intervention rules out the cheap explanation: giving CFAR the same integrated mainlobe energy makes it worse, not better, on every condition. Guard-region access contributes a small tolerant-detection benefit and simultaneously broadens the response, hurting exact centring. Most usefully, the frozen arm's localisation deficit is largely attributable to its flat 3×3 supervision rather than to learned detection as such — concentrating the label removes most of it at negligible detection cost. Together these narrow the field to an adaptive learned decision rule without uniquely identifying its mechanism, which the experiments were not designed to do.

RQ3 — whether it survives transfer. It does not. Evaluated zero-shot on the CARRADA test split, with no retraining, fine-tuning or real-data recalibration, the synthetic-trained network and CA-CFAR are **statistically indistinguishable** in exact dense-mask recall (0.8721 against 0.8838; sequence-balanced difference -0.0136, 95 % CI [-0.0269, +0.0045]), while the network emits 1.9 times more unmatched detections per frame. The classical detector does not reverse the result; the learned advantage simply disappears, and the learned arm pays for parity with output density.

RQ4 — whether the training domain was the limiting factor. Substantially, yes. Rebuilding the synthetic training background from CARRADA *training*-split statistics alone — with no real frame entering neural optimisation or threshold calibration — raised exact test recall from 0.8721 to 0.9449 (+0.0629, 95 % CI [+0.0353, +0.0879]), positive on every test sequence, and closed the near-range deficit. But it did so at 4.7 times the output density, and burden-matched analysis showed CA-CFAR still ahead once output was equalised. Domain alignment recovered recall; it did not by itself establish superiority.

RQ5 — what real supervision adds. Fine-tuned on CARRADA training sequences, with the operating point fixed on held-out training sequences under a burden constraint before the evaluation split was opened, the same network reached exact recall 0.9611 against CA-CFAR's 0.8725 on previously unscored validation sequences — a difference of +0.0554 (95 % CI [+0.0024, +0.1210]) at a slightly *lower* unmatched-detection burden (11.17 against 11.64). Against the domain-informed synthetic arm, fine-tuning established *no* recall gain (+0.0022, CI [-0.0090, +0.0138]); what it changed was the recall–output trade-off. At CA-CFAR's own burden of 11.64 detections per frame the fine-tuned arm reads 0.9613 against the synthetic-only arm's 0.8569 and CA-CFAR's 0.8725.

What this means for CFAR versus deep learning

This study does not support replacing CFAR universally with deep learning. CA-CFAR provided a strong training-free baseline and transferred to a new real domain without adaptation of any kind. TinyCNN demonstrated a higher observed performance ceiling, but that ceiling depended on the training distribution. Controlled synthetic gains were lost under zero-shot real transfer, substantially recovered through target-domain-informed simulation, and converted into a recall–efficiency advantage only after supervised real-domain adaptation. **The practical choice is therefore conditional: when representative training data are unavailable, CFAR remains a robust default; when the deployment domain can be represented during training, a learned local detector can exceed it.**

Two qualifications belong with that statement.

Recall alone would have been misleading at every stage. Read on recall only, the five stages say: the learned detector wins, then ties, then wins large, then wins. Read with output burden beside it, they say something more useful — that the learned detector's standing depends on how much output it is allowed to emit, and that the one stage where it leads on both axes at once is the stage where it received real supervision.

The final comparison gives the learned detector something CA-CFAR does not get. The fine-tuned network is supervised with the same dense segmentation support the primary metric scores against. That is legitimate supervised adaptation — it is what the experiment set out to test — but it aligns the network's output geometry with the truth geometry, and the classical detector has no equivalent opportunity. The defensible statement is that *under supervised target-domain adaptation, the learned detector outperformed CA-CFAR on this evaluation*, not that it is inherently the better detector. The interval also only narrowly excludes zero: its lower bound is +0.0024 on 6 sequences, with the fine-tuned arm ahead on 4 of 6, so substantial sequence-level uncertainty remains.

Limitations

- **One simulator, one real dataset, one architecture.** Nothing here supports a claim about learned detectors in general.
- **No false-alarm measurement on real data.** CARRADA annotates the objects that were labelled, not every scatterer present, so the learned arm's denser output is an output-burden measure and can never be priced against recall as a false-alarm rate.
- **The domain intervention changed several properties jointly.** It supports a causal statement about the combined intervention and about nothing narrower; no individual background property has been isolated. It is also a *partial* domain-gap reduction: the Doppler correlation structure was deliberately left unmatched, because correcting it would have required changing the range–Doppler processing window that also defines the target point spread function and the guard geometry.
- **The attribution experiments run on validation data.** They explain the frozen benchmark; they do not extend it, and no arm they identify as better was itself frozen and scored on held-out data.
- **The fine-tune was deliberately small and had not converged.** Every seed reached the ten-epoch ceiling with its development loss still falling, so the final result is a lower bound on that configuration rather than its best achievable

behaviour.

- **The post-processing protocol was revised during the work**, on synthetic development data, and the real data re-scored under it. Appendix A records what changed.

Future work

The results point to specific next experiments rather than general ambitions.

- **Separate what the domain intervention bundled**, with one ablation per property — multi-look-like smoothness, ridge modelling, range non-stationarity, correlation structure — to convert a combined causal statement into individual ones.
- **Measure how much real data is actually needed.** The fine-tuning result used all fifteen fitting sequences; a sample-efficiency curve over increasing fractions of real labelled data would answer the question a practitioner actually faces.
- **Prefer domain randomisation to calibration against one dataset.** Randomising background statistics over a plausible range would test whether the recovered transfer generalises or is specific to CARRADA, and a second real dataset would test the same thing directly.
- **Make output burden part of the objective.** Burden mattered as much as recall at every stage, yet it was only ever measured, never optimised. A burden-aware loss or calibration criterion follows naturally.
- **Adapt with weaker supervision**, since dense mask supervision is expensive and it is open whether weak, self-supervised or unsupervised adaptation recovers any of the same benefit.
- **Carry the improved supervision design through a complete benchmark.** The label ablation showed concentrated supervision substantially improves localisation at negligible detection cost, but that arm was never frozen, never scored on held-out data and never taken to real measurements.

Larger architectures are deliberately *not* the first recommendation. The behaviour that dominated every stage of this project — domain sensitivity, operating-point dependence, the recall-versus-burden trade — was visible at 5,041 parameters, and is better understood there than hidden inside a larger model.

Final statement

CFAR is the stronger training-free detector: cheap, interpretable, and transferable to a real sensor it had never seen with no change at all. A small learned detector demonstrated the higher observed performance ceiling in this study, but reached it only when the deployment domain was represented during training. The practical question is therefore not whether deep learning beats CFAR, but whether enough domain knowledge and representative training data are available to justify that ceiling. Where they are, it is worth the effort. Where they are not, CFAR is still the right answer.

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Appendix A

Protocol Revisions and Sensitivity Checks

Every substantive choice in this project — ground truth, eligibility, matching, metrics, the statistical plan, the threshold rule and the primary classical comparator — was written down and frozen before the corresponding evaluation data were scored. Two things changed afterwards, and both affect how the final evidence should be read. They are recorded here in full. Everything else that supports them — freeze records with their hashes, the file manifests that protect completed experiments from later modification, the commit chain, environment fingerprints and the complete corrections log kept during development — is retained in the accompanying project repository, along with the raw detector outputs from which every table and figure in this report is derived.

A.1 Revision of the post-processing protocol

What happened

The frozen configuration fixed the non-maximum-suppression radius at 5 (11×11 cells) before the synthetic benchmark was scored, on the basis of a development observation. When that justification was checked against a controlled sweep it did not hold. The sweep was therefore run properly, on development data only, under a selection rule written and frozen *before* any measurement was taken, and it selected radius 2: the smallest radius tested that produced no duplicate in-gate detections. **Every result in this report is at radius 2.**

What was re-measured, and on what data

- **The radius was selected on synthetic development data alone** — the tuning split for selection, the validation split for confirmation. The final synthetic holdout was not used, and **no real data of any kind entered the choice**.
- **The synthetic benchmark was re-scored on a freshly generated holdout** of 3,200 frames, produced from a new seed after the radius was fixed, and verified to share no frame content with any previously generated split. The superseded test split was not re-used, so the revised benchmark is a genuine one-shot read rather than a second look at data whose answer was already known. This is why Chapters 3 and 5 describe a new holdout rather than a re-scored one.
- **The real-data evaluation was re-run** under the revised protocol, on the same official CARRADA test sequences, with thresholds re-derived from synthetic validation data alone.
- **The three attribution experiments were re-scored** at radius 2 from their original checkpoints. No model was retrained, fine-tuned, re-initialised or re-selected; because a wider window's local maxima are a strict subset of a narrower one's, a smaller radius cannot be recovered from a stored candidate list, so inference was regenerated and the cached CFAR and TinyCNN baselines were rebuilt at the same radius. Split, seeds, epoch, operating-point definition, matching rule, tolerance gates, bootstrap seed and Bonferroni correction are unchanged. No CARRADA data of any split was opened, and the held-out synthetic split was not read.

What did not change, and what the reader should conclude

The detectors are untouched. The same eight checkpoints at epoch 21 were loaded with the same weights. The CFAR window geometry, the threshold multipliers, the input transform, the label definition, the matching rule, the operating-point definition, the tolerance ladder and the statistical treatment are all as specified in Chapter 4. **One post-processing constant changed, and nothing else.**

The real-data evaluation of Chapter 7 therefore remains zero-shot in the sense that matters: no real measurement was used for training, fine-tuning, threshold calibration or radius selection. It is **not**, however, the original untouched run. It is a re-run of the frozen protocol with one post-processing field revised, performed after the original result was known. The original radius-5 outputs remain on disk beside the revised ones, together with the frozen selection rule.

Why the change matters more on real data than on synthetic data

The synthetic conclusions are almost unaffected by the radius, and the reason is geometric. A simulated target is a point scatterer whose response spans about two bins, so any reasonable suppression window contains exactly one maximum per target and shrinking it changes little. A CARRADA road user is an extended region whose annotated support has a median extent of roughly 18×13 bins — *larger than the suppression window itself.* There the window operates inside a single object, and which local maximum survives is close to arbitrary. That is why the real-data comparison moved and the synthetic one did not, and Chapter 7 reports it as a finding in its own right rather than as an inconvenience.

A.2 A defect in the threshold-derivation code

What happened

`predict_points` declares its suppression radius as a default argument, and a Python default binds once, when the function is defined — not when it is called. The module-level constant it binds to is the original radius. The routine that collects a whole split calls `predict_points` without passing a radius, so it always extracted candidates at the *original* radius regardless of what a caller set afterwards. The revised-radius copy sets the constant on a different module, which reaches the per-frame helpers but not that collector.

Consequence. The zero-shot thresholds published as radius-2 were in fact derived at radius 5, which is why they are byte-identical to the radius-5 set. Candidate extraction during the evaluation itself was unaffected, because that path passes the radius explicitly. The reported evaluation therefore ran at radius 2 with thresholds derived at radius 5. Because the smaller radius keeps a superset of candidates, the rule-correct threshold would have been slightly higher, and the learned arm operated slightly above its intended output level.

Sensitivity analysis

The size of the effect was measured rather than argued. The domain intervention of Section 8.2 derived its thresholds *both* ways and ran the comparison twice. The primary difference was $+0.0629$ under the original code path and $+0.0625$ under the corrected one — the same conclusion, the same sign, the same significance.

Conclusion: the defect is real and the interpretation is materially unchanged. It is recorded because the published thresholds were not derived under the

rule they were described by, not because it changes any result in this report.

A.3 Chronology of the follow-up experiments

Chapters 5 to 7 report the work as originally planned: the synthetic benchmark, the attribution experiments, and one evaluation of the CARRADA test split under a protocol frozen beforehand. The two experiments of Chapter 8 came afterwards, in response to that result, and are reported as follow-up work rather than folded back into it.

- **Stage 4** (Section 8.2) calibrated a revised synthetic background from the 18 official CARRADA *training* sequences and evaluated the resulting networks on the CARRADA test split — a split whose original zero-shot result was already known, which is why Stage 4 is described as a domain intervention and not as a fresh external validation. No real frame entered neural optimisation or threshold calibration.
- **Stage 5** (Section 8.3) fine-tuned on CARRADA training sequences, fixed its operating point on three held-out training sequences, and then read the official CARRADA *validation* split once. Those sequences had never been scored by any detector in this project, which is why they are described as *previously unscored* rather than unseen: the domain audit of Section 7.4 read their frames to measure background statistics, with no detector involved and no label used.
- **The CARRADA test split was not opened during Stage 5**, and the validation split was not opened during Stages 1–4.

Appendix B

Supplementary Tables and Figures

These tables and figures give the breakdowns the results chapters summarise — among them Figure B.3 on window-geometry selection, Figure B.5 on annotated object support by class, and Figure B.6 on zero-shot recall by stratum. The tables are generated directly from the same artefacts the chapters read, by the script `make_appendix_tables.py`, so no value here is transcribed by hand and none can drift from the main text.

Nothing already shown clearly in the main body is repeated. What is added is the detail those chapters compress: every CFAR variant rather than the per-condition best, both matching gates rather than the primary one, per-condition localisation rather than the pooled figure, the per-sequence and per-seed real-data results, and the full stratum breakdown.

All synthetic values are from the held-out split at the final suppression radius of 2; all real-data values are from the revised evaluation described in Appendix A.

B.1 Supporting figures

CARRADA also publishes a bounding box per object, and it is tempting to treat it as the extent of the dense mask. It is not: the box is strictly smaller than the dense support on both axes in every one of the 8,750 instances, so the two describe different annotation constructs. Quoting one while meaning the other would understate object extent by roughly a quarter.

B.2 Supporting tables

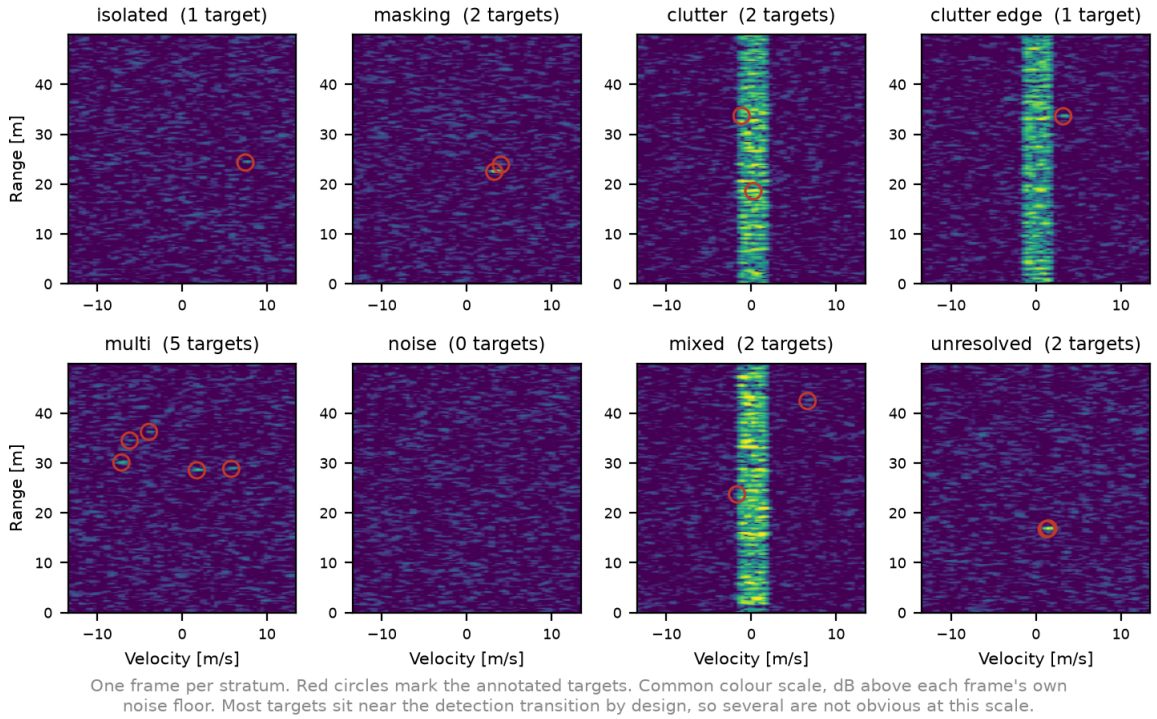


Figure B.1: The frames of Figure 3.3 at full map scale, with red circles marking the annotated targets. Several targets are not obvious by eye, which is by design: the benchmark places them near the detection transition.

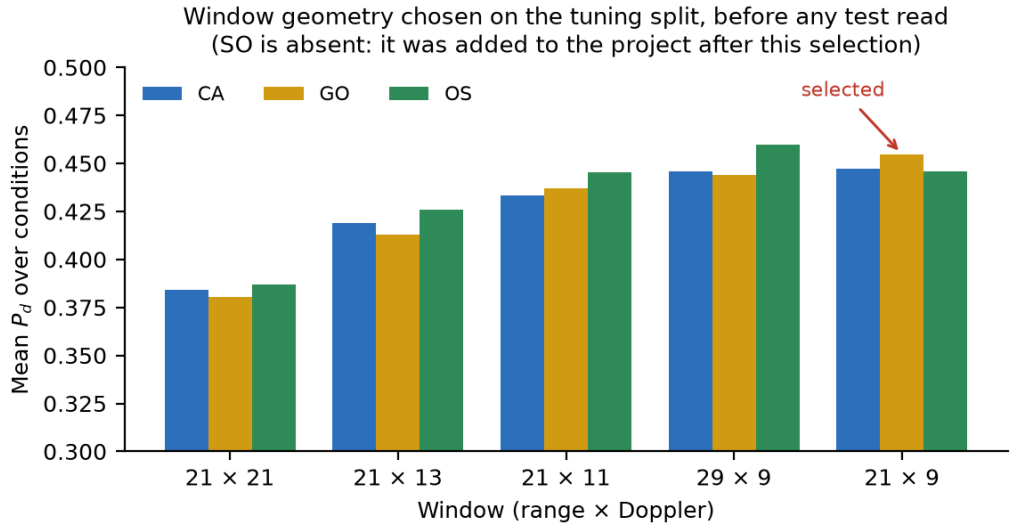
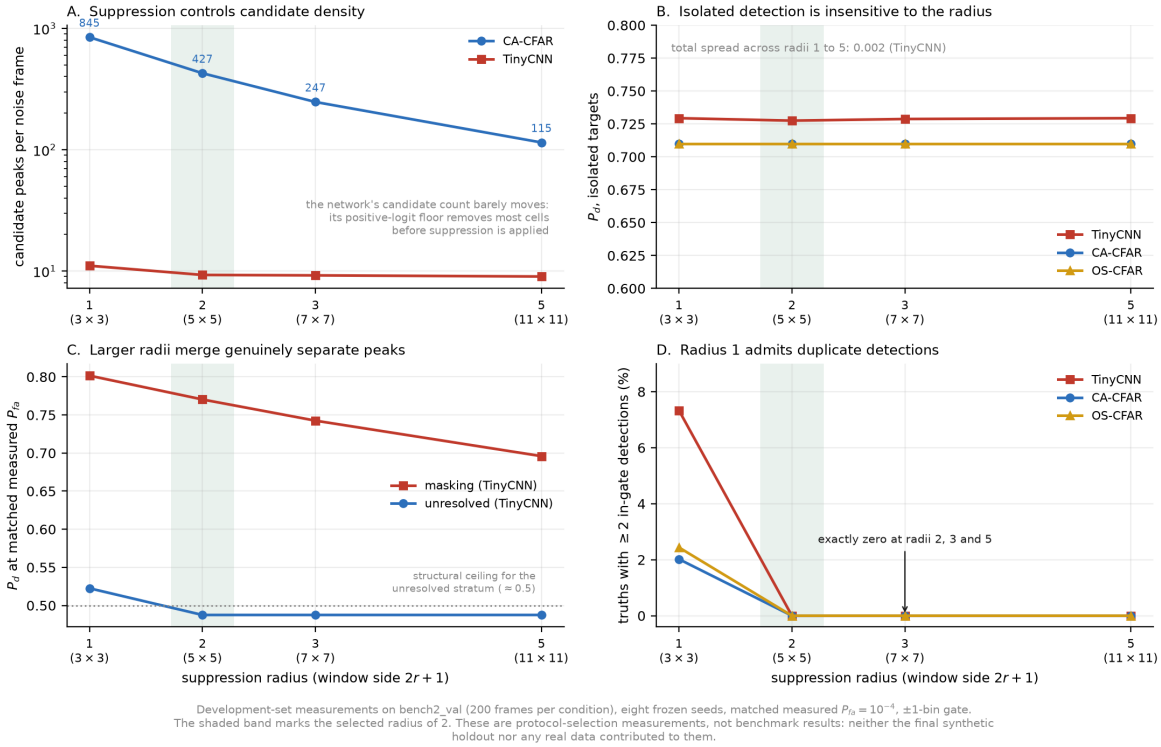


Figure B.2: Mean detection probability across conditions for each candidate CFAR window, measured on `bench2_tune` and fixed before any read of the test split. Only CA, GO and OS appear, for the reason given in Section 4.1.



Development-set measurements on bench2_val (200 frames per condition), eight frozen seeds, matched measured $P_{fa} = 10^{-4}$, ±1-bin gate.
The shaded band marks the selected radius of 2. These are protocol-selection measurements, not benchmark results: neither the final synthetic holdout nor any real data contributed to them.

Figure B.3: The full suppression-radius sweep, on **development data only**. **A**: candidate density on pure-noise frames, the quantity the radius directly controls — going from $r = 5$ to $r = 1$ raises CA-CFAR's count from 115 to 845 per frame. **B**: detection probability on isolated targets, essentially flat across radii. **C**: the masking and unresolved strata, where a larger window merges peaks belonging to different targets; the unresolved stratum reads 0.5225 at $r = 1$, lifted by the duplicate mechanism of panel D. **D**: the fraction of truths receiving two or more detections inside their gate, non-zero only at radius 1. Shading marks the selected radius. **These are protocol-selection measurements, not benchmark results**: no held-out synthetic data and no real data of any kind entered this choice.

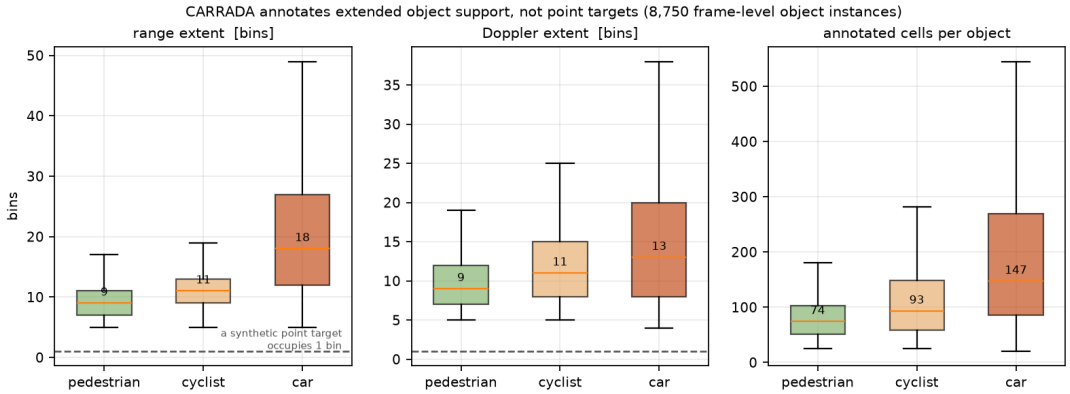


Figure B.4: Extent of CARRADA's annotated dense support by class, over all 8,750 frame-level object instances. A synthetic point target occupies one bin; this difference is why the synthetic point-target metric cannot transfer (Section 7.2).



Figure B.5: Zero-shot recall by class, zero-Doppler proximity and range block on the CARRADA test split, with counts annotated. The range blocks are discussed in Section 7.5; class recall is confounded by mask size.

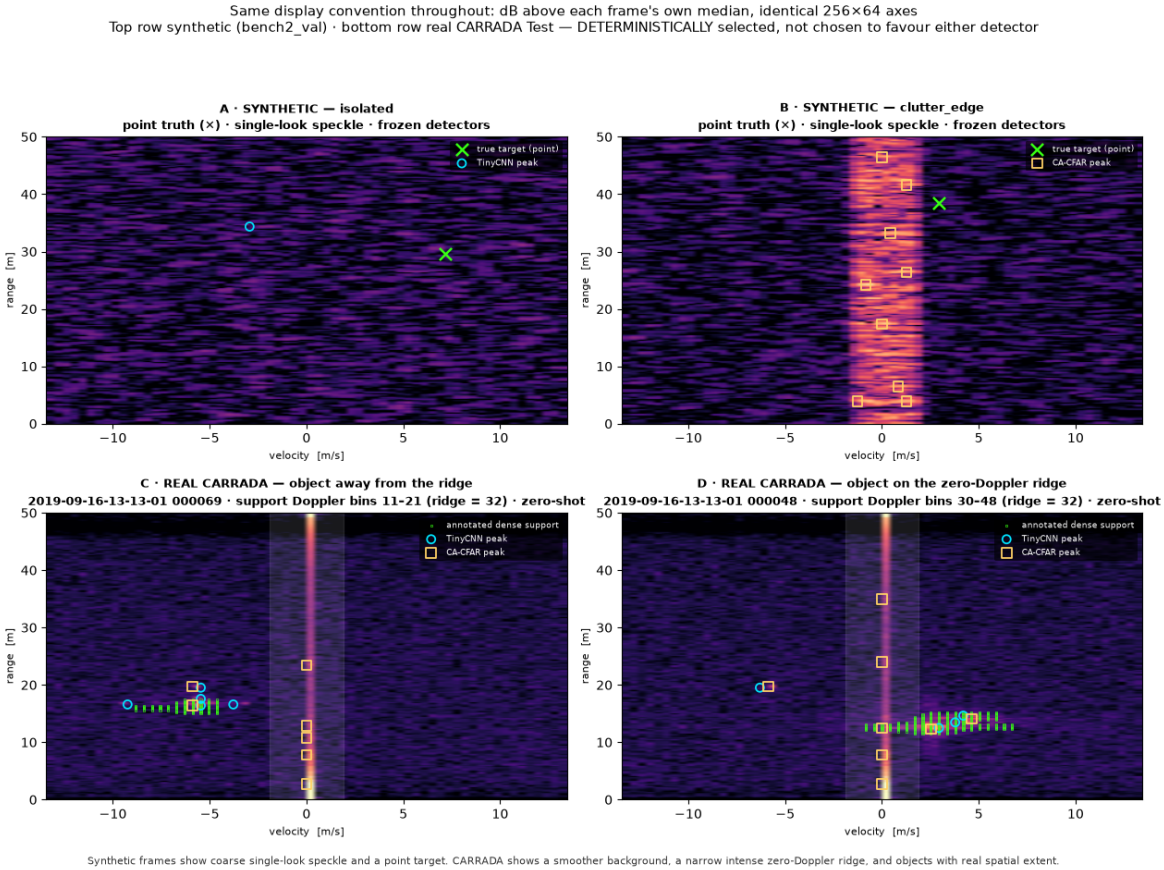


Figure B.6: Synthetic frames (top) and real CARRADA frames (bottom) under one display convention, with the annotated truth and both detectors' peaks. Frames were selected deterministically on annotation geometry alone, never on which detector performed better. Illustrative, not evidence.

Table B.1: Detection probability for every arm on every scored condition, at both matching gates, on the held-out split at matched measured P_{fa}. The main text reports only the best CFAR variant per condition; this table gives all four. TinyCNN values are means over 8 seeds.

Condition	Gate	CA-CFAR	GO-CFAR	OS-CFAR	SO-CFAR	TinyCNN
isolated	1	0.6550	0.6500	0.6500	0.6425	0.7041
	3	0.6600	0.6525	0.6525	0.6450	0.7144
masking	1	0.5350	0.5350	0.6987	0.6875	0.7758
	3	0.5363	0.5375	0.7000	0.6900	0.7903
clutter	1	0.1605	0.1631	0.1529	0.0599	0.1635
	3	0.1847	0.1834	0.1745	0.0739	0.1769
clutter edge	1	0.3676	0.3824	0.4232	0.1765	0.6904
	3	0.3676	0.3824	0.4232	0.1797	0.6942
multi-target	1	0.6979	0.6895	0.6918	0.6878	0.7290
	3	0.6973	0.6895	0.6923	0.6884	0.7331
mixed	1	0.3994	0.4178	0.3841	0.2319	0.5175
	3	0.4065	0.4208	0.3912	0.2370	0.5211
unresolved	1	0.4725	0.4700	0.4725	0.4725	0.4805
	3	0.4725	0.4700	0.4725	0.4725	0.4836

Table B.2: Combined localisation RMSE in bins, per condition, at the primary gate. Measured over each detector's own true positives, so the averages are not taken over identical target populations — see Section 5.6.

Condition	CA-CFAR	GO-CFAR	OS-CFAR	SO-CFAR	TinyCNN
isolated	0.528	0.508	0.511	0.503	0.936
masking	0.560	0.627	0.545	0.612	0.941
clutter	0.436	0.484	0.465	0.700	1.110
clutter edge	0.660	0.683	0.524	0.451	0.950
multi-target	0.521	0.529	0.521	0.536	0.913
mixed	0.613	0.597	0.572	0.647	0.930
unresolved	0.466	0.478	0.445	0.469	0.768
pooled	**0.544**	**0.559**	**0.517**	**0.534**	**0.921**

Table B.3: Achieved per-cell false-alarm rate as a ratio to the design rate of 10^{-4}, per condition and arm, including the target-free noise stratum. A value of 1.000 is an exactly matched operating point. This is Figure 5.1 in numerical form.

Condition	CA-CFAR	GO-CFAR	OS-CFAR	SO-CFAR	TinyCNN
clutter	0.996	0.993	0.991	0.987	0.996
clutter edge	0.998	0.994	0.996	1.002	0.996
isolated	1.009	0.998	0.992	0.999	1.001
masking	1.012	0.998	1.006	0.983	1.001
mixed	0.987	0.997	0.983	0.985	0.998
multi-target	1.004	1.002	0.989	0.994	0.997
noise	0.986	1.008	1.006	0.987	0.999
unresolved	1.013	0.992	1.005	0.990	0.996

Table B.4: The full benchmark at the ±3-bin robustness gate. The main text reports only the homogeneous-clutter row of this table. Same protocol, same frames, same 2,000 paired frame resamples, Bonferroni-corrected across 7 conditions.

Condition	**CFAR**	**Diff.**	**95 % CI**	p	**Sig.**
clutter edge	OS	+0.2710	[+0.2363, +0.3072]	0.000	yes
mixed	GO	+0.1002	[+0.0805, +0.1204]	0.000	yes
masking	OS	+0.0903	[+0.0727, +0.1089]	0.000	yes
isolated	CA	+0.0544	[+0.0306, +0.0797]	0.000	yes
multi-target	CA	+0.0358	[+0.0261, +0.0461]	0.000	yes
unresolved	CA	+0.0111	[+0.0045, +0.0187]	0.001	yes
clutter	CA	-0.0078	[−0.0283, +0.0136]	0.476	no

Table B.5: Exact dense-mask object recall per CARRADA test sequence, for every detector arm. The sequence-balanced mean difference is −0.0136 with a 95 % interval of [−0.0269, +0.0045] over 10000 bootstrap resamples. This is the numerical form of the right-hand panel of Figure 7.2.

Sequence	n	**TinyCNN**	**CA**	**Diff.**	**GO**	**SO**	**OS**
`2019-09-16-13-13-01`	150	0.9283	0.9533	-0.0250	0.9600	0.9600	0.9667
`2019-09-16-13-18-33`	212	0.9853	0.9953	-0.0100	0.9953	1.0000	1.0000
`2020-02-28-12-13-54`	523	0.8387	0.8700	-0.0313	0.9426	0.9598	0.9579
`2020-02-28-12-23-30`	376	0.8112	0.7846	+0.0266	0.8218	0.9654	0.9521
`2020-02-28-13-08-51`	224	0.8677	0.8795	-0.0117	0.9286	0.9732	0.9732
`2020-02-28-13-14-35`	142	0.9349	0.9648	-0.0299	0.9507	0.9648	0.9718

Table B.6: Per-seed exact dense-mask object recall on CARRADA, against the primary classical comparator's 0.8838. No seed was selected, dropped or ensembled; the reported figure is the mean of all 8. Standard deviation 0.0166.

Seed	**Recall**	**− CA-CFAR**	**Relative to CA**
0	0.8537	-0.0301	below
1	0.8783	-0.0055	below
2	0.8709	-0.0129	below
3	0.8752	-0.0086	below
4	0.8457	-0.0381	below
5	0.8697	-0.0141	below
6	0.8851	+0.0012	above
7	0.8980	+0.0141	above

Table B.7: Every pre-specified CARRADA test stratum, with its difference column. Section 7.5 discusses the range blocks; class comparisons are confounded by mask size and the cyclist row rests on two sequences only.

Stratum	n	**TinyCNN**	**CA-CFAR**	**Diff.**	**OS-CFAR**
Class					
car	513	0.9627	0.9747	-0.0119	0.9825
cyclist	159	0.8868	0.8742	+0.0126	0.9686
pedestrian	955	0.8209	0.8366	-0.0157	0.9571
Range block [bins]					
10-60	533	0.8455	0.9475	-0.1020	0.9719
60-110	853	0.8589	0.8300	+0.0289	0.9555
110-160	111	0.9809	0.9279	+0.0529	1.0000
160-210	79	0.9921	0.9114	+0.0807	0.9873
210-246	51	0.9485	0.9804	-0.0319	0.9804
Zero-Doppler					
away zero	262	0.9509	0.9580	-0.0072	0.9656
near zero	1365	0.8570	0.8696	-0.0126	0.9663